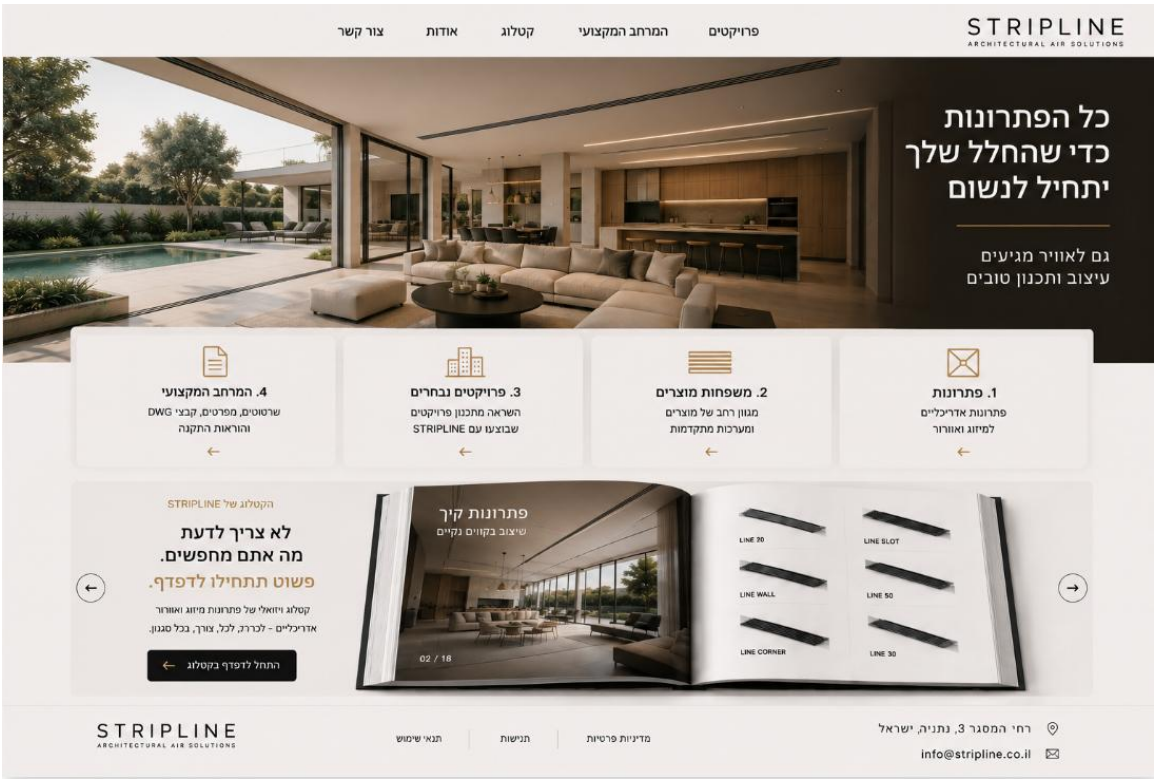
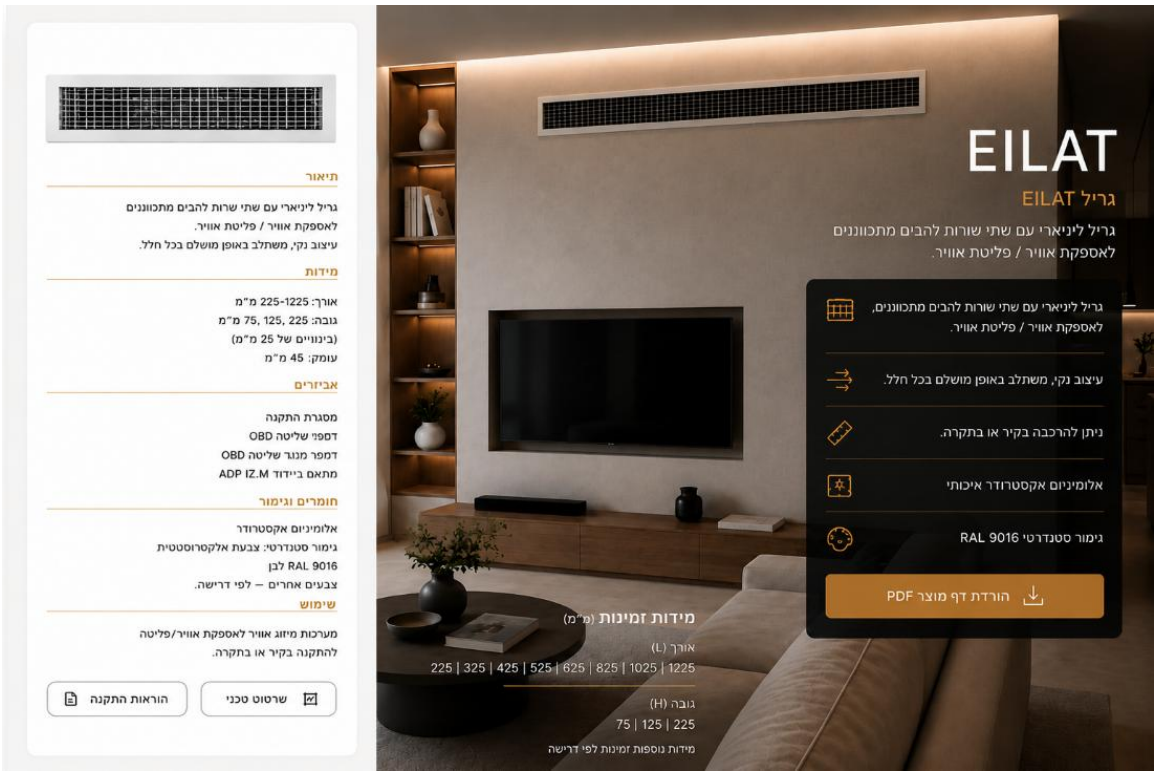
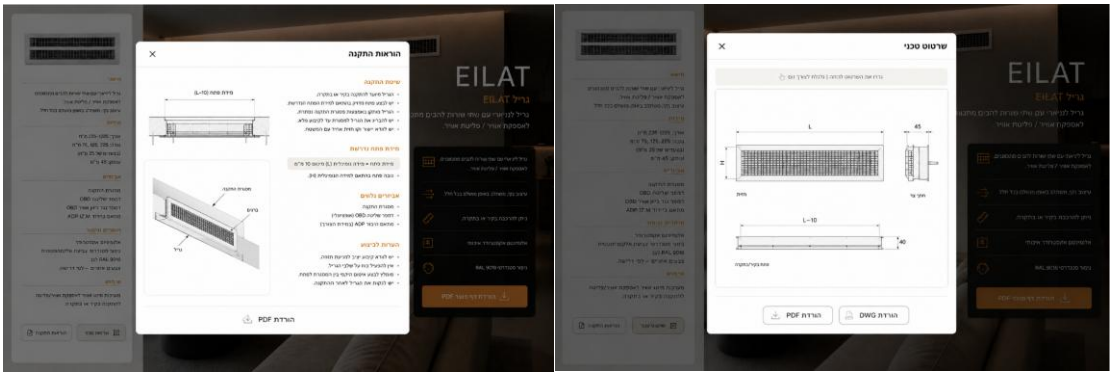
## עמוד הבית




## דף מוצר



התנהגות כפתורים




PDFדף מוצר מלא בהורדת

Ram air flow

רם אספקת אוויר - פתרונות דקורטיביים לפתחי מיזוג

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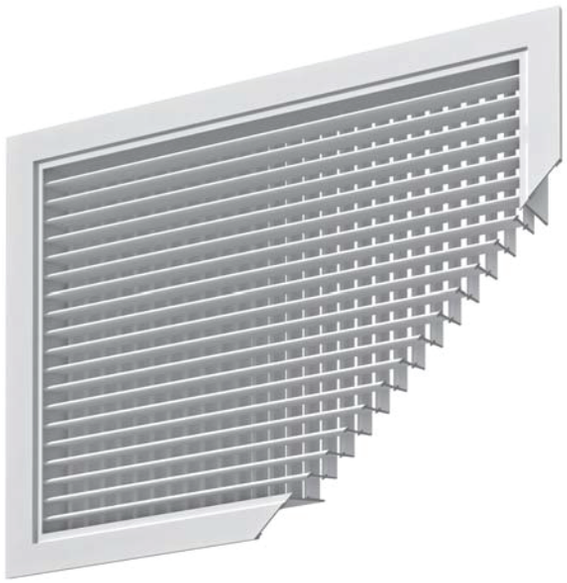
# Double Deflection Grille DD/DD1



DD

## Description
Aluminium ventilation grille equipped with individually adjustable blades that allow air flow direction. This can be provided with two rows of adjustable blades - called double deflection grille **DD** or a single row of adjustable blades - called **DD1**.

## Application
Supply/ exhaust of the air flow

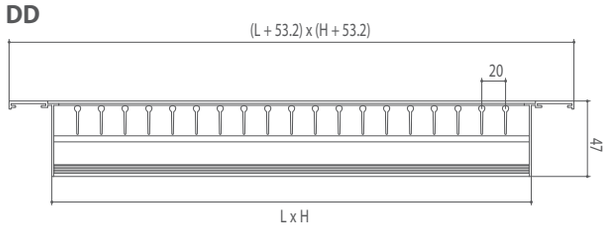
## Characteristics
The front row blades are mounted perpendicular to the length **"L"**.
The grille is provided with mounting holes (standard).
A masked settlement frame **CC** can be provided (optional)
The mounting of the grille into the settlement frame is carried out using a clip mechanism to ease the mounting/ removal process.
Specially designed for paneled ceiling insertion (600 x 600 mm), the 595 x 595 mm (exterior) grille is provided without mounting holes.
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.
The grilles with length more than 500 mm are provided with intermediate shaft for reinforcement.

## Accessories
Settlement frame made of galvanized sheet metal **CC**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

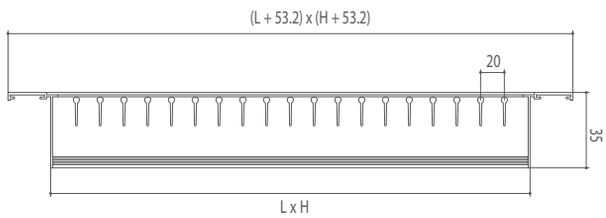
## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design

**DD**




**DD1**




## Assembly Methods
Visible mounting holes on grille frame (standard).
Settlement frame (optional)

## Selection table DD/DD1

| Air flow ($m^3/h$) | LxH | 200 x 100 | 300 x100 | 300 x 150 | 400 x 100 | 500 x 100 | 300 x 200 | 400 x 150 | 600 x 100 | 400 x 200 | 500 x 150 | 800x100 | 500x200 | 600 x 150 | 600 x200 | 800 x 150 | 500 x 300 | 800 x 200 | 600 x 300 | 800 x 300 | 1000 x 300 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ak($m^2$) | 0.011 | 0.016 | 0.024 | 0.024 | 0.03 | 0.04 | 0.04 | 0.04 | 0.049 | 0.049 | 0.049 | 0.06 | 0.06 | 0.078 | 0.078 | 0.105 | 0.105 | 0.132 | 0.16 | 0.21 |
| **150** | Veff[m/s] | 3.8 | 2.6 | | | | | | | | | | | | | | | | | | |
| | ΔPt [pa] | 10.0 | 5.0 | | | | | | | | | | | | | | | | | | |
| | X [m] | 4.1 | 3.4 | | | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | - | - | | | | | | | | | | | | | | | | | | |
| **200** | Veffm/s] | 5.0 | 3.5 | 2.3 | 2.3 | | | | | | | | | | | | | | | | |
| | ΔPt [pa] | 18.0 | 8.0 | 4.0 | 4.0 | | | | | | | | | | | | | | | | |
| | X [m] | 5.5 | 4.5 | 3.7 | 3.7 | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | 19.0 | - | - | - | | | | | | | | | | | | | | | | |
| **300** | Veff[m/s] | 7.6 | 5.2 | 3.5 | 3.5 | 2.8 | 2.1 | 2.1 | 2.1 | | | | | | | | | | | | |
| | ΔPt [pa] | 40.0 | 19.0 | 8.0 | 8.0 | 5.0 | 3.0 | 3.0 | 3.0 | | | | | | | | | | | | |
| | X [m] | 8.2 | 6.8 | 5.6 | 5.6 | 5.0 | 4.3 | 4.3 | 4.3 | | | | | | | | | | | | |
| | NR [dB(A)] | 28.0 | 21.0 | - | - | - | - | - | - | | | | | | | | | | | | |
| **400** | Veff[m/s] | | 6.9 | 4.6 | 4.6 | 3.7 | 2.8 | 2.8 | 2.8 | 2.3 | 2.3 | 2.3 | | | | | | | | | |
| | ΔPt [pa] | | 3.4 | 15.0 | 15.0 | 10.0 | 5.0 | 5.0 | 5.0 | 4.0 | 4.0 | 4.0 | | | | | | | | | |
| | X [m] | | 9.1 | 7.4 | 7.4 | 6.0 | 5.7 | 5.7 | 5.7 | 5.2 | 5.2 | 5.2 | | | | | | | | | |
| | NR [dB(A)] | | 28.0 | 20.0 | 20.0 | 16.0 | - | - | - | - | - | - | | | | | | | | | |
| **500** | Veff[m/s] | | | 5.8 | 5.8 | 4.6 | 3.5 | 3.5 | 3.5 | 2.8 | 2.8 | 2.8 | 2.3 | 2.3 | | | | | | | |
| | ΔPt [pa] | | | 23.0 | 23.0 | 15.0 | 8.0 | 8.0 | 8.0 | 6.0 | 6.0 | 6.0 | 4.0 | 4.0 | | | | | | | |
| | X [m] | | | 9.3 | 9.3 | 8.3 | 7.2 | 7.2 | 7.2 | 6.5 | 6.5 | 6.5 | 5.9 | 5.9 | | | | | | | |
| | NR [dB(A)] | | | 25.0 | 25.0 | 21.0 | - | - | - | - | - | - | - | - | | | | | | | |
| **700** | Veff[m/s] | | | | | 6.5 | 4.9 | 4.9 | 4.9 | 4.0 | 4.0 | 4.0 | 3.2 | 3.2 | 2.5 | 2.5 | | | | | |
| | ΔPt [pa] | | | | | 29.0 | 17.0 | 17.0 | 17.0 | 11.0 | 11.0 | 11.0 | 7.0 | 7.0 | 4.0 | 4.0 | | | | | |
| | X [m] | | | | | 12.0 | 10.0 | 10.0 | 10.0 | 9.1 | 9.1 | 9.1 | 8.2 | 8.2 | 7.2 | 7.2 | | | | | |
| | NR [dB(A)] | | | | | 29.0 | 23.0 | 23.0 | 23.0 | 19.0 | 19.0 | 19.0 | 15.0 | 15.0 | - | - | | | | | |
| **900** | Veff[m/s] | | | | | | 6.2 | 6.2 | 6.2 | 5.1 | 5.1 | 5.1 | 4.2 | 4.2 | 3.2 | 3.2 | 2.4 | 2.4 | | | |
| | ΔPt [pa] | | | | | | 27.0 | 27.0 | 27.0 | 18.0 | 18.0 | 18.0 | 12.0 | 12.0 | 7.0 | 7.0 | 4.0 | 4.0 | | | |
| | X [m] | | | | | | 13.0 | 13.0 | 13.0 | 12.0 | 12.0 | 12.0 | 11.0 | 11.0 | 9.2 | 9.2 | 8.0 | 8.0 | | | |
| | NR [dB(A)] | | | | | | 29.0 | 29.0 | 29.0 | 25.0 | 25.0 | 25.0 | 21.0 | 21.0 | 16.0 | 16.0 | - | - | | | |
| **1200** | Veff[m/s] | | | | | | | | | 6.8 | 6.8 | 6.8 | 5.5 | 5.5 | 4.3 | 4.3 | 3.2 | 3.2 | 2.5 | 2.1 | |
| | ΔPt [pa] | | | | | | | | | 32.0 | 32.0 | 32.0 | 22.0 | 22.0 | 13.0 | 13.0 | 7.0 | 7.0 | 4.0 | 3.0 | |
| | X [m] | | | | | | | | | 16.0 | 16.0 | 16.0 | 14.0 | 14.0 | 12.0 | 12.0 | 11.0 | 11.0 | 9.5 | 8.6 | |
| | NR [dB(A)] | | | | | | | | | 32.0 | 32.0 | 32.0 | 22.0 | 22.0 | 23.0 | 23.0 | 17.0 | 17.0 | - | - | |
| **1500** | Veff[m/s] | | | | | | | | | | | | 6.9 | 6.9 | 5.3 | 5.3 | 4.0 | 4.0 | 3.2 | 2.6 | 2.0 |
| | ΔPt [pa] | | | | | | | | | | | | 34.0 | 34.0 | 20.0 | 20.0 | 11.0 | 11.0 | 7.0 | 5.0 | 3.0 |
| | X [m] | | | | | | | | | | | | 18.0 | 18.0 | 15.0 | 15.0 | 13.0 | 13.0 | 12.0 | 11.0 | 9.4 |
| | NR [dB(A)] | | | | | | | | | | | | 34.0 | 34.0 | 29.0 | 29.0 | 23.0 | 23.0 | 18.0 | - | - |

## Annotation

- Ak [m²] - Free surface of the grille
- Veff[m/s] - Air velocity in grille
- Δ Pt [pa] - Pressure loss
- X [m] - Air jet length
- NR [dB(A)] - Noise level without the room attenuation
- Vk [m/s] - Speed at which airflow length was calculated "x" [m] - Vk [m/s] = 0.375 m/s

NR < 25 | 25≤NR≤35

## Correction coefficients for other air velocities

| Vk[m/s] | | 0.25 | 0.375 | 0.5 |
|---|---|---|---|---|
| x[m] | A | 1.5 | 1 | 0.75 |
| | B | 1.05 | 0.7 | 0.53 |

- A = with Coanda effect
- B = without Coanda effect

## Order codes

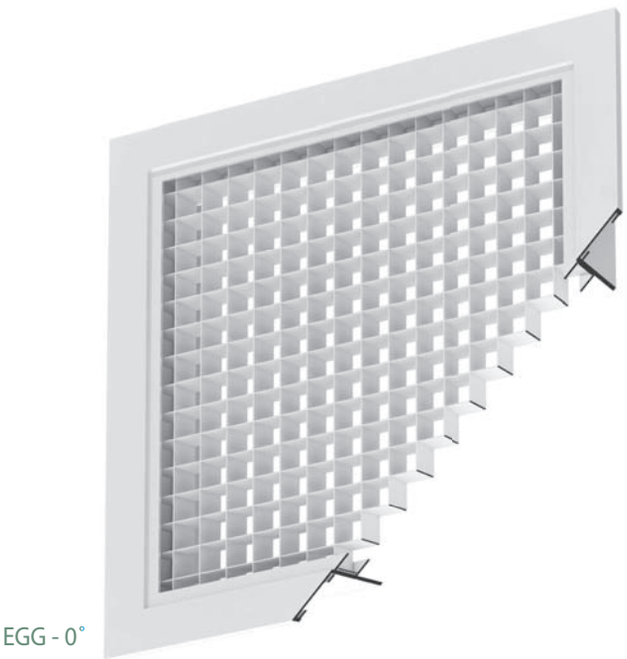
**Type** + **OBD** + **CC** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: DD / DD-1
- **OBD**: Air control damper
- **CC**: Settlement frame
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions**: **L X H**

| L X H | L X H |
|---|---|
| 200 x 100 | 500 x 300 |
| 300 x 100 | 600 x 100 |
| 300 x 150 | 600 x 150 |
| 300 x 200 | 600 x 200 |
| 400 x 100 | 600 x 300 |
| 400 x 150 | 800 x 100 |
| 400 x 200 | 800 x 150 |
| 500 x 100 | 800 x 200 |
| 500 x 150 | 800 x 300 |
| 500 x 200 | 1000 x 300 |

- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

# Perforated Grille EGG/EGGD



EGG - 0˚

## Description

Rectangular ventilation grille with grid that allows the evacuation of high air flow rates with minimum pressure loss **EGG/EGGD**.

## Application

Exhaust and recirculation of the air flow.

## Characteristics

The grid measures 12.5 x 12.5mm, at 0 or 45 degrees.
Grille is provided with countersunk holes for mounting (standard) **EGG/EGGD.**
Specifically designed for integration into the ceiling box dimensions 600 x 600 mm, the 595 x 595 mm grille (external mesurement) is provided without mounting holes.
Optionally for **EGG,** a settlement frame can be used for concealed mounting and mounting holes on inlet connection for **EGGD**.
The grille is mounted into the settlement frame using a clip mechanism, allowing fast detachment/installation.
**EGGD** is opened via a click type system (open/close by pressing).
Optionally, an air filter **FIL** can be mounted on the inlet connection of the grille.

## Accessories

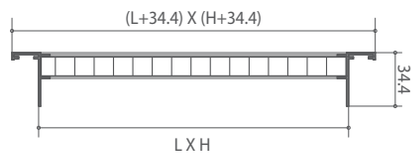
Air filter **FIL**
Air control damper with opposable blades **OBD**
Settlement frame made of galvanized metal sheet **CC**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing

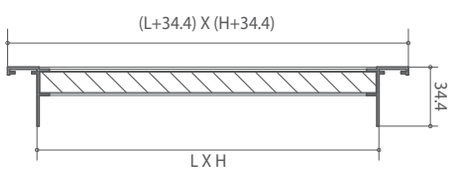
Aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design

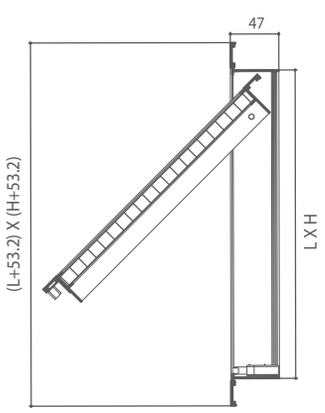
**EGG - 0˚**




**EGG - 45˚**




**EGGD-0˚**




## Assembly Methods

Mounting screws (standard)
Settlement frame (optional) **EGG.**
For **EGGD,** with screws on the inlet connection of the grille (optional).

## Selection table EGG/EGGD

| Air flow ($m^3/h$) | LxH Ak [$m^2$] | 200 x 100 0.0135 | 300 x 150 0.0324 | 400 x 200 0.0675 | 300 x 300 0.0729 | 500 x 300 0.1296 | 600 x 300 0.1539 | 800 x 300 0.206 | 600 x 600 0.3249 | 1000 x 600 0.55575 |
|---|---|---|---|---|---|---|---|---|---|---|
| **100** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | 2.1<br>< 20<br>7.0 | | | | | | | | |
| **200** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | 4.1<br>29.0<br>27.0 | 1.7<br>< 20<br>5.0 | | | | | | | |
| **300** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | 6.2<br>40.0<br>61.0 | 2.6<br>21.0<br>11.0 | 1.2<br>< 20<br>2.0 | 1.1<br>< 20<br>2.0 | | | | | |
| **400** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | 8.2<br>48.0<br>108.0 | 3.4<br>29.0<br>19.0 | 1.6<br>< 20<br>4.0 | 1.5<br>< 20<br>4.0 | | | | | |
| **500** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | 4.3<br>35.0<br>29.0 | 2.1<br>20.0<br>7.0 | 1.9<br>< 20<br>6.0 | 1.1<br>< 20<br>2.0 | | | | |
| **600** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | 5.1<br>40.0<br>42.0 | 2.5<br>25.0<br>10.0 | 2.3<br>23.0<br>8.0 | 1.3<br>< 20<br>3.0 | 1.1<br>< 20<br>2.0 | | | |
| **700** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | 6.0<br>44.0<br>58.0 | 2.9<br>29.0<br>13.0 | 2.7<br>27.0<br>11.0 | 1.5<br>< 20<br>4.0 | 1.3<br>< 20<br>3.0 | | | |
| **800** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | 6.9<br>48.0<br>75.0 | 3.3<br>33.0<br>17.0 | 3.0<br>31.0<br>15.0 | 1.7<br>< 20<br>5.0 | 1.4<br>< 20<br>3.0 | 1.1<br>< 20<br>2.0 | | |
| **900** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | 7.7<br>51.0<br>95.0 | 3.7<br>36.0<br>22.0 | 3.4<br>34.0<br>19.0 | 1.9<br>22.0<br>6.0 | 1.6<br>< 20<br>4.0 | 1.2<br>< 20<br>2.0 | | |
| **1000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | 4.1<br>39.0<br>27.0 | 3.8<br>37.0<br>23.0 | 2.1<br>25.0<br>7.0 | 1.8<br>21.0<br>5.0 | 1.3<br>< 20<br>3.0 | | |
| **1500** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | 6.2<br>50.0<br>61.0 | 5.7<br>48.0<br>52.0 | 3.2<br>36.0<br>17.0 | 2.7<br>32.0<br>12.0 | 2.0<br>26.0<br>7.0 | 1.3<br>< 20<br>3.0 | |
| **2000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | 7.6<br>56.0<br>93.0 | 4.3<br>44.0<br>29.0 | 3.6<br>40.0<br>21.0 | 2.7<br>34.0<br>12.0 | 1.7<br>24.0<br>5.0 | 1.0<br>< 20<br>2.0 |
| **3000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 6.4<br>55.0<br>66.0 | 5.4<br>51.0<br>47.0 | 4.0<br>45.0<br>26.0 | 2.6<br>35.0<br>11.0 | 1.5<br>24.0<br>4.0 |
| **4000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | 7.2<br>59.0<br>83.0 | 5.4<br>53.0<br>47.0 | 3.4<br>43.0<br>19.0 | 2.0<br>32.0<br>6.0 |
| **5000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | 6.7<br>59.0<br>73.0 | 4.3<br>49.0<br>29.0 | 2.5<br>38.0<br>10.0 |
| **6000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | | 5.1<br>54.0<br>42.0 | 3.0<br>43.0<br>14.0 |
| **7000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | | 6.0<br>59.0<br>57.0 | 3.5<br>47.0<br>20.0 |
| **8000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | | | 4.0<br>51.0<br>26.0 |
| **9000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | | | 4.5<br>54.0<br>32.0 |
| **10000** | Veff<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | | | 5.0<br>57.0<br>40.0 |

### Annotation

- Ak [m²] - Free surface of the grille
- Veff [m/s] - Air velocity inside the grille
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

NR < 35 | 35≤NR≤45 | NR>45

## Order codes

Type - 0/45° + FIL + OBD + CC + ADP IZ/N / Dimensions / L X H / O

- **EGG/EGGD**
- Air filter
- Air control damper
- Settlement frame
- Insulated/ non-insulated adapter
- Dimensions
  - L X H: 200 x 100, 300 x 150, 400 x 200, 300 x 300, 500 x 300
  - L X H: 600 x 300, 800 x 300, 600 x 600, 1000 x 600
- Other dimensions on request
- **0** Painted white RAL 9016
  **R** Other RAL paints
  Natural anodised

# EILAT Grille

## Description
Rectangular ventilation grille with two rows of adjustable blades and operating lever.

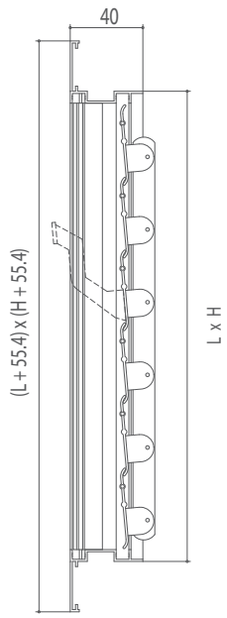
## Application
Supply/ exhaust of the air flow

## Characteristics
The front row blades are mounted perpendicular to the length "L" and are individually adjustable.
Additional blades are mounted perpendicular to the "H" (high) and allow adjustment of air flow through the operating lever.
The action lever operates on a maximum length of 500 mm. For larger dimensions, additional operating lever will be mounted.
The grille is provided with mounting holes (standard).

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Standard mounting, with screws.

## Selection table EILAT

| Air flow ($m^3/h$) | LxH | 200 x 100 | 300 x100 | 300 x 150 | 400 x 100 | 500 x 100 | 300 x 200 | 400 x 150 | 600 x 100 | 400 x 200 | 500 x 150 | 800x100 | 500x200 | 600 x 150 | 600 x200 | 800 x 150 | 500 x 300 | 800 x 200 | 600 x 300 | 800 x 300 | 1000 x 300 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ak($m^2$) | 0.011 | 0.016 | 0.024 | 0.024 | 0.03 | 0.04 | 0.04 | 0.04 | 0.049 | 0.049 | 0.049 | 0.06 | 0.06 | 0.078 | 0.078 | 0.105 | 0.105 | 0.132 | 0.16 | 0.21 |
| 150 | Veff[m/s] | 3.8 | 2.6 | | | | | | | | | | | | | | | | | | |
| | ΔPt [pa] | 10.0 | 5.0 | | | | | | | | | | | | | | | | | | |
| | X [m] | 4.1 | 3.4 | | | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | - | - | | | | | | | | | | | | | | | | | | |
| 200 | Veffm/s] | 5.0 | 3.5 | 2.3 | 2.3 | | | | | | | | | | | | | | | | |
| | ΔPt [pa] | 18.0 | 8.0 | 4.0 | 4.0 | | | | | | | | | | | | | | | | |
| | X [m] | 5.5 | 4.5 | 3.7 | 3.7 | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | 19.0 | - | - | - | | | | | | | | | | | | | | | | |
| 300 | Veff[m/s] | 7.6 | 5.2 | 3.5 | 3.5 | 2.8 | 2.1 | 2.1 | 2.1 | | | | | | | | | | | | |
| | ΔPt [pa] | 40.0 | 19.0 | 8.0 | 8.0 | 5.0 | 3.0 | 3.0 | 3.0 | | | | | | | | | | | | |
| | X [m] | 8.2 | 6.8 | 5.6 | 5.6 | 5.0 | 4.3 | 4.3 | 4.3 | | | | | | | | | | | | |
| | NR [dB(A)] | 28.0 | 21.0 | - | - | - | - | - | - | | | | | | | | | | | | |
| 400 | Veff[m/s] | | 6.9 | 4.6 | 4.6 | 3.7 | 2.8 | 2.8 | 2.8 | 2.3 | 2.3 | 2.3 | | | | | | | | | |
| | ΔPt [pa] | | 3.4 | 15.0 | 15.0 | 10.0 | 5.0 | 5.0 | 5.0 | 4.0 | 4.0 | 4.0 | | | | | | | | | |
| | X [m] | | 9.1 | 7.4 | 7.4 | 6.0 | 5.7 | 5.7 | 5.7 | 5.2 | 5.2 | 5.2 | | | | | | | | | |
| | NR [dB(A)] | | 28.0 | 20.0 | 20.0 | 16.0 | - | - | - | - | - | - | | | | | | | | | |
| 500 | Veff[m/s] | | | 5.8 | 5.8 | 4.6 | 3.5 | 3.5 | 3.5 | 2.8 | 2.8 | 2.8 | 2.3 | 2.3 | | | | | | | |
| | ΔPt [pa] | | | 23.0 | 23.0 | 15.0 | 8.0 | 8.0 | 8.0 | 6.0 | 6.0 | 6.0 | 4.0 | 4.0 | | | | | | | |
| | X [m] | | | 9.3 | 9.3 | 8.3 | 7.2 | 7.2 | 7.2 | 6.5 | 6.5 | 6.5 | 5.9 | 5.9 | | | | | | | |
| | NR [dB(A)] | | | 25.0 | 25.0 | 21.0 | - | - | - | - | - | - | - | - | | | | | | | |
| 700 | Veff[m/s] | | | | | 6.5 | 4.9 | 4.9 | 4.9 | 4.0 | 4.0 | 4.0 | 3.2 | 3.2 | 2.5 | 2.5 | | | | | |
| | ΔPt [pa] | | | | | 29.0 | 17.0 | 17.0 | 17.0 | 11.0 | 11.0 | 11.0 | 7.0 | 7.0 | 4.0 | 4.0 | | | | | |
| | X [m] | | | | | 12.0 | 10.0 | 10.0 | 10.0 | 9.1 | 9.1 | 9.1 | 8.2 | 8.2 | 7.2 | 7.2 | | | | | |
| | NR [dB(A)] | | | | | 29.0 | 23.0 | 23.0 | 23.0 | 19.0 | 19.0 | 19.0 | 15.0 | 15.0 | - | - | | | | | |
| 900 | Veff[m/s] | | | | | | 6.2 | 6.2 | 6.2 | 5.1 | 5.1 | 5.1 | 4.2 | 4.2 | 3.2 | 3.2 | 2.4 | 2.4 | | | |
| | ΔPt [pa] | | | | | | 27.0 | 27.0 | 27.0 | 18.0 | 18.0 | 18.0 | 12.0 | 12.0 | 7.0 | 7.0 | 4.0 | 4.0 | | | |
| | X [m] | | | | | | 13.0 | 13.0 | 13.0 | 12.0 | 12.0 | 12.0 | 11.0 | 11.0 | 9.2 | 9.2 | 8.0 | 8.0 | | | |
| | NR [dB(A)] | | | | | | 29.0 | 29.0 | 29.0 | 25.0 | 25.0 | 25.0 | 21.0 | 21.0 | 16.0 | 16.0 | - | - | | | |
| 1200 | Veff[m/s] | | | | | | | | | 6.8 | 6.8 | 6.8 | 5.5 | 5.5 | 4.3 | 4.3 | 3.2 | 3.2 | 2.5 | 2.1 | |
| | ΔPt [pa] | | | | | | | | | 32.0 | 32.0 | 32.0 | 22.0 | 22.0 | 13.0 | 13.0 | 7.0 | 7.0 | 4.0 | 3.0 | |
| | X [m] | | | | | | | | | 16.0 | 16.0 | 16.0 | 14.0 | 14.0 | 12.0 | 12.0 | 11.0 | 11.0 | 9.5 | 8.6 | |
| | NR [dB(A)] | | | | | | | | | 32.0 | 32.0 | 32.0 | 22.0 | 22.0 | 23.0 | 23.0 | 17.0 | 17.0 | - | - | |
| 1500 | Veff[m/s] | | | | | | | | | | | | 6.9 | 6.9 | 5.3 | 5.3 | 4.0 | 4.0 | 3.2 | 2.6 | 2.0 |
| | ΔPt [pa] | | | | | | | | | | | | 34.0 | 34.0 | 20.0 | 20.0 | 11.0 | 11.0 | 7.0 | 5.0 | 3.0 |
| | X [m] | | | | | | | | | | | | 18.0 | 18.0 | 15.0 | 15.0 | 13.0 | 13.0 | 12.0 | 11.0 | 9.4 |
| | NR [dB(A)] | | | | | | | | | | | | 34.0 | 34.0 | 29.0 | 29.0 | 23.0 | 23.0 | 18.0 | - | - |

NR < 25 | 25≤NR≤35

## Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without room attenuation
* Vk [m/s] - Speed at which airflow length was calculated
  "x" [m] - Vk [m/s] = 0.375 m/s

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **EILAT**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L X H** / **L X H**

| L X H | L X H |
|---|---|
| 200 x 100 | 500 x 300 |
| 300 x 100 | 600 x 100 |
| 300 x 150 | 600 x 150 |
| 300 x 200 | 600 x 200 |
| 400 x 100 | 600 x 300 |
| 400 x 150 | 800 x 100 |
| 400 x 200 | 800 x 150 |
| 500 x 100 | 800 x 200 |
| 500 x 150 | 800 x 300 |
| 500 x 200 | 1000 x300 |

- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

# Gravitational Grille NRD/NRDD

NRD

## Description
Rectangular aluminum grille for air overpressure with mobile blades (normally closed) .

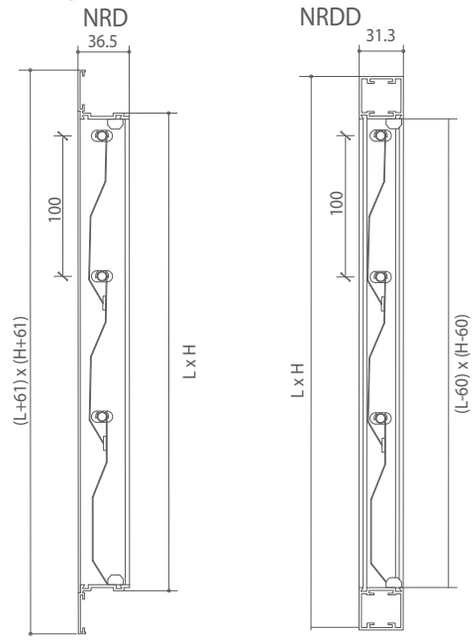
## Application
Exhaust/ supply of the airflow

## Characteristics
The standard grille is produced for air flow exhaust **NRD/NRDD**
Upon request, if the grid is used for introduction of air, the blades will be placed in reverse.
Plastic bearing sections to avoid friction.
Sealing gasket.
Central reinforcement for lengths over 500 mm.
Provided with mounting holes.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Anodised frame with natural aluminum blades
Plastic bearing sections

## Technical design




## Assembly Methods
Countersinking holes on the visible frame of the grille (standard) **NRD**

## Selection table NRD

| Air velocity inside the grille Veff(m/s) | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Pressure loss ΔPt (Pa) | 10 | 18 | 25 | 30 | 36 | 44 | 52 |

| Discharge area AK [m²] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Height (mm) | Length (mm) | | | | | | | | |
| | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 |
| **200** | 0.03 | 0.05 | 0.07 | 0.08 | 0.10 | 0.12 | 0.14 | 0.15 | 0.17 |
| 300 | 0.05 | 0.08 | 0.10 | 0.13 | 0.16 | 0.19 | 0.21 | 0.24 | 0.27 |
| 400 | 0.07 | 0.10 | 0.14 | 0.18 | 0.22 | 0.25 | 0.29 | 0.33 | 0.37 |
| 500 | 0.08 | 0.13 | 0.18 | 0.23 | 0.27 | 0.32 | 0.37 | 0.42 | 0.46 |
| 600 | 0.10 | 0.16 | 0.22 | 0.27 | 0.33 | 0.39 | 0.45 | 0.50 | 0.56 |
| 700 | 0.12 | 0.19 | 0.25 | 0.32 | 0.39 | 0.46 | 0.52 | 0.59 | 0.66 |
| 800 | 0.14 | 0.21 | 0.29 | 0.37 | 0.45 | 0.52 | 0.60 | 0.68 | 0.76 |
| 900 | 0.15 | 0.24 | 0.33 | 0.42 | 0.50 | 0.59 | 0.68 | 0.77 | 0.85 |
| 1000 | 0.17 | 0.27 | 0.37 | 0.46 | 0.56 | 0.66 | 0.76 | 0.85 | 0.95 |

## Selection table NRD - depending on air flow for velocity Veff = 3 m/s

| Air flow [mc/h] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Height (mm) | Length (mm) | | | | | | | | |
| | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 |
| **200** | 324 | 540 | 756 | 864 | 1080 | 1296 | 1512 | 1620 | 1836 |
| 300 | 540 | 864 | 1080 | 1404 | 1728 | 2052 | 2268 | 2592 | 2916 |
| 400 | 756 | 1080 | 1512 | 1944 | 2376 | 2700 | 3132 | 3564 | 3996 |
| 500 | 864 | 1404 | 1944 | 2484 | 2916 | 3456 | 3996 | 4536 | 4968 |
| 600 | 1080 | 1728 | 2376 | 2916 | 3564 | 4212 | 4860 | 5400 | 6048 |
| 700 | 1296 | 2052 | 2700 | 3456 | 4212 | 4968 | 5616 | 6372 | 7128 |
| 800 | 1512 | 2268 | 3132 | 3996 | 4860 | 5616 | 6480 | 7344 | 8208 |
| 900 | 1620 | 2592 | 3564 | 4536 | 5400 | 6372 | 7344 | 8316 | 9180 |
| 1000 | 1836 | 2916 | 3996 | 4968 | 6048 | 7128 | 8208 | 9180 | 10260 |

## Annotation

* Veff[m/s] - Air velocity inside the grille
* ΔPt [pa] - Pressure loss
* Ak [m²] - Free surface

Note: Optimal velocity for net dimensions selection is between 2 and 4 m/s.
By applying the formula Ak[m²] = Air flow [mc/h] / 3600[s] / Veff [m/s] and using the selection table, the dimensions of the grille can be determined.

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **L X H**

**NRD**
**NRDD**

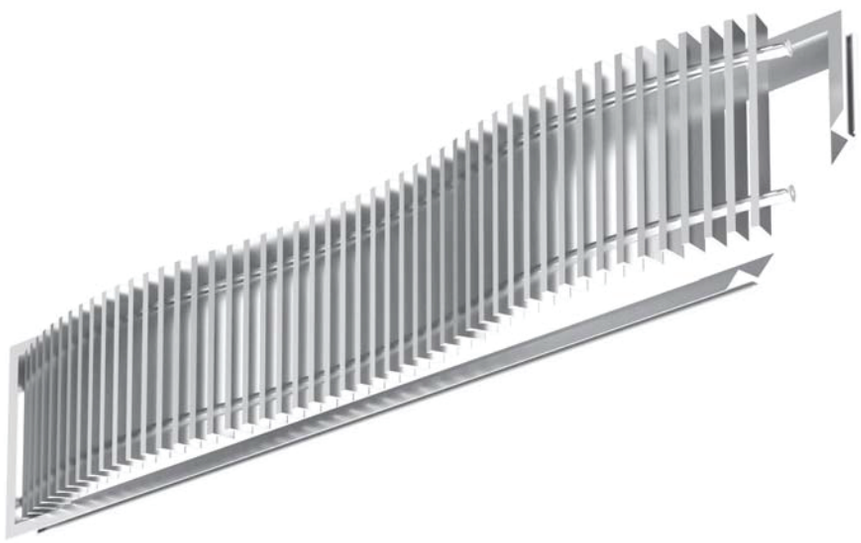
Insulated/Non-insulated adapter

**L X H**

200 x 200
300 x 300
400 x 400
500 x 500
600 x 600
700 x 700
800 x 800
900 x 900
1000 x 1000

Other dimensions on request

# Floor Linear Diffuser LBDF



## Description
Rectangular aluminum grille for floor mounting

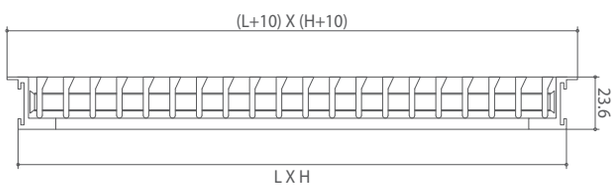
## Application
Various air supply/ exhaust systems (eg. floor convector).

## Characteristics
15 degree shaped blades perpendicular to the length "L" of the grille
The grille consists of an outdoor frame and a removable inner rolling core.
Removable core allows easy installation and easy cleaning.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Extruded aluminum with natural anodised finishing

## Technical design



## Selection table LBDF

| Supply | | Veff [m/s] | 2 |
|---|---|---|---|
| | | ΔPt [pa] | 4 |
| | | NR [dB(A)] | 21 |
| Length (mm) | Height (mm) | Air flow ($m^3$/h) | Ak (mp) |
| 450 | 100 | 100 | 0.014 |
| 550 | | 125 | 0.017 |
| 650 | | 150 | 0.021 |
| 800 | | 180 | 0.025 |
| 1050 | | 250 | 0.035 |
| 1300 | | 300 | 0.042 |
| 450 | 150 | 180 | 0.025 |
| 550 | | 230 | 0.032 |
| 650 | | 270 | 0.038 |
| 800 | | 350 | 0.049 |
| 1050 | | 450 | 0.063 |
| 1300 | | 570 | 0.079 |
| 450 | 200 | 270 | 0.038 |
| 550 | | 330 | 0.046 |
| 650 | | 400 | 0.056 |
| 800 | | 500 | 0.069 |
| 1050 | | 670 | 0.093 |
| 1300 | | 840 | 0.117 |

| Exhaust | | Veff [m/s] | 3 |
|---|---|---|---|
| | | ΔPt [pa] | 7.5 |
| | | NR [dB(A)] | 24 |
| Length (mm) | Height (mm) | Air flow ($m^3$/h) | Ak (mp) |
| 450 | 100 | 150 | 0.014 |
| 550 | | 180 | 0.017 |
| 650 | | 220 | 0.021 |
| 800 | | 270 | 0.025 |
| 1050 | | 370 | 0.035 |
| 1300 | | 460 | 0.042 |
| 450 | 150 | 270 | 0.025 |
| 550 | | 350 | 0.032 |
| 650 | | 410 | 0.038 |
| 800 | | 510 | 0.049 |
| 1050 | | 690 | 0.063 |
| 1300 | | 860 | 0.079 |
| 450 | 200 | 400 | 0.038 |
| 550 | | 500 | 0.046 |
| 650 | | 600 | 0.056 |
| 800 | | 750 | 0.069 |
| 1050 | | 1010 | 0.093 |
| 1300 | | 1250 | 0.117 |

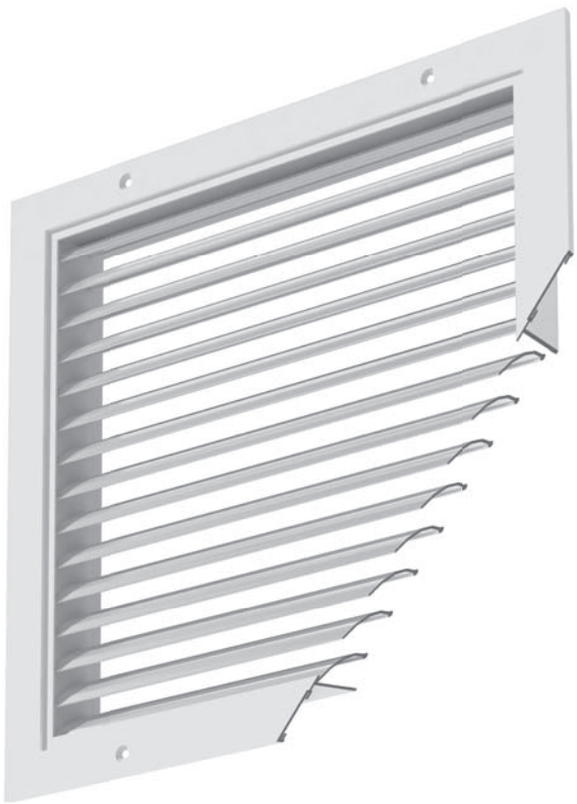
## Annotation

* Veff[m/s] - Air velocity in grille
* Δ Pt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation
* Ak [$m^2$] - Free surface of the grille

## Order codes

| Type | + | ADP IZ/N | / | Dimensions |
|---|---|---|---|---|
| LBDF | | Insulated/Non-insulated adapter | | On request |

# Aspiration Grille SD



## Description
Aluminum grille for air exhaust/ recirculation with fixed blades at 45 degrees **SD**.

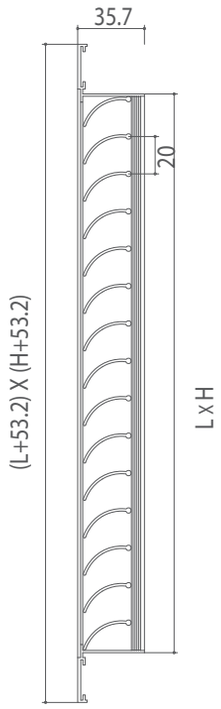
## Application
Exhaust/recirculation of the air.

## Characteristics
The grille is provided with fixed blades and countersinking holes for mounting, except size 595 x 595 mm (paneled ceiling).
Optionally, can be used a settlement frame for concealed mounting.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

## Accessories
Air filter **FIL**
Air control damper with opposable blades **OBD**.
Settlement frame **CC** made of galvanized steel sheet.
Insulated/Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design




## Assembly Method
With screws (standard).
Settlement frame for mounting (optional).

# Aspiration Grille SDD

## Description
Aluminum grille for air exhaust/ recirculation with fixed blades at 45 degrees **SDD**.

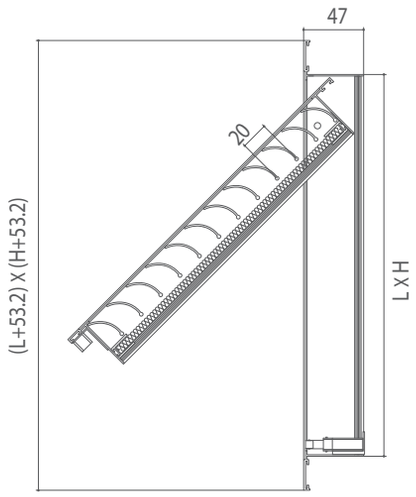
## Application
Exhaust/recirculation of the air

## Characteristics
The grille is provided with fixed blades and countersinking holes for mounting, except size 595 x 595 mm (paneled ceiling).
Optionally can be used a settlement frame for concealed mounting or mounting holes on inlet connection.
**SDD** is provided with access door and air filter.
The opening method of **SDD** consists of a click type system (open/close by pressing).
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

## Accessories
Air control damper with opposable blades **OBD.**
Settlement frame **CC** made of galvanized steel sheet.
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design




## Assembly Method
Mounting with screws on the visible frame (standard)
Mounting with screws on the inlet connection of the grille (optional).

## Selection table SD/SDD

| Air flow | L x H | 200 x 100 | 300 x 150 | 400 x 200 | 300 x 300 | 500 x 300 | 600 x 300 | 800 x 300 | 600 x 600 | 1000 x 600 |
|---|---|---|---|---|---|---|---|---|---|---|
| ($m^3$/h) | Ak [$m^2$] | 0.0076 | 0.0189 | 0.0345 | 0.0416 | 0.0721 | 0.0915 | 0.1173 | 0.1759 | 0.2934 |
| **60** | Veff[m/s] | 2.2 | | | | | | | | |
| | ΔPt [pa] | 5.0 | | | | | | | | |
| | NR [dB(A)] | 22.0 | | | | | | | | |
| **80** | Veff[m/s] | 2.9 | | | | | | | | |
| | ΔPt [pa] | 9.0 | | | | | | | | |
| | NR [dB(A)] | 29.0 | | | | | | | | |
| **100** | Veff[m/s] | 3.7 | | | | | | | | |
| | ΔPt [pa] | 14.0 | | | | | | | | |
| | NR [dB(A)] | 35.0 | | | | | | | | |
| **140** | Veff[m/s] | 5.1 | 2.1 | | | | | | | |
| | ΔPt [pa] | 27.0 | 2.0 | | | | | | | |
| | NR [dB(A)] | 44.0 | 16.0 | | | | | | | |
| **180** | Veff[m/s] | 6.6 | 2.6 | | | | | | | |
| | ΔPt [pa] | 45.0 | 3.0 | | | | | | | |
| | NR [dB(A)] | 51.0 | 22.0 | | | | | | | |
| **200** | Veff[m/s] | 7.3 | 2.9 | 1.6 | | | | | | |
| | ΔPt [pa] | 56.0 | 4.0 | 3.0 | | | | | | |
| | NR [dB(A)] | 53.0 | 25.0 | 22.0 | | | | | | |
| **300** | Veff[m/s] | | 4.4 | 2.4 | 2.0 | | | | | |
| | ΔPt [pa] | | 9.0 | 6.0 | 2.0 | | | | | |
| | NR [dB(A)] | | 36.0 | 32.0 | 22.0 | | | | | |
| **400** | Veff[m/s] | | 5.9 | 3.2 | 2.7 | | | | | |
| | ΔPt [pa] | | 16.0 | 12.0 | 4.0 | | | | | |
| | NR [dB(A)] | | 43.0 | 40.0 | 30.0 | | | | | |
| **500** | Veff[m/s] | | 7.3 | 4.0 | 3.3 | 1.9 | | | | |
| | ΔPt [pa] | | 25.0 | 18.0 | 7.0 | 2.0 | | | | |
| | NR [dB(A)] | | 49.0 | 46.0 | 36.0 | 24.0 | | | | |
| **600** | Veff[m/s] | | | | 4.0 | 2.3 | 1.8 | 1.4 | | |
| | ΔPt [pa] | | | | 11.0 | 3.0 | 2.0 | 1.0 | | |
| | NR [dB(A)] | | | | 41.0 | 29.0 | 24.0 | 18.0 | | |
| **800** | Veff[m/s] | | | | 5.3 | 3.1 | 2.4 | 1.9 | | |
| | ΔPt [pa] | | | | 19.0 | 6.0 | 4.0 | 2.0 | | |
| | NR [dB(A)] | | | | 48.0 | 36.0 | 32.0 | 26.0 | | |
| **1000** | Veff[m/s] | | | | 6.7 | 3.9 | 3.0 | 2.4 | 1.6 | |
| | ΔPt [pa] | | | | 30.0 | 10.0 | 6.0 | 4.0 | 1.0 | |
| | NR [dB(A)] | | | | 54.0 | 42.0 | 38.0 | 32.0 | 21.0 | |
| **1400** | Veff[m/s] | | | | | 5.4 | 4.3 | 3.3 | 2.2 | |
| | ΔPt [pa] | | | | | 20.0 | 13.0 | 7.0 | 2.0 | |
| | NR [dB(A)] | | | | | 51.0 | 46.0 | 40.0 | 29.0 | |
| **1800** | Veff[m/s] | | | | | | 5.5 | 4.3 | 2.8 | |
| | ΔPt [pa] | | | | | | 22.0 | 12.0 | 4.0 | |
| | NR [dB(A)] | | | | | | 53.0 | 47.0 | 36.0 | |
| **2000** | Veff[m/s] | | | | | | | 4.7 | 3.2 | |
| | ΔPt [pa] | | | | | | | 15.0 | 5.0 | |
| | NR [dB(A)] | | | | | | | 50.0 | 39.0 | |
| **3000** | Veff[m/s] | | | | | | | | 4.7 | 2.8 |
| | ΔPt [pa] | | | | | | | | 13.0 | 1.0 |
| | NR [dB(A)] | | | | | | | | 49.0 | 22.0 |
| **4000** | Veff[m/s] | | | | | | | | | 3.8 |
| | ΔPt [pa] | | | | | | | | | 1.0 |
| | NR [dB(A)] | | | | | | | | | 30.0 |

NR < 25 | 25≤NR≤35 | NR>35

## Annotation

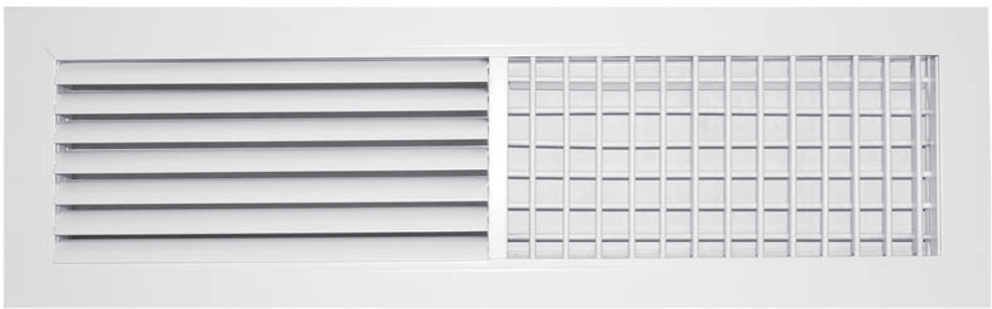
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the grille
* Δ Pt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

**Type** + **FIL** + **OBD** + **CC** + **ADP IZ/N** / **Dimensions** / **L x H** / **O**

- **Type**: **SD** **SDD**
- **FIL**: Air filter (for SD)
- **OBD**: Air control damper
- **CC**: Settlement frame
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L x H** 200 x 100, 300 x 150, 400 x 200, 300 x 300, 500 x 300, 600 x 300, 800 x 300, 600 x 600, 1000 x 600
- **L x H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

# Combined Grille SD-DD



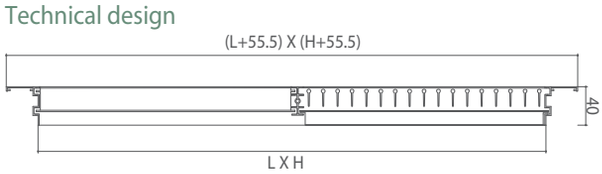
## Description
Aluminum rectangular grill that consists in simple deflection grille **SD** combined with a double deflection grille **DD**

## Application
Supply/ exhaust of the airflow

## Characteristics
The DD grille blades are individually adjustable and allow airflow direction and the SD grille blades are fixed.
Grille is provided with countersunk holes for mounting.
A settlement frame CC is recommended for concealed installation
The settlement frame can only be used if the SD blades are perpendicular to the length scale L
The air flow can be adjusted by mounting an air control damper OBD on the grille inlet connection.

## Accessories
Air control damper with opposable blades **OBD**
Settlement frame made of galvanized metal sheet **CC**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design



## Assembly Methods
Screws for mounting (standard)
Settlement frame for mounting (optional)

## Order codes

| Type | + | OBD | + | CC | + | ADP IZ/N | / | L X H | / | O |
|---|---|---|---|---|---|---|---|---|---|---|
| **SD-DD** | | Air control damper | | Settlement frame | | Indulated/Non-insulated adapter | | On request | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Transfer Grille TG / TGS

TG

## Description
Aluminum transfer grille for door mounting **TG/TGS**.

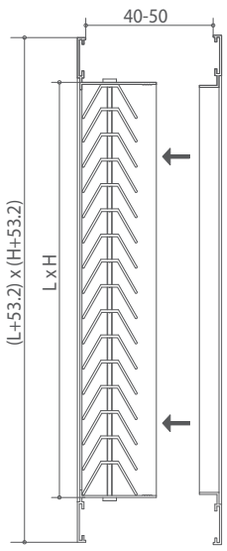
## Application
Designed to allow air transfer and pressure equalization between rooms

## Characteristics
Grid transfer is provided with "V"-shaped blades with no opposite visibility.
**TG** is used for doors with thickness of 40-50 mm, and **TGS** for doors with thickness of 20 mm.
The standard version of this product is provided with mounting holes.

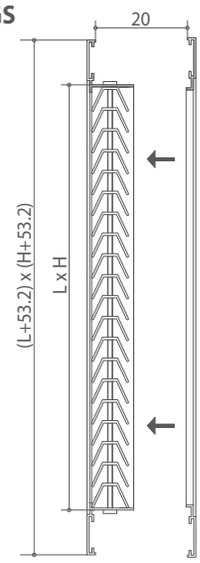
## Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design

**TG**




**TGS**




## Assembly Method
Standard mounting, with screws.

## Selection table TG

| Air flow ($m^3/h$) | L x H | 200 X 100 | 300 X 100 | 400 X 200 | 300 X 300 | 400 X 300 | 500 X 300 | 600 X 300 | 600 X 400 |
|---|---|---|---|---|---|---|---|---|---|
| | Ak [$m^2$] | 0.0099 | 0.0245 | 0.0447 | 0.0507 | 0.0681 | 0.0856 | 0.1031 | 0.138 |
| 50 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 1.4<br>30.0<br>10.0 | | | | | | | |
| 60 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 1.7<br>35.0<br>14.0 | 0.7<br>13.0<br>2.0 | | | | | | |
| 70 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 2.0<br>39.0<br>19.0 | 0.8<br>16.0<br>3.0 | | | | | | |
| 80 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 2.2<br>42.0<br>24.0 | 0.9<br>20.0<br>4.0 | | | | | | |
| 90 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 2.5<br>45.0<br>31.0 | 1.0<br>23.0<br>5.0 | 0.6<br>8.0<br>2.0 | | | | | |
| 100 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 2.8<br>48.0<br>38.0 | 1.1<br>25.0<br>6.0 | 0.6<br>10.0<br>2.0 | | | | | |
| 120 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | 3.4<br>52.0<br>55.0 | 1.4<br>30.0<br>9.0 | 0.7<br>15.0<br>3.0 | 0.7<br>12.0<br>2.0 | | | | |
| 140 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 1.6<br>34.0<br>12.0 | 0.9<br>19.0<br>4.0 | 0.8<br>16.0<br>3.0 | 0.6<br>8.0<br>2.0 | | | |
| 160 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 1.8<br>37.0<br>16.0 | 1.0<br>22.0<br>5.0 | 0.9<br>19.0<br>4.0 | 0.7<br>12.0<br>2.0 | | | |
| 180 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 2.0<br>40.0<br>20.0 | 1.1<br>25.0<br>6.0 | 1.0<br>22.0<br>5.0 | 0.7<br>15.0<br>3.0 | 0.6<br>9.0<br>2.0 | | |
| 200 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 2.3<br>43.0<br>25.0 | 1.2<br>28.0<br>8.0 | 1.1<br>25.0<br>6.0 | 0.8<br>17.0<br>3.0 | 0.6<br>12.0<br>2.0 | | |
| 250 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 2.8<br>48.0<br>39.0 | 1.6<br>33.0<br>12.0 | 1.4<br>30.0<br>9.0 | 1.0<br>23.0<br>5.0 | 0.8<br>17.0<br>3.0 | 0.7<br>13.0<br>2.0 | |
| 300 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | 3.4<br>53.0<br>57.0 | 1.9<br>38.0<br>17.0 | 1.6<br>35.0<br>13.0 | 1.2<br>27.0<br>7.0 | 1.0<br>22.0<br>5.0 | 0.8<br>17.0<br>3.0 | |
| 350 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | 2.2<br>42.0<br>23.0 | 1.9<br>38.0<br>18.0 | 1.4<br>31.0<br>10.0 | 1.1<br>26.0<br>6.0 | 0.9<br>21.0<br>4.0 | 0.7<br>14.0<br>2.0 |
| 400 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | 2.5<br>45.0<br>30.0 | 2.2<br>42.0<br>24.0 | 1.6<br>34.0<br>13.0 | 1.3<br>29.0<br>8.0 | 1.1<br>24.0<br>6.0 | 0.8<br>17.0<br>3.0 |
| 500 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | 3.1<br>50.0<br>47.0 | 2.7<br>47.0<br>37.0 | 2.0<br>40.0<br>20.0 | 1.6<br>34.0<br>13.0 | 1.3<br>30.0<br>9.0 | 1.0<br>23.0<br>5.0 |
| 600 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | 3.3<br>52.0<br>53.0 | 2.4<br>45.0<br>29.0 | 1.9<br>39.0<br>19.0 | 1.6<br>34.0<br>13.0 | 1.2<br>27.0<br>7.0 |
| 700 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | 2.9<br>48.0<br>40.0 | 2.3<br>43.0<br>25.0 | 1.9<br>38.0<br>17.0 | 1.4<br>31.0<br>10.0 |
| 800 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | 3.3<br>52.0<br>52.0 | 2.6<br>46.0<br>33.0 | 2.2<br>41.0<br>23.0 | 1.6<br>34.0<br>13.0 |
| 900 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | | 2.9<br>49.0<br>42.0 | 2.4<br>44.0<br>29.0 | 1.8<br>37.0<br>16.0 |
| 1000 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | | 3.2<br>52.0<br>52.0 | 2.7<br>47.0<br>36.0 | 2.0<br>40.0<br>20.0 |
| 1200 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | | | 3.2<br>52.0<br>51.0 | 2.4<br>44.0<br>29.0 |
| 1400 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | | | | 2.8<br>48.0<br>39.0 |
| 1600 | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [Pa] | | | | | | | | 3.2<br>51.0<br>51.0 |

## Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

NR < 35 | 35≤NR≤45 | NR>45

## Order codes

**Type** / **Dimensions** / **L x H** / **O**

**Type**
**TG**
**TGS**

**Dimensions**
LxH
200 x 100
300 x 150
400 x 200
300 x 300
400 x 300
600 x 200
500 x 300
600 x 300
600 x 400

**L x H**
Other dimensions on request

**O**
**0** Painted white RAL 9016
**R** Other RAL paints
Natural anodised

# Exterior Grille WPL/WPLS

## Description
Weatherproof rectangular aluminum ventilation grille, providing protection against ingress of rain, snow, leaves, birds and insects.

## Application
Supply/ exhaust of the airflow

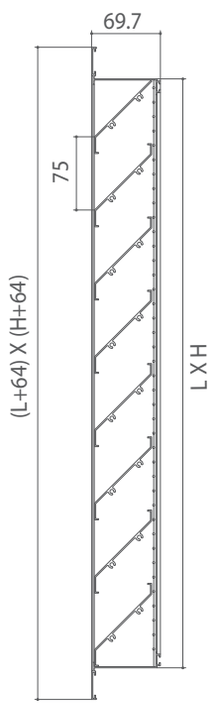
## Characteristics
The grille is provided with weatherproof blades, galvanized steel wire mesh and mounting holes for fixing.
Wire mesh used for **WPL** grille is 10 x 10 mm and 6.3 mm x 6.3 mm for **WPLS**
Available sizes:
**WPL** any size from 800 x 800 mm to 2000 x 2000 mm.
**WPLS** any size from 100 x 100 mm to 800 x 800 mm.
Modular execution for large dimensions

## Accessories
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

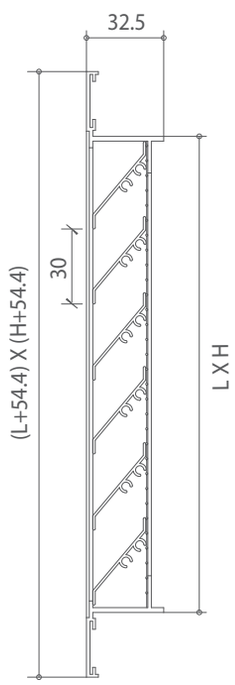
## Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum.

## Technical design

**WPL**




**WPLS**




## Assembly Method
Standard mounting, with screws.

## Selection table WPL

| Air flow ($m^3/h$) | Ak [$m^2$] | 0.30 | 0.40 | 0.50 | 0.60 | 0.80 | 1.00 | 1.5 | 2.00 | 2.5 | 3.00 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **3000** | Veff [m/s]<br>ΔPt [pa] | 2.9<br>10 | 2.2<br>12.0 | | | | | | | | |
| **4000** | Veff [m/s]<br>ΔPt [pa] | 4<br>15 | 2.9<br>25.0 | 2.4<br>14.0 | | | | | | | |
| **5000** | Veff [m/s]<br>ΔPt [pa] | 4.4<br>25 | 3.6<br>40.0 | 2.9<br>24.0 | 2.4<br>15.0 | | | | | | |
| **6000** | Veff [m/s]<br>ΔPt [pa] | 5.6<br>30 | 4.2<br>54.0 | 3.3<br>35.0 | 2.9<br>24.0 | 2.1<br>8 | | | | | |
| **7000** | Veff [m/s]<br>ΔPt [pa] | 6.5<br>28 | 4.9<br>77.0 | 4.0<br>48.0 | 3.5<br>37.0 | 2.4<br>15.0 | | | | | |
| **8000** | Veff [m/s]<br>ΔPt [pa] | | | 4.6<br>63.0 | 3.9<br>53.0 | 2.8<br>24.0 | 2.3<br>10 | | | | |
| **9000** | Veff [m/s]<br>ΔPt [pa] | | | 5.4<br>90.0 | 4.5<br>63.0 | 3.2<br>34.0 | 2.7<br>20.0 | | | | |
| **10000** | Veff [m/s]<br>ΔPt [pa] | | | | 4.9<br>77.0 | 3.5<br>37.0 | 2.8<br>24.0 | | | | |
| **20000** | Veff [m/s]<br>ΔPt [pa] | | | | | | 5.4<br>90.0 | 3.8<br>45.0 | 2.8<br>23.0 | 2.3<br>14.0 | |
| **30000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | 5.7<br>92.0 | 4.0<br>53.0 | 3.3<br>35.0 | 2.8<br>23.0 |
| **40000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | | 5.4<br>90.0 | 4.5<br>63.0 | 3.8<br>45.0 |

| Discharge area Ak [$m^2$] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Height (mm) | Length (mm) | | | | | | | | |
| | 800 | 900 | 1000 | 1100 | 1200 | 1400 | 1600 | 1800 | 2000 |
| **800** | 0.36 | 0.40 | 0.45 | 0.50 | 0.55 | 0.64 | 0.74 | 0.83 | 0.93 |
| **900** | 0.40 | 0.45 | 0.51 | 0.56 | 0.61 | 0.72 | 0.83 | 0.93 | 1.04 |
| **1000** | 0.45 | 0.51 | 0.57 | 0.63 | 0.69 | 0.81 | 0.94 | 1.06 | 1.18 |
| **1100** | 0.51 | 0.57 | 0.64 | 0.71 | 0.78 | 0.91 | 1.05 | 1.08 | 1.32 |
| **1200** | 0.55 | 0.62 | 0.70 | 0.77 | 0.84 | 0.99 | 1.14 | 1.28 | 1.43 |
| **1400** | 0.66 | 0.74 | 0.83 | 0.92 | 1.01 | 1.18 | 1.36 | 1.53 | 1.71 |
| **1600** | 0.75 | 0.85 | 0.95 | 1.05 | 1.15 | 1.36 | 1.56 | 1.76 | 1.96 |
| **1800** | 0.85 | 0.96 | 1.08 | 1.19 | 1.30 | 1.53 | 1.76 | 1.98 | 2.21 |
| **2000** | 0.95 | 1.08 | 1.21 | 1.34 | 1.47 | 1.72 | 1.98 | 2.23 | 2.49 |

## Selection table WPLS

| Air flow ($m^3/h$) | Ak [$m^2$] | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.08 | 0.10 | 0.15 | 0.20 | 0.25 | 0.30 | 0.40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **200** | Veff [m/s]<br>ΔPt [pa] | 5.6<br>50 | 2.8<br>8.0 | | | | | | | | | | | |
| **300** | Veff [m/s]<br>ΔPt [pa] | | 4.2<br>28.0 | 2.8<br>8.0 | | | | | | | | | | |
| **400** | Veff[m/s]<br>ΔPt [pa] | | 5.6<br>50.0 | 3.9<br>26.0 | 2.8<br>8.0 | 2.3<br>3.0 | | | | | | | | |
| **500** | Veff [m/s]<br>ΔPt [pa] | | | 4.8<br>38.0 | 3.6<br>20.0 | 2.8<br>8 | 2.3<br>3.0 | | | | | | | |
| **600** | Veff [m/s]<br>ΔPt [pa] | | | 5.8<br>53.0 | 4.3<br>29.0 | 3.4<br>18.0 | 3.0<br>14.0 | 2.2<br>2 | | | | | | |
| **700** | Veff [m/s]<br>ΔPt [pa] | | | | 5<br>40 | 4.0<br>25.0 | 3.4<br>18.0 | 2.5<br>7.0 | 2.2<br>2 | | | | | |
| **800** | Veff [m/s]<br>ΔPt [pa] | | | | 5.8<br>53.0 | 4.7<br>35.0 | 3.8<br>25 | 2.8<br>8 | 2.4<br>4.0 | | | | | |
| **1000** | Veff [m/s]<br>ΔPt [pa] | | | | | 5.5<br>50 | 4.8<br>38.0 | 3.5<br>18.0 | 2.8<br>8.0 | | | | | |
| **2000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | | 5.8<br>53.0 | 3.8<br>25.0 | 2.8<br>8.0 | 2.4<br>4.0 | | |
| **3000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | | | 5.8<br>53.0 | 4.4<br>30 | 3.5<br>19.0 | 2.8<br>8.0 | 2.2<br>2.0 |
| **4000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | | | | 5.6<br>51.0 | 4.6<br>32.0 | 3.8<br>25 | 2.8<br>8.0 |
| **5000** | Veff [m/s]<br>ΔPt [pa] | | | | | | | | | | | 5.5<br>50.0 | 4.8<br>37.0 | 3.5<br>18.0 |

| Discharge area Ak [$m^2$] | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Height (mm) | Length (mm) | | | | | | | |
| | 150 | 200 | 300 | 400 | 500 | 600 | 700 | 800 |
| **100** | 0.001 | 0.008 | 0.012 | 0.017 | 0.024 | 0.028 | 0.040 | 0.050 |
| **150** | 0.013 | 0.017 | 0.025 | 0.030 | 0.040 | 0.050 | 0.060 | 0.070 |
| **200** | 0.018 | 0.020 | 0.030 | 0.040 | 0.060 | 0.070 | 0.080 | 0.090 |
| **300** | 0.027 | 0.035 | 0.050 | 0.070 | 0.090 | 0.100 | 0.120 | 0.140 |
| **400** | 0.036 | 0.045 | 0.070 | 0.090 | 0.120 | 0.140 | 0.170 | 0.190 |
| **500** | 0.045 | 0.060 | 0.090 | 0.120 | 0.150 | 0.180 | 0.210 | 0.240 |
| **600** | 0.050 | 0.070 | 0.100 | 0.130 | 0.180 | 0.210 | 0.250 | 0.290 |
| **700** | 0.060 | 0.080 | 0.110 | 0.160 | 0.210 | 0.250 | 0.300 | 0.330 |
| **800** | 0.070 | 0.090 | 0.130 | 0.180 | 0.240 | 0.300 | 0.340 | 0.400 |

## Annotation

* Ak [$m^2$] - Free surface of the grille
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

Note: For different discharge areas than those recorded in selection tables, air velocity and pressure drop will be obtained by interpolation.

## Order codes

**Type** + **FIL** + **ADP IZ/N** / **Dimensions** / **L x H** / **O**

**Type**
- **WPL**
- **WPLS**

**FIL**
- Air filter

**ADP IZ/N**
- Insulated/Non-insulated adapter

**Dimensions**

| WPL | WPLS |
|---|---|
| 800 x 800 | 150 x 150 |
| 900 X 900 | 200 x 200 |
| 1000 X 1000 | 300 x 300 |
| 1100 X 1100 | 400 x 400 |
| 1200 X 1200 | 500 x 500 |
| 1400 X 1400 | 600 x 600 |
| 1600 X 1600 | 700 x 700 |
| 1800 X 1800 | 800 x 800 |
| 2000 X 2000 | |

**L x H**
- Other dimensions on request

**O**
- **0** Painted white RAL 9016
- **R** Other RAL paints
- Natural anodised

# Exterior Grille With Mobile Blades WPLH

## Description
Weatherproof rectangular aluminum grille with drive system/ control for the blades. Ensures protection against intrusion of rain, snow, leaves and birds.

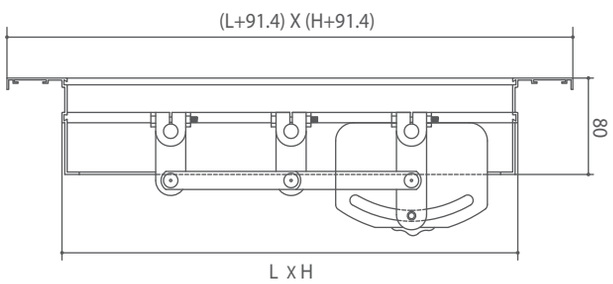
## Application
Supply/ exhaust of the airflow

## Characteristics
The grille is equipped with weatherproof blades, drive system for adjustment/ locking of the blades, wire mesh and countersinking holes for fixing.
The drive allows adjustment and locking of the blades in the desired position by operating the control lever.
Galvanized steel wire mesh with grids of 10 x 10 mm.

## Materials and finishing
Natural anodised aluminum

## Assembly Method
Standard mounting, with screws.

## Technical design




## Order codes

| Type | / | Dimensions | / | 0 |
|---|---|---|---|---|
| **WPLH** | | On request | | Natural anodised |

# Circular Exterior Grille WPLSR, WPLR

WPLSR

## Description
Weatherproof circular aluminum grille. Ensures protection against intrusion of rain, snow, leaves, birds and insects.

## Application
Supply/exhaust of the airflow

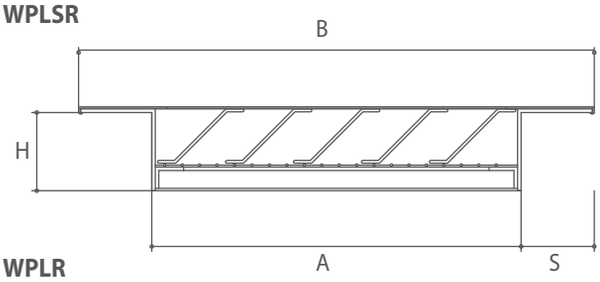
## Characteristics
The grille is equipped with weatherproof blades, wire mesh and countersinking holes for fixing.
The wire mesh is available in multiple sizes (grid: 1.5 x1.5, 6.3 x 6.3 and 10 x 10 mm).

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.

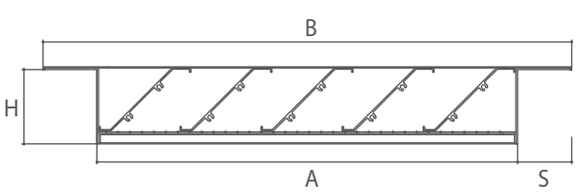
## Assembly Method
Standard mounting, with screws.

## Technical design

**WPLSR**




**WPLR**



| Type | Ø | A | B | S | H | Mesh dimension (mm) |
|---|---|---|---|---|---|---|
| WPLSR | 125 | 121 | 181 | 30 | 33 | 1.5 x 1.5 |
| | 160 | 156 | 216 | 30 | 33 | |
| | 200 | 196 | 256 | 30 | 33 | |
| | 250 | 246 | 306 | 30 | 33 | |
| | 315 | 295 | 375 | 40 | 33 | 6.3 x 6.3 |
| | 400 | 380 | 460 | 40 | 33 | |
| | 450 | 430 | 510 | 40 | 33 | |
| | 500 | 480 | 560 | 40 | 33 | |
| | 630 | 610 | 690 | 40 | 33 | |
| | 710 | 690 | 770 | 40 | 33 | |
| WPLR | 800 | 782 | 882 | 50 | 70 | 10 x 10 |
| | 1000 | 982 | 1082 | 50 | 70 | |

## Selection table WPLSR, WPLR

| Air flow | Diameter (mm) | 125 | 160 | 200 | 250 | 315 | 400 | 450 | 500 | 630 | 710 | 800 | 1000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ($m^3/h$) | Ak($m^2$) | 0.006 | 0.01 | 0.015 | 0.024 | 0.038 | 0.063 | 0.08 | 0.1 | 0.16 | 0.2 | 0.27 | 0.43 |
| 60 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | 2.6<br>34.0<br>25.0 | | | | | | | | | | | |
| 100 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | 4.3<br>37.0<br>29.0 | 2.8<br>34.0<br>27.0 | 1.9<br><20<br>4.0 | | | | | | | | | |
| 150 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | 4.2<br>40.0<br>36.0 | 2.8<br>34.0<br>25.0 | | | | | | | | | |
| 200 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | 3.7<br>40.0<br>40.0 | 2.3<br>34.0<br>18.0 | 2.0<br>20.0<br>9.0 | | | | | | | |
| 400 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | 4.6<br>38.0<br>30.0 | 2.9<br>34.0<br>30.0 | 1.8<br>30.0<br>12.0 | 1.4<br>23.0<br>7.0 | 1.1<br><16<br>4.0 | | | | |
| 700 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | 5.1<br>54.0<br>97.0 | 3.1<br>34.0<br>34.0 | 2.4<br>30.0<br>20.0 | 1.9<br>30.0<br>12.5 | 1.2<br>19.0<br>5.0 | | | |
| 800 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | 3.5<br>44.0<br>45.0 | 2.8<br>34.0<br>26.0 | 2.2<br>32.0<br>18.0 | 1.4<br>24.0<br>7.0 | 1.1<br><15<br>4.0 | | |
| 1000 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | 4.4<br>50.0<br>68.0 | 3.5<br>45.0<br>43.0 | 2.8<br>34.0<br>27.0 | 1.7<br>29.0<br>10.0 | 1.4<br>22.0<br>6.0 | 1.0<br><15<br>1.5 | |
| 1600 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | | | 4.4<br>51.0<br>70.0 | 2.8<br>34.0<br>25.0 | 2.2<br>30.0<br>14.0 | 1.6<br>25.0<br>4.0 | 1.0<br><15<br>1.5 |
| 2000 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | | | | 3.5<br>45.0<br>40.0 | 2.8<br>34.0<br>25.0 | 2.1<br>30.0<br>6.0 | 1.3<br>22.0<br>2.5 |
| 2500 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | | | | 4.3<br>52.0<br>63.0 | 3.5<br>45.0<br>36.0 | 2.6<br>34.0<br>9.0 | 1.6<br>28.0<br>3.5 |
| 4000 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | | | | | | 4.1<br>48.0<br>23.0 | 2.6<br>34.0<br>9.0 |
| 6000 | Veff [m/s]<br>NR [dB(A)]<br>△Pt [pa] | | | | | | | | | | | | 3.9<br>48.0<br>20.0 |

## Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the grille
* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss

## Order codes

**Type** / **Ø** / **0**

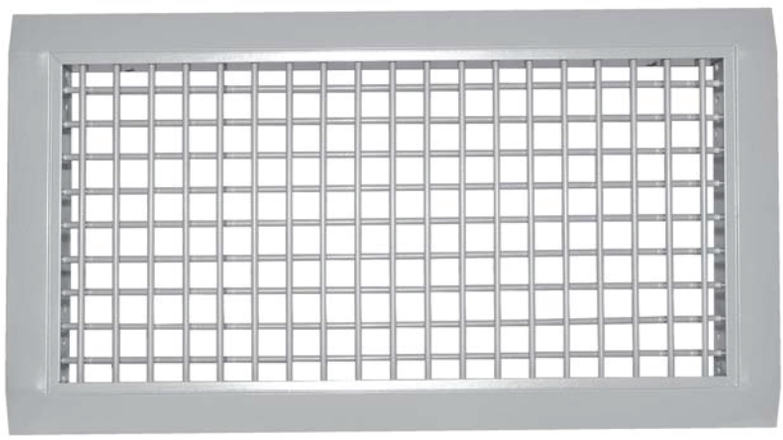
**WPLSR**
**WPLR**

WPLSR: 125, 160, 200, 250, 315, 400, 450, 500, 630, 710

WPLR: 800, 1000

**0** Painted white RAL 9016
**R** Other RAL paints

# Ventilation Grille for Circular Air Ducts RDD/RDD1



RDD

## Description
Rectangular aluminum grille for mounting on circular air ducts.

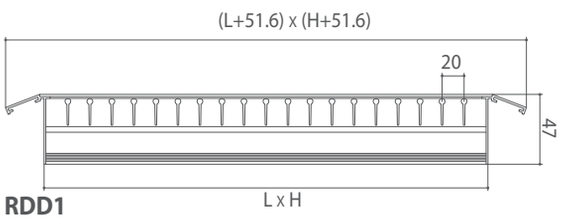
## Application
Supply/ exhaust of the air flow.

## Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
**RDD** has two rows of adjustable blades and **RDD1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

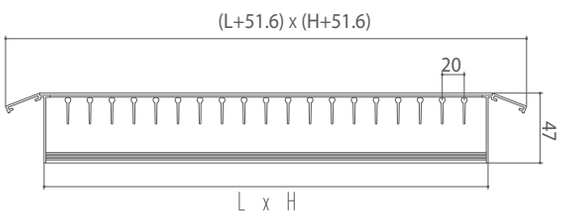
## Accessories
Air control damper with opposable blades **OBD.**

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request

## Technical design
**RDD**




**RDD1**




| Grill Dimensions | Ø tube | |
|---|---|---|
| | min. | max. |
| 425 x 75 | 160 | 400 |
| 525 x 75 | 160 | 400 |
| 625 x 75 | 160 | 400 |
| 425 x 125 | 315 | 900 |
| 525 x 125 | 315 | 900 |
| 625 x 125 | 315 | 900 |
| 425 x 225 | 630 | 1400 |
| 525 x 225 | 630 | 1400 |
| 625 x 225 | 630 | 1400 |
| 825 x 225 | 630 | 1400 |
| 1025 x 225 | 630 | 1400 |

## Assembly Method
Standard mounting, with screws

## Selection table RDD/RDD1

| Air flow m³/h | LxH | 425 x 75 | 525 x 75 | 625 x 75 | 425 x 125 | 525 x 125 | 625 x 125 | 425 x 225 | 525 x 225 | 625 x 225 | 825 x 225 | 1025 x 225 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ak[m²] | 0.013 | 0.016 | 0.019 | 0.025 | 0.031 | 0.037 | 0.049 | 0.061 | 0.073 | 0.097 | 0.122 |
| **200** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 5.1<br>24.0<br>4.3<br>12.1 | 4.6<br>20.0<br>3.5<br>8.0 | | | | | | | | | |
| **250** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 6.4<br>29.0<br>5.3<br>18.8 | 5.7<br>25.0<br>4.3<br>12.4 | 5.3<br>22.0<br>3.7<br>8.8 | | | | | | | | |
| **300** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 7.6<br>33.0<br>6.4<br>27.1 | 6.9<br>29.0<br>5.2<br>17.9 | 6.3<br>26.0<br>4.4<br>12.7 | 5.5<br>21.0<br>3.3<br>7.3 | | | | | | | |
| **350** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 8.9<br>37.0<br>7.5<br>36.9 | 8.0<br>33.0<br>6.1<br>24.4 | 7.4<br>30.0<br>5.1<br>17.3 | 6.4<br>24.0<br>3.9<br>10.0 | 5.8<br>20.0<br>3.1<br>6.5 | | | | | | |
| **400** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 10.2<br>40.0<br>8.5<br>48.2 | 9.2<br>36.0<br>6.9<br>31.8 | 8.4<br>33.0<br>5.8<br>22.6 | 7.3<br>28.0<br>4.4<br>13.0 | 6.6<br>23.0<br>3.6<br>8.5 | 6.0<br>20.0<br>3.0<br>6.0 | | | | | |
| **450** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 11.5<br>43.0<br>9.6<br>61.0 | 10.3<br>39.0<br>7.8<br>40.3 | 9.5<br>36.0<br>6.6<br>18.6 | 8.3<br>31.0<br>5.0<br>16.5 | 7.4<br>26.0<br>4.0<br>10.7 | 6.8<br>23.0<br>3.4<br>7.5 | | | | | |
| **500** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | 11.5<br>42.0<br>8.7<br>49.7 | 10.5<br>38.0<br>7.3<br>35.3 | 9.2<br>33.0<br>5.6<br>20.4 | 8.2<br>29.0<br>4.5<br>13.2 | 7.6<br>25.0<br>3.8<br>9.3 | 6.6<br>20.0<br>2.8<br>5.3 | | | | |
| **600** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | 12.6<br>43.0<br>8.8<br>50.8 | 11.0<br>38.0<br>6.7<br>29.3 | 9.9<br>33.0<br>5.4<br>19.1 | 9.1<br>30.0<br>4.5<br>13.4 | 7.9<br>24.0<br>3.4<br>7.6 | 7.1<br>20.0<br>2.7<br>4.9 | | | |
| **700** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | 12.9<br>41.0<br>7.8<br>39.9 | 11.5<br>37.0<br>6.3<br>26.0 | 10.6<br>34.0<br>5.3<br>18.2 | 9.2<br>28.0<br>4.0<br>0.4 | 8.2<br>24.0<br>3.2<br>6.7 | 7.5<br>20.0<br>2.7<br>4.7 | | |
| **800** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | 14.7<br>45.0<br>8.9<br>52.1 | 13.2<br>40.0<br>7.2<br>33.9 | 12.1<br>37.0<br>6.0<br>23.8 | 10.5<br>31.0<br>4.5<br>3.6 | 9.4<br>27.0<br>3.6<br>8.8 | 8.6<br>24.0<br>3.0<br>6.1 | | |
| **900** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | 14.8<br>43.0<br>8.1<br>42.9 | 13.6<br>40.0<br>6.8<br>30.1 | 11.8<br>34.0<br>5.1<br>17.2 | 10.6<br>30.0<br>4.1<br>11.1 | 9.7<br>26.0<br>3.4<br>7.7 | 8.4<br>21.0<br>2.6<br>4.4 | |
| **1000** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | 15.1<br>42.0<br>7.5<br>37.2 | 13.1<br>37.0<br>5.7<br>21.2 | 11.8<br>33.0<br>4.6<br>13.7 | 10.8<br>29.0<br>3.8<br>9.6 | 9.3<br>23.0<br>2.9<br>5.4 | |
| **1200** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | 15.7<br>41.0<br>6.8<br>30.5 | 14.1<br>37.0<br>5.5<br>19.7 | 12.9<br>33.0<br>4.6<br>13.8 | 11.2<br>28.0<br>3.4<br>7.8 | 10.0<br>23.0<br>2.7<br>4.9 |
| **1400** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | 18.4<br>45.0<br>7.9<br>41.6 | 16.5<br>41.0<br>6.4<br>26.8 | 15.1<br>37.0<br>5.3<br>18.7 | 13.1<br>32.0<br>4.0<br>10.6 | 11.6<br>27.0<br>3.2<br>6.7 |
| **1600** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | 18.8<br>44.0<br>7.3<br>35.0 | 17.2<br>40.0<br>6.1<br>24.5 | 14.9<br>35.0<br>4.6<br>13.9 | 13.3<br>30.0<br>3.6<br>8.8 |
| **1800** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | 19.4<br>43.0<br>6.8<br>31.0 | 16.8<br>38.0<br>5.2<br>17.5 | 15.0<br>33.0<br>4.1<br>11.1 |
| **2000** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | | 18.7<br>40.0<br>5.7<br>21.6 | 16.6<br>36.0<br>4.6<br>13.7 |
| **2500** | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | | | 20.8<br>41.0<br>5.7<br>21.4 |

Vk = 0.2 (m/s)

NR < 25 | 25≤NR≤35 | NR>35

### Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

### Order codes

| Type | + | OBD | / | Dimensions | / | L x H | / | 0 |
|---|---|---|---|---|---|---|---|---|
| RDD / RDD1 | | Air control damper | | L x H: 425 x 75, 525 x 75, 625 x 75 / 425 x 125, 525 x 125, 625 x 125 / 425 x 225, 525 x 225, 625 x 225, 825 x 225, 1025 x 225 | | Other dimensions on request | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Ventilation Grille for Circular Air Ducts RDR/RDR1

RDR

## Description
Rectangular aluminum grille for mounting on circular air ducts.

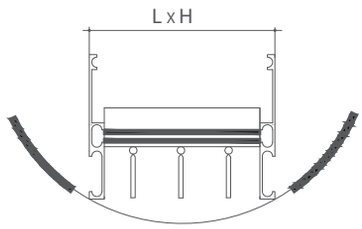
## Application
Supply/ exhaust of the air flow.

## Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
**RDR** has two rows of adjustable blades and **RDR1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

## Accessories
Air control damper with opposable blades **OBD**.

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request .

## Technical design
**RDR**




| Grill Dimension | Ø tube |
|---|---|
| 400 x 100 | 200 |
| 500 x 100 | 200 |
| 600 x 100 | 250 |
| 400 x 150 | 250 |
| 500 x 150 | 315 |
| 600 x 150 | 400 |
| 400 x 200 | 315 |
| 500 x 200 | 400 |
| 600 x 200 | 400 |
| 800 x 200 | 500 |
| 1000 x 200 | 630 |

## Assembly Method
Standard mounting, with screws.

## Selection table RDR/RDR1

| Air flow m³/h | LxH | 400x100 | 500x100 | 600x100 | 400x150 | 500x150 | 600x150 | 400x200 | 500x200 | 600x200 | 800x200 | 1000x200 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ak[m²] | 0.013 | 0.016 | 0.019 | 0.025 | 0.031 | 0.037 | 0.033 | 0.042 | 0.05 | 0.067 | 0.083 |
| 350 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 9.3<br>-<br>7.48<br>16.0 | 8.0<br>-<br>6.08<br>10.0 | 7.1<br>-<br>5.12<br>7.0 | 5.9<br>-<br>3.89<br>5.0 | | | | | | | |
| 400 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 10.7<br>-<br>8.55<br>21.0 | 9.2<br>-<br>6.94<br>13.0 | 8.1<br>-<br>5.85<br>9.0 | 6.7<br>-<br>4.44<br>6.0 | 5.8<br>-<br>3.58<br>4.0 | | 5.6<br>-<br>3.37<br>3.0 | | | | |
| 450 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 12.0<br>21.0<br>9.62<br>27.0 | 10.3<br>-<br>7.81<br>17.0 | 9.1<br>-<br>6.58<br>12.0 | 7.6<br>-<br>5.00<br>8.0 | 6.5<br>-<br>4.03<br>5.0 | | 6.3<br>-<br>3.79<br>4.0 | | | | |
| 500 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 13.3<br>25.0<br>10.7<br>33.0 | 11.5<br>19.0<br>8.7<br>21.0 | 10.2<br>-<br>7.3<br>14.0 | 8.4<br>-<br>5.6<br>10.0 | 7.3<br>-<br>4.5<br>6.0 | 6.4<br>-<br>3.8<br>4.0 | 6.9<br>-<br>4.2<br>5.0 | 6.0<br>-<br>3.3<br>3.0 | | | |
| 600 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 16.0<br>30.0<br>12.8<br>47.0 | 13.8<br>24.0<br>10.4<br>30.0 | 12.2<br>19.0<br>8.8<br>21.0 | 10.1<br>-<br>6.7<br>14.0 | 8.7<br>-<br>5.4<br>9.0 | 7.7<br>-<br>4.5<br>6.0 | 8.3<br>-<br>5.1<br>8.0 | 7.2<br>-<br>4.0<br>5.0 | 6.4<br>-<br>3.3<br>3.0 | 5.2<br>-<br>2.5<br>2.0 | |
| 700 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 18.7<br>35.0<br>15.0<br>64.0 | 16.1<br>29.0<br>12.2<br>41.0 | 14.2<br>24.0<br>10.2<br>28.0 | 11.8<br>19.0<br>7.8<br>19.0 | 10.2<br>-<br>6.3<br>12.0 | 9.0<br>-<br>5.3<br>8.0 | 9.7<br>-<br>5.9<br>10.0 | 8.4<br>-<br>4.6<br>7.0 | 7.4<br>-<br>3.9<br>5.0 | 6.1<br>-<br>2.9<br>3.0 | 5.3<br>-<br>2.3<br>2.0 |
| 800 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 21.3<br>39.0<br>17.1<br>84.0 | 18.4<br>33.0<br>13.9<br>53.0 | 16.2<br>28.0<br>11.7<br>37.0 | 13.5<br>23.0<br>8.9<br>25.0 | 11.6<br>-<br>7.2<br>16.0 | 10.3<br>-<br>6.0<br>11.0 | 11.1<br>-<br>6.7<br>14.0 | 9.6<br>-<br>5.3<br>9.0 | 8.5<br>-<br>4.4<br>6.0 | 7.0<br>-<br>3.3<br>3.0 | 6.0<br>-<br>2.7<br>2.0 |
| 900 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | 24.0<br>42.0<br>19.2<br>106.0 | 20.7<br>36.0<br>15.6<br>67.0 | 18.3<br>32.0<br>13.2<br>46.0 | 15.2<br>21.0<br>10.0<br>32.0 | 13.1<br>-<br>8.1<br>20.0 | 11.6<br>-<br>6.8<br>14.0 | 12.5<br>-<br>7.6<br>17.0 | 10.8<br>-<br>6.0<br>11.0 | 9.5<br>-<br>5.0<br>8.0 | 7.9<br>-<br>3.7<br>4.0 | 6.8<br>-<br>3.0<br>3.0 |
| 1000 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | 22.9<br>40.0<br>17.4<br>83.0 | 20.3<br>35.0<br>14.6<br>57.0 | 16.8<br>30.0<br>11.1<br>40.0 | 14.5<br>24.0<br>9.0<br>25.0 | 12.8<br>19.0<br>7.5<br>17.0 | 13.9<br>22.0<br>8.4<br>21.0 | 12.0<br>-<br>6.6<br>13.0 | 10.6<br>-<br>5.6<br>9.0 | 8.7<br>-<br>4.1<br>5.0 | 7.5<br>-<br>3.3<br>3.0 |
| 1200 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | 24.4<br>40.0<br>17.5<br>83.0 | 20.2<br>35.0<br>13.3<br>57.0 | 17.4<br>29.0<br>10.8<br>36.0 | 15.4<br>25.0<br>9.0<br>25.0 | 16.7<br>27.0<br>10.1<br>31.0 | 14.4<br>21.0<br>7.9<br>19.0 | 12.7<br>-<br>6.7<br>13.0 | 10.5<br>-<br>5.0<br>7.0 | 9.0<br>-<br>4.0<br>5.0 |
| 1500 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | 30.5<br>47.0<br>21.9<br>129.0 | 25.3<br>42.0<br>16.7<br>89.0 | 21.8<br>36.0<br>13.4<br>56.0 | 19.3<br>31.0<br>11.3<br>39.0 | 20.8<br>34.0<br>12.6<br>48.0 | 17.9<br>28.0<br>9.9<br>30.0 | 15.9<br>23.0<br>8.3<br>21.0 | 13.1<br>-<br>6.2<br>12.0 | 11.3<br>-<br>5.0<br>7.0 |
| 1750 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | 25.4<br>41.0<br>15.7<br>77.0 | 22.5<br>36.0<br>13.1<br>53.0 | 24.3<br>39.0<br>14.7<br>65.0 | 20.9<br>33.0<br>11.6<br>41.0 | 18.5<br>28.0<br>9.7<br>28.0 | 15.3<br>20.0<br>7.3<br>16.0 | 13.2<br>-<br>5.9<br>10.0 |
| 2000 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | 25.7<br>40.0<br>15.0<br>69.0 | 27.8<br>43.0<br>16.8<br>85.0 | 23.9<br>37.0<br>13.2<br>53.0 | 21.2<br>32.0<br>11.1<br>37.0 | 17.5<br>24.0<br>8.3<br>20.0 | 15.0<br>-<br>6.7<br>13.0 |
| 2500 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | 29.9<br>44.0<br>16.5<br>84.0 | 26.5<br>39.0<br>13.9<br>58.0 | 21.8<br>31.0<br>10.4<br>32.0 | 18.8<br>25.0<br>8.4<br>20.0 |
| 3000 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | 31.8<br>44.0<br>16.7<br>83.0 | 26.2<br>37.0<br>12.4<br>46.0 | 22.6<br>31.0<br>10.0<br>29.0 |
| 3500 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | 37.1<br>49.0<br>19.4<br>113.0 | 30.6<br>41.0<br>14.5<br>63.0 | 26.0<br>35.0<br>11.7<br>40.0 |
| 4000 | X [m]<br>NR [dB(A)]<br>Veff(m/s)<br>ΔPt [pa] | | | | | | | | | | | 30.0<br>40.0<br>13.4<br>52.0 |

Vk = 0.25 (m/s) Room height = 2.7 m

NR < 25 | 25≤NR≤35 | NR>35

### Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

### Order codes

| Type | + | OBD | / | Dimensions | / | L x H | / | 0 |
|---|---|---|---|---|---|---|---|---|
| RDR / RDR1 | | Air control damper | | L x H<br>400 x 100<br>500 x 100<br>600 x 100<br>400 x 150<br>500 x 150<br>600 x 150<br>400 x 200<br>500 x 200<br>600 x 200<br>800 x 200<br>1000 x 200 | | Other dimensions on request | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Decorative Grille D-SS1/D-SS2

## Description
Supply/exhaust aluminum grille with fixed blades in 1 or 2 directions **D-SS1/ D-SS2**.

## Application
Supply/ exhaust of the air flow

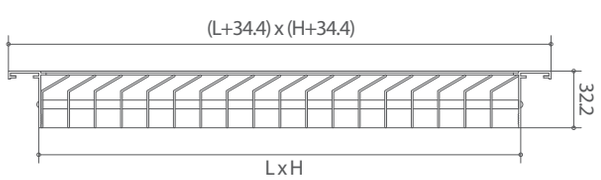
## Characteristics
Fixed blades in 1 or 2 directions mounted parallel to length "L"
The grille is provided with mounting holes (standard).
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.

## Accessories
Settlement frame made of galvanized sheet metal **CC**
Attachment spring **A**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

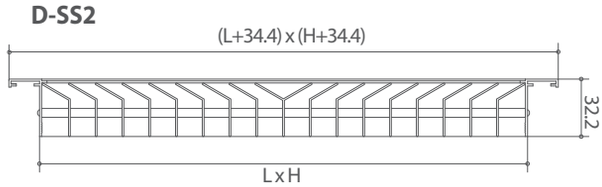
## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

## Technical design

**D-SS1**




**D-SS2**




## Assembly Methods
Screws mounting (standard)
Settlement frame (optional)
Attachment spring (optional)

## Selection table D-SS1 / D-SS2

L/H>2.1

| Air flow ($m^3/h$) | Dimensions Ak [$m^2$] | 500x200 0.05 | 600x200 0.06 | 800x200 0.08 | 1000x200 0.10 | 1200x200 800x300 0.12 | 1000x300 0.15 | 1200x300 1000x400 0.20 | 1200x400 0.25 | 1200x500 0.30 |
|---|---|---|---|---|---|---|---|---|---|---|
| **200** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | <3<br>1.1<br><20<br><4 | <3<br>0.9<br><20<br><4 | | | | | | | |
| **250** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.0<br>1.4<br><20<br><4 | <3<br>1.2<br><20<br><4 | <3<br>0.9<br><20<br><4 | | | | | | |
| **300** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.5<br>1.7<br><20<br><4 | 3.3<br>1.4<br><20<br><4 | 3.0<br>1.0<br><20<br><4 | <3<br>0.8<br><20<br><4 | | | | | |
| **350** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.5<br>1.9<br><20<br><4 | 4.0<br>1.6<br><20<br><4 | 3.8<br>1.2<br><20<br><4 | 3.0<br>1.0<br><20<br><4 | <3<br>0.8<br><20<br><4 | | | | |
| **400** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.0<br>2.2<br><20<br>4.0 | 4.5<br>1.9<br><20<br><4 | 4.0<br>1.4<br><20<br><4 | 3.5<br>1.1<br><20<br><4 | 3.0<br>0.9<br><20<br><4 | | | | |
| **450** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 5.0<br>2.1<br><20<br><4 | 4.0<br>1.6<br><20<br><4 | 3.5<br>1.3<br><20<br><4 | 3.0<br>1.0<br><20<br><4 | | | | |
| **500** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 5.5<br>2.3<br><20<br>4.5 | 5.0<br>1.7<br><20<br><4 | 4.5<br>1.4<br><20<br><4 | 4.0<br>1.2<br><20<br><4 | | | | |
| **600** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 6.5<br>2.8<br>20.0<br>6.0 | 6.0<br>2.1<br><20<br><4 | 5.3<br>1.7<br><20<br><4 | 5.0<br>1.4<br><20<br><4 | | | | |
| **700** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 9.0<br>3.2<br>25.0<br>8.5 | 8.0<br>2.4<br><20<br>4.5 | 7.0<br>1.9<br><20<br><4 | 6.3<br>1.6<br><20<br><4 | | | | |
| **800** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | 8.0<br>2.8<br>20.0<br>6.0 | 7.5<br>2.2<br><20<br>4.0 | 7.0<br>1.9<br><20<br><4 | | | | |
| **900** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | 9.5<br>3.1<br>23.0<br>8.0 | 8.5<br>2.5<br><20<br>5.0 | 8.0<br>2.1<br><20<br><4 | | | | |
| **1000** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 9.0<br>2.8<br>20.0<br>6.0 | 8.0<br>2.3<br><20<br>4.5 | | | | |
| **1200** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 12.5<br>3.3<br>28.0<br>9.0 | 11.0<br>2.8<br>20.0<br>6.0 | | | | |
| **1400** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 14.0<br>3.9<br>28.0<br>12.0 | 13.0<br>3.2<br>25.0<br>8.5 | 12.0<br>2.6<br><20<br>5.5 | | | |
| **1600** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 15.0<br>3.7<br>25.0<br>10.0 | 13.5<br>3.0<br>22.0<br>7.0 | 12.0<br>2.2<br><20<br>4.0 | 11.0<br>1.8<br><20<br><4 | |
| **1800** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | >15<br>4.2<br>30.0<br>14.0 | 15.0<br>3.3<br>28.0<br>9.0 | 13.0<br>2.5<br><20<br>5.0 | 11.5<br>2.0<br><20<br><4 | |
| **2000** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | >15<br>4.6<br>32.0<br>16.0 | >15<br>3.7<br>25.0<br>10.0 | 14.0<br>2.8<br>20.0<br>6.0 | 12.5<br>2.2<br><20<br>4.0 | |
| **2500** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | >15<br>5.8<br>37.0<br>25.0 | >15<br>4.6<br>32.0<br>16.0 | >15<br>3.5<br>25.0<br>9.5 | >15<br>2.8<br>20.0<br>6.0 | 15.0<br>2.3<br><20<br>4.5 |
| **3000** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | >15<br>6.9<br>43.0<br>38.0 | >15<br>5.6<br>33.0<br>20.0 | >15<br>4.2<br>30.0<br>14.0 | >15<br>3.3<br>28.0<br>9.0 | >15<br>2.8<br>20.0<br>6.0 |
| **3500** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | >15<br>8.1<br>47.0<br>50.0 | >15<br>6.5<br>40.0<br>30.0 | >15<br>4.9<br>32.0<br>18.0 | >15<br>3.9<br>28.0<br>12.0 | >15<br>3.2<br>25.0<br>8.5 |
| **3600** | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | | | >15<br>5.0<br>37.0<br>24.0 | >15<br>4.0<br>28.0<br>13.0 | >15<br>3.3<br>28.0<br>9.0 |

## Selection table D-SS1 , D-SS2

L/H<2.1

| Air flow ($m^3/h$) | Dimensions | 400x200 | 500x300 | 600x300 | 600x400 | 800x400 |
|---|---|---|---|---|---|---|
| | Ak [$m^2$] | 0.04 | 0.07 | 0.08 | 0.12 | 0.15 |
| 200 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | <1<br>1.4<br><20<br><4 | <1<br>0.8<br><20<br><4 | | | |
| 250 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | <1<br>1.7<br><20<br><4 | <1<br>1.0<br><20<br><4 | <1<br>0.9<br><20<br><4 | | |
| 300 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 1.0<br>2.1<br><20<br><4 | <1<br>1.2<br><20<br><4 | <1<br>1.0<br><20<br><4 | <1<br>0.7<br><20<br><4 | |
| 350 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 1.2<br>2.4<br><20<br>4.5 | <1<br>1.4<br><20<br><4 | <1<br>1.2<br><20<br><4 | <1<br>0.8<br><20<br><4 | <1<br>0.6<br><20<br><4 |
| 400 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 1.4<br>2.8<br>20.0<br>6.0 | 1.0<br>1.6<br><20<br><4 | <1<br>1.4<br><20<br><4 | <1<br>0.9<br><20<br><4 | <1<br>0.7<br><20<br><4 |
| 450 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 1.6<br>3.1<br>23.0<br>8.0 | 1.2<br>1.8<br><20<br><4 | 1.0<br>1.6<br><20<br><4 | <1<br>1.0<br><20<br><4 | <1<br>0.8<br><20<br><4 |
| 500 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 1.8<br>3.5<br>25.0<br>9.5 | 1.4<br>2.0<br><20<br><4 | 1.2<br>1.7<br><20<br><4 | <1<br>1.2<br><20<br><4 | <1<br>0.9<br><20<br><4 |
| 600 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 1.8<br>2.4<br><20<br>4.5 | 1.5<br>2.1<br><20<br><4 | 1.2<br>1.4<br><20<br><4 | <1<br>1.1<br><20<br><4 |
| 700 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 1.9<br>2.8<br>20.0<br>6.0 | 1.8<br>2.4<br><20<br>4.5 | 1.5<br>1.6<br><20<br><4 | 1.2<br>1.3<br><20<br><4 |
| 800 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 2.1<br>3.2<br>25.0<br>8.5 | 2.0<br>2.8<br>20.0<br>6.0 | 1.8<br>1.9<br><20<br><4 | 1.4<br>1.5<br><20<br><4 |
| 900 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | 2.2<br>3.1<br>23.0<br>8.0 | 1.9<br>2.1<br><20<br><4 | 1.5<br>1.7<br><20<br><4 |
| 1000 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | 2.5<br>3.5<br>25.0<br>9.5 | 1.9<br>2.3<br><20<br>4.5 | 1.8<br>1.9<br><20<br><4 |
| 1200 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | 2.9<br>4.2<br>30.0<br>14.0 | 2.4<br>2.8<br>20.0<br>6.0 | 2.1<br>2.2<br><20<br>4.0 |
| 1400 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 2.8<br>3.2<br>25.0<br>8.5 | 2.4<br>2.6<br><20<br>5.5 |
| 1600 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 3.2<br>3.7<br>25.0<br>10.0 | 2.8<br>3.0<br>22.0<br>7.0 |
| 1800 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 3.6<br>4.2<br>30.0<br>14.0 | 3.1<br>3.3<br>28.0<br>9.0 |
| 2000 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 4.0<br>4.6<br>32.0<br>16.0 | 3.5<br>3.7<br>25.0<br>10.0 |
| 2500 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 4.3<br>4.6<br>32.0<br>16.0 |
| 3000 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 5.0<br>5.6<br>33.0<br>20.0 |
| 3500 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 6.5<br>6.5<br>40.0<br>30.0 |
| 3600 | X [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 7.0<br>6.7<br>42.0<br>33.0 |

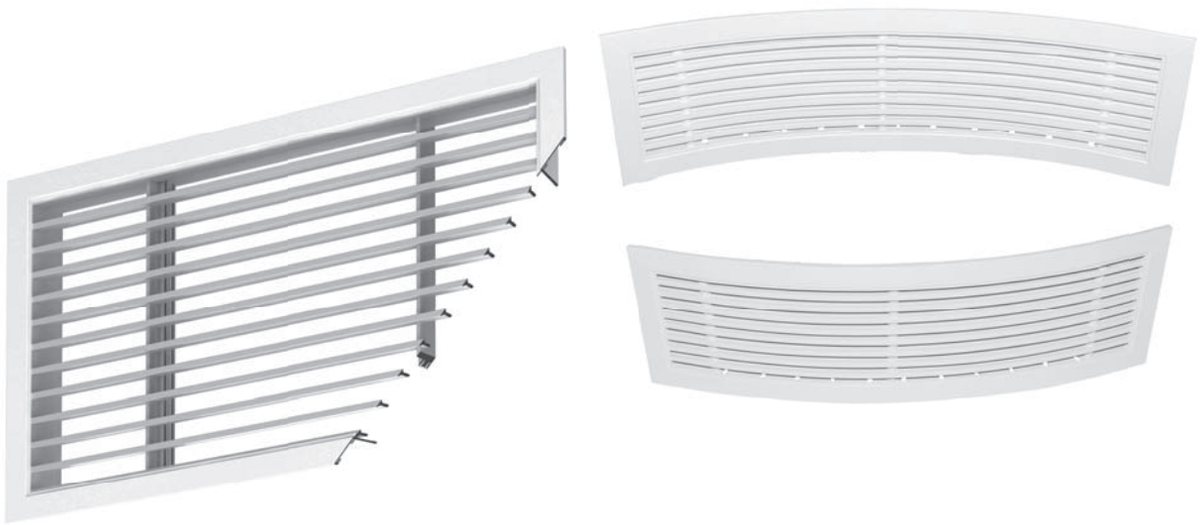
## Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.5 m/s - Speed at which airflow length was calculated "x" [m]

## Order codes

| Type | + | CC | + | A (Attachment spring) | + | FIL | + | OBD | + | ADP IZ/N | / | Dimensions | / | O |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D-SS1 / D-SS2 | | Settlement frame | | No. of springs depending on grille dimensions | | Air filter | | Air control damper | | Insulated/ Non-insulated adapter | | On request | | **O** Painted white RAL 9016<br>**R** Other RAL paints<br>Natural anodised |

# Linear Decorative Grille LBD



### Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

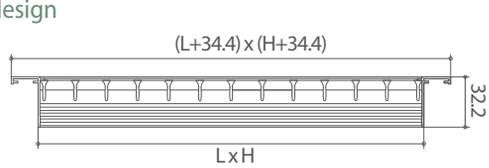
### Application
Supply/exhaust of the air flow.

### Characteristics
0 or 15 degree fixed blades, parallel with length "L".
The grille is provided with mounting holes (standard)
The air flow can be adjusted by mounting an air control damper OBD on the inlet connection.
On demand, the grille can be curved (concave/convex)

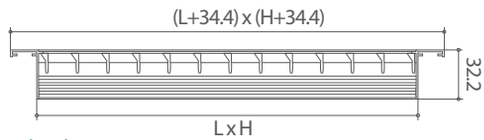
### Accessories
Settlement frame made of galvanized sheet metal **CC**
Mounting spring **A.**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

### Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

### Technical design
**LBD - 0°**

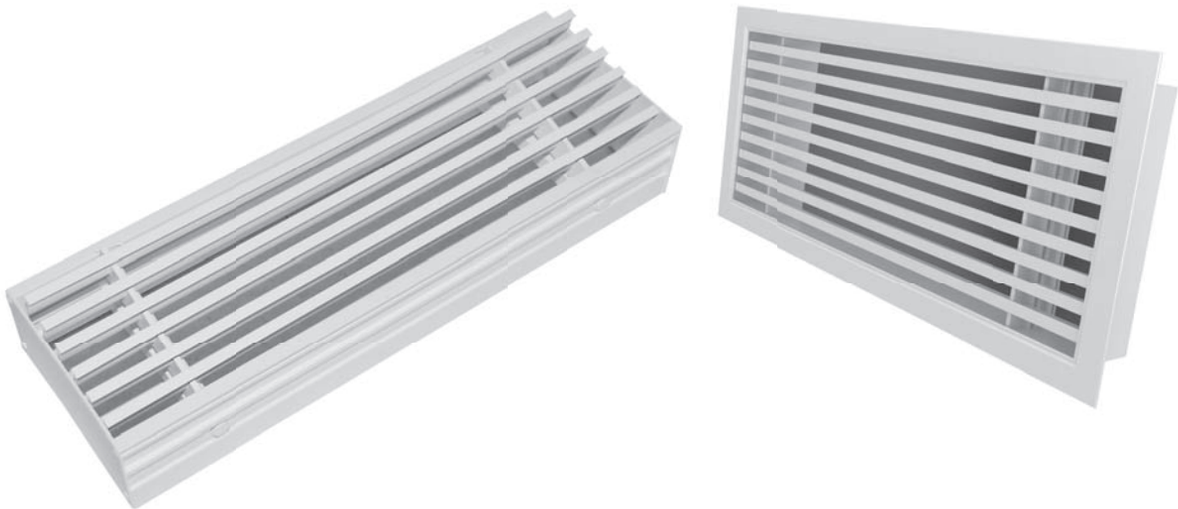



**LBD - 15°**




### Assembly Methods
Screws (standard).
Settlement frame (optional)
Mounting spring (optional).

# Linear Decorative Grille LBDB/LBDD



## Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

## Application
Supply/exhaust of the air flow.

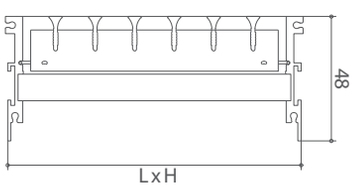
## Characteristics
**LBDB**
- 0 or 15 degree fixed blades, parallel with length "L".
- Removable blades system
- Encased mounting into wall/ ceiling

**LBDD**
- Two rows of blades
- Frontal fixed blades at 0°/15°, parallel with lenght "L"
- Individually adjustable blades that allow air flow direction
- Provided with mounting holes (standard).

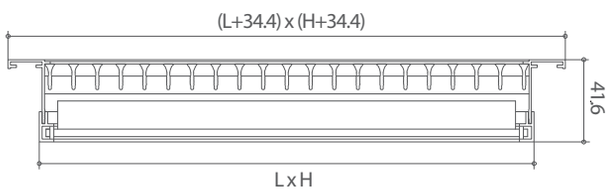
## Accessories
Air control damper with opposable blades **OBD (LBDB)**
Mounting spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design
**LBDB**




**LBDD**




## Assembly Methods
LBDB - With screws on the inlet connection of the grille
LBDD - With screws on the frame (standard).
Mounting spring

## Selection table LBD / LBDB / LBDD

| Air flow m³/h | L x H | 250x100 | 300x100 | 400x100 | 500x100 | 600x100 | 800x100 | 1000x100 | 1500x100 | 400x150 | 600x150 | 1000x150 | 600x200 | 800x200 | 1000x200 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ak(m²) | 0.011 | 0.013 | 0.019 | 0.024 | 0.029 | 0.041 | 0.05 | 0.075 | 0.033 | 0.045 | 0.081 | 0.064 | 0.085 | 0.12 |
| 140 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | 3.0<br><20<br>6.0<br>3.2 | | | | | | | | | | | | | |
| 200 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | 4.2<br>25.0<br>10.0<br>4.6 | 3.8<br>21.0<br>10.0<br>4.0 | 3.2<br><20<br>5.0<br>2.8 | | | | | | | | | | | |
| 300 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | 6.3<br>35.0<br>30.0<br>6.9 | 5.8<br>30.0<br>18.0<br>6.0 | 5.0<br>24.0<br>10.0<br>4.2 | 4.3<br>18.0<br>7.0<br>3.3 | | | | | | | | | | |
| 400 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | 8.5<br>43.0<br>50.0<br>9.3 | 7.8<br>38.0<br>35.0<br>7.9 | 6.5<br>32.0<br>18.0<br>5.6 | 5.8<br>25.0<br>10.0<br>4.4 | 5.3<br>22.0<br>9.0<br>3.8 | 4.6<br><20<br>5<br>2.6 | | | 5.1<br>25<br>8<br>3.4 | | | | | |
| 600 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | 12.0<br>53.0<br>118.0<br>13.9 | 11.5<br>48.0<br>80.0<br>11.9 | 10.0<br>42.0<br>40.0<br>8.3 | 8.7<br>35.0<br>30.0<br>6.7 | 6.0<br>33.0<br>20.0<br>5.7 | 7<br>26<br>12<br>4.0 | 6<br>25<br>7<br>3.1 | | 8<br>35<br>16<br>5.1 | 6.3<br>25<br>8<br>3.7 | | | | |
| 800 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | 15.0<br>51.0<br>75.0<br>11.1 | 12.0<br>45.0<br>48.0<br>8.9 | 10.5<br>45.0<br>35.0<br>7.7 | 9.3<br>33<br>20<br>5.3 | 8.5<br>32<br>15<br>4.2 | 6.5<br>20<br>6<br>3.2 | 11<br>42<br>32<br>6.7 | 8.5<br>30<br>15<br>4.9 | 6.8<br>21<br>5<br>2.7 | 7<br>25<br>8<br>3.5 | | |
| 1000 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | | | 13.0<br>45.0<br>55.0<br>9.6 | 11<br>42<br>30<br>6.6 | 10.2<br>37<br>20<br>5.2 | 8.3<br>28<br>9<br>4.0 | 13<br>46<br>48<br>8.4 | 11<br>38<br>22<br>6.2 | 8<br>27<br>8<br>3.3 | 9<br>30<br>12<br>4.3 | 7.8<br>25<br>8<br>3.3 | 7<br><20<br>3<br>2.3 |
| 1400 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | | | | 16.5<br>50<br>55<br>9.3 | 14<br>45<br>37<br>7.3 | 12<br>37<br>18<br>5.6 | | 15<br>48<br>40<br>8.6 | 12<br>35<br>15<br>4.7 | 12<br>40<br>22<br>6.1 | 11<br>33<br>12<br>4.6 | 11<br>33<br>12<br>3.2 |
| 2000 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | | | | | | | | | 16<br>45<br>30<br>6.7 | 18<br>50<br>42<br>8.7 | 15<br>43<br>25<br>6.5 | 14<br>35<br>17<br>4.6 |
| 2500 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | | | | | | | | | | | 20<br>48<br>37<br>8.2 | 18<br>45<br>25<br>5.8 |
| 3000 | X(m)<br>NR (dB)<br>ΔPt [pa]<br>Veff (m/s) | | | | | | | | | | | | | | 21<br>48<br>35<br>6.9 |

NR < 35 | 35≤NR≤45 | NR>45

## Annotation

* Ak [m²] - Free surface
* X [m] - Air jet length
* Vk = 0.2 [m/s] - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

## Order codes

**Type - Blade's angle** + **CC** + **A** (Mounting spring) + **FIL** + **OBD** + **ADP IZ/N** / **Dimensions** / **O**

- **LBD/LBDB / LBDD -0°/15°**
- Settlement frame (LBD)
- No. of springs depending on grille dimensions
- Air filter (LBD)
- Air control damper (LBD/LBDB)
- Insulated/ Non-insulated adapter
- On request
- **0** Painted white RAL 9016
  **R** Other RAL paints
  Natural anodised (LBD)

# Circular Ceiling Diffuser RCD/RCD-R/RCDH/RCD-RH

RCD-R

RCDH

## Description
Ceiling diffusers with circular or square front panel and adjustable circular opening.

## Application
Supply/ exhaust of the air flow.

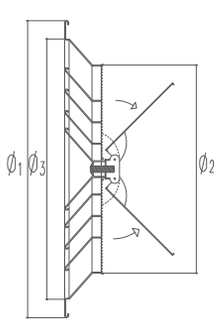
## Characteristics
Recommended difference between air temperature and indoor air is ± 10 K .
Conic shape with removable central core.
Depending on the type of roof and front panel, the following alternatives are possible:
RCD - for low ceiling with circular front panel
RCD-R - for low ceiling with square front panel 595 X 595 mm.
RCDH - for high ceiling with circular front panel
RCD-RH - for high ceiling with square front panel 595 X 595 mm.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**

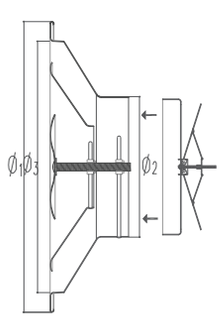
## Materials and finishing
Aluminum material electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design

**RCD**




**RCDH**




## Available dimensions

RCDH / RCD / RCD-RH / RCD-R

| Diffuser type | Ø2 | Ø1 | Ø3 |
|---|---|---|---|
| RCDH | 100 | 250 | 170 |
| | 160 | 335 | 280 |
| | 200 | 423 | 360 |
| | 250 | 517 | 445 |
| | 315 | 640 | 555 |
| | 355 | 730 | 640 |
| | 400 | 776 | 680 |
| | 450 | 825 | 735 |
| | 500 | 915 | 820 |
| | 630 | 1045 | 943 |
| RCD | 150 | 258 | 208 |
| | 200 | 308 | 255 |
| | 250 | 358 | 305 |
| | 300 | 408 | 355 |
| | 350 | 458 | 405 |
| RCD-RH | 160 | □595X595 | 280 |
| | 200 | | 360 |
| | 250 | | 445 |
| | 315 | | 555 |
| RCD-R | 200 | □595X595 | 395 |
| | 250 | | 395 |
| | 300 | | 395 |

## Selection table RCD/RCD-R/RCDH/RCD-RH

### RCDH/RCD-RH - high ceiling

| Air flow m³/h | Connection size Ø(mm) | 160 | 200 | 250 | 315 | 355 | 400 | 450 | 500 | 630 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Ak [m²] | 0.015 | 0.027 | 0.04 | 0.056 | 0.074 | 0.09 | 0.011 | 0.013 | 0.019 |
| 200 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | 1.4<br>25.0<br>3.7<br>14.0 | | | | | | | | |
| 250 | X [m]<br>NR [dB(A)]<br>Veffm/s]<br>ΔPt [pa] | 1.7<br>30.0<br>4.4<br>20.0 | 1.3<br>18.0<br>2.5<br>6.6 | | | | | | | |
| 300 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | 2.1<br>36.0<br>5.6<br>33.0 | 1.5<br>23.0<br>3.0<br>10.0 | 1.3<br>15.0<br>2.0<br>4.2 | | | | | | |
| 400 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | 2.8<br>45<br>7.2<br>54 | 2.1<br>30.0<br>4.0<br>17.0 | 1.7<br>22.0<br>2.7<br>7.7 | 1.5<br>3.0<br>2.0<br>4.2 | | | | | |
| 500 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | 2.6<br>36.0<br>5.0<br>26.0 | 2.1<br>27.0<br>3.4<br>12.0 | 1.8<br>20.0<br>2.4<br>6.0 | | | | | |
| 600 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | 3.2<br>42.0<br>6.0<br>38.0 | 2.6<br>33.0<br>4.0<br>17.0 | 2.2<br>25.0<br>3.0<br>10.0 | 1.9<br>20.0<br>2.3<br>5.0 | | | | |
| 800 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | 3.5<br>41.0<br>5.6<br>33.0 | 3.0<br>34.0<br>4.0<br>17.0 | 2.6<br>25.0<br>3.0<br>9.0 | 2.3<br>20.0<br>2.5<br>6.5 | | | |
| 1000 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | 4.5<br>50.0<br>7.0<br>52.0 | 3.7<br>40.0<br>5.0<br>26.0 | 3.3<br>34.0<br>3.8<br>16.0 | 3.0<br>28.0<br>3.0<br>9.0 | 2.7<br>22.0<br>2.5<br>6.5 | | |
| 1250 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | 4.6<br>47.0<br>6.2<br>40.0 | 4.1<br>42.0<br>4.8<br>23.0 | 3.8<br>35.0<br>3.8<br>16.0 | 3.3<br>26.0<br>3.0<br>9.0 | 3.0<br>24.0<br>2.6<br>6.9 | |
| 1500 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | 5.0<br>47.0<br>5.5<br>32.0 | 4.5<br>41.0<br>4.6<br>22.0 | 4.0<br>35.0<br>3.8<br>16.0 | 3.6<br>29.0<br>3.1<br>10.0 | |
| 1750 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | | | 4.8<br>40.0<br>4.4<br>20.0 | 4.3<br>34.0<br>3.6<br>14.0 | 3.5<br>25.0<br>2.6<br>6.9 |
| 2000 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | | | | 4.9<br>38.0<br>4.1<br>19.0 | 4.1<br>30.0<br>2.9<br>8.0 |
| 2500 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | | | | 6.0<br>45.0<br>5.0<br>29.0 | 5.1<br>37.0<br>3.6<br>14.0 |

### RCD/RCD-R - low ceiling

| Air flow m³/h | Connection size Ø(mm) | 160 | 200 | 250 | 315 | 355 |
|---|---|---|---|---|---|---|
| | Ak [m²] | 0.011 | 0.02 | 0.031 | 0.046 | 0.047 |
| 150 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | 1.8<br>-<br>3.9<br>1.0 | 1.3<br>-<br>2.1<br>6.0 | | | |
| 200 | X [m]<br>NR [dB(A)]<br>Veffm/s]<br>ΔPt [pa] | 2.3<br>27.0<br>5.0<br>22.0 | 1.8<br>-<br>2.8<br>7.8 | 1.4<br>-<br>1.9<br>- | | |
| 300 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | 3.4<br>41.0<br>7.5<br>50.0 | 2.6<br>23.0<br>4.1<br>15.0 | 2.1<br>-<br>2.7<br>7.2 | 1.7<br>-<br>1.9<br>- | |
| 400 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | 3.4<br>33.0<br>5.7<br>28.0 | 2.7<br>20.0<br>3.6<br>12.0 | 2.3<br>-<br>2.5<br>5.5 | 1.9<br>7.0<br>2.4<br>3.2 |
| 500 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | 4.3<br>41.0<br>7.0<br>45.0 | 3.5<br>27.0<br>4.5<br>18.0 | 2.8<br>15.0<br>3.1<br>8.0 | 2.0<br>10.0<br>2.8<br>4.8 |
| 600 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | 4.1<br>34.0<br>5.5<br>26.0 | 3.3<br>22.0<br>3.7<br>14.0 | 2.8<br>20.0<br>3.5<br>7.0 |
| 800 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | 5.5<br>45.0<br>7.5<br>50.0 | 4.6<br>32.0<br>5.0<br>22.0 | 3.5<br>30.0<br>4.8<br>12.0 |
| 1000 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | 5.7<br>40.0<br>6.2<br>35.0 | 4.5<br>37.0<br>6.0<br>20.0 |
| 1200 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | 6.7<br>46.0<br>7.5<br>50.0 | 5.4<br>40.0<br>7.0<br>28.0 |
| 1400 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | 7.1<br>50.0<br>10.2<br>39.2 |
| 1600 | X [m]<br>NR [dB(A)]<br>Veff[m/s]<br>ΔPt [pa] | | | | | 7.8<br>57.0<br>11.6<br>50.3 |

Correction coefficients for other velocities inside sitting area.

| Vk[m/s] | 0.2 | 0.25 | 0.375 | 0.5 |
|---|---|---|---|---|
| x[m] | 1.25 | 1 | 0.75 | 0.5 |

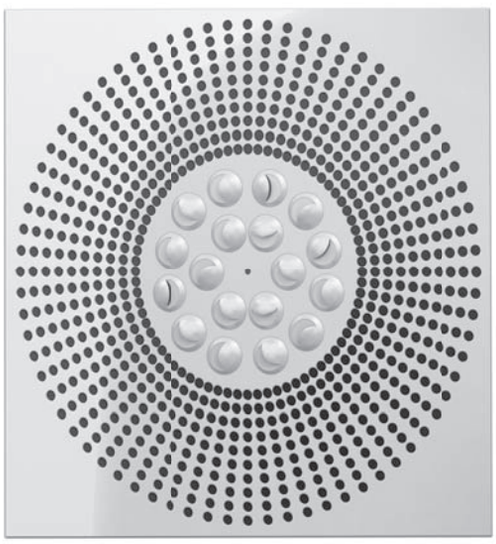
## Annotation

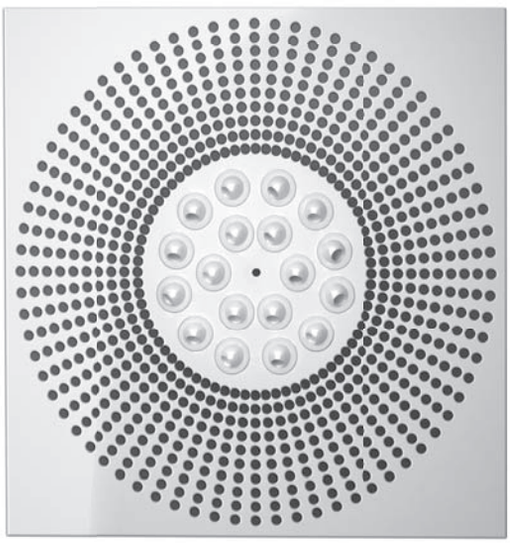
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.25 [m/s] - velocity at which the air jet length was calculated "x" [m]

## Order codes

| Type | + | ADP IZ/N | / | Dimensions | / | 0 |
|---|---|---|---|---|---|---|
| | | | | Ø | | |
| **RCDH/RCD**<br>**RCD-RH/RCD-R** | | Insulated/ Non-insulated adapter | | high ceiling: 160, 200, 250, 315, 355, 400, 450, 500, 630<br>low ceiling: 160, 200, 250, 315, 355 | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Ceiling Diffuser PGP/PGR





## Description
Ceiling diffuser with maneuverable disks/ nozzles and perforated metal sheet, designed for use in crowded areas.

## Application
Air supply.

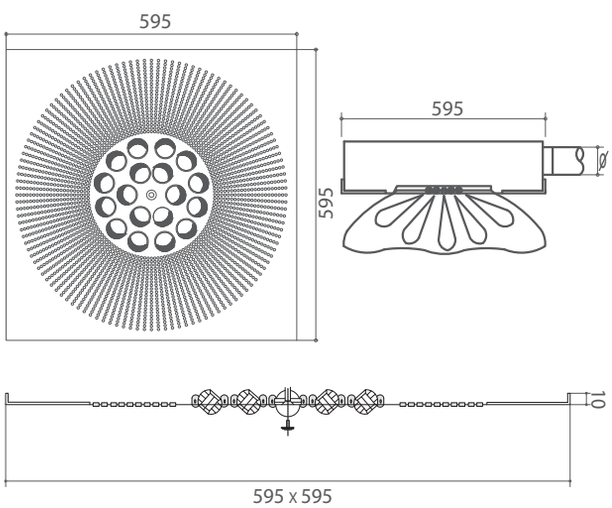
## Characteristics
The speaker is equipped with Ø5 mm circular perforations radially arranged and 18 individually maneuverable disks/ nozzles , located in the center of the front.
Specially designed for paneled ceiling size 600 x 600 mm.
The diffuser is supplied with gasket seals, fastening screw and plastic cover for concealment.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene disks/ nozzles.

## Technical design

**PGP** **PGR**




## Assembly Methods
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm on mounting traverse of the adapter.

## Selection table PGP/PGR

### PGP

| Q m³/h | 595 X 595 Ak [m²] | 0.03536 |
|---|---|---|
| 175 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 1.37<br><20<br>6<br>0.9 |
| 200 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 1.57<br><20<br>8<br>1 |
| 250 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 1.96<br>24<br>13<br>1.3 |
| 300 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 2.36<br>29<br>19<br>1.5 |
| 350 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 2.75<br>34<br>25<br>1.8 |
| 400 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 3.14<br>38<br>33<br>2 |
| 450 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 3.54<br>41<br>42<br>2.3 |
| 500 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 3.93<br>44<br>52<br>2.5 |

ΔT=-10K
H(room height)=2,84 m
Vk=0,2 m/s

### PGR

| Q m³/h | 595 X 595 Ak [m²] | 0.0485 |
|---|---|---|
| 400 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 2.29<br>31<br>15<br>2 |
| 600 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 3.44<br>44<br>33<br>5 |
| 800 | Veff (m/s)<br>NR[dB(A)]<br>Pt(Pa)<br>X(m) | 4.58<br>53<br>58<br>7 |

ΔT=-10K
Vk=0,25 m/s

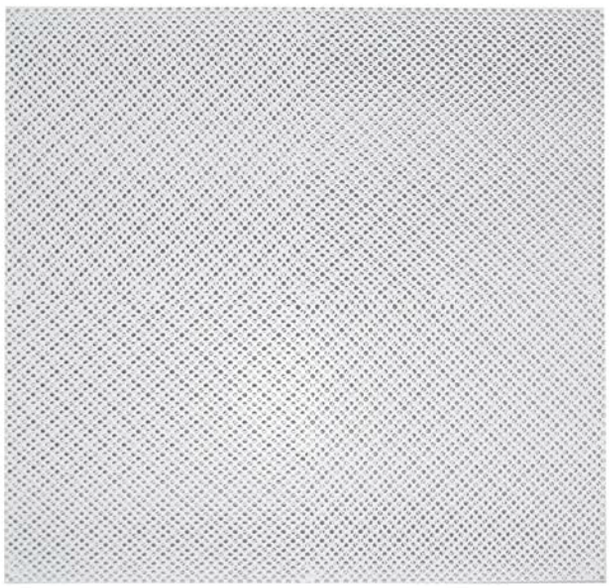
## Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* Vk = velocity at which the air jet length was calculated "x" [m]

## Order codes

| Type | + | ADP IZ/N | / | Dimensions | / | Nozzles/ disks color | / | 0 |
|---|---|---|---|---|---|---|---|---|
| **PGR / PGP** | | Insulated/Non-insulated adapter | | L X H<br>595 X 595 | | PGR - white RAL 9016<br>PGP - white RAL9016 / black RAL9005 | | **0** Vopsit alb RAL 9016<br>**R** Vopsit RAL… |

# Perforated Ceiling Diffuser PFD/PF



PFD

## Description
Rectangular ceiling diffuser with circular perforations for exhaust/ supply of the air flow **PFD/PF.**

## Application
For air supply - **PFD**, for air exhaust **PF.**

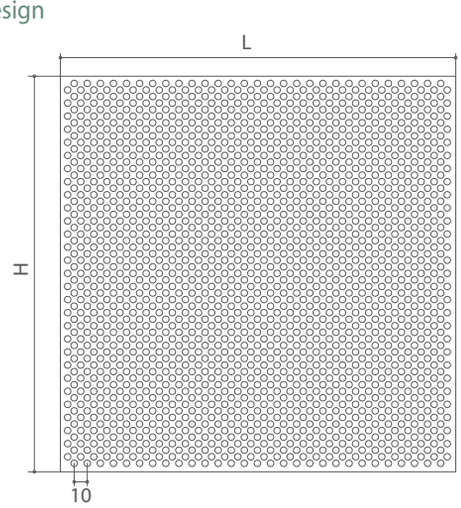
## Characteristics
PFD/PF is projected for paneled ceiling mounting (600 x 600 mm).
Circular perforations of 5 mm diameter.
The PFD diffuser is provided with an air jet orientation system in 1, 2, 3 or 4 directions, without air flow deviations or pressure loss.
The diffuser connects to the ventilation duct through the adapter.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

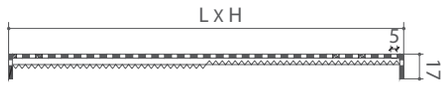
## Materials and finishing
Galvanized steel sheet painted in electrostatic field (white RAL 9016).
Other RAL colors available on request.

## Technical design

**PFD/PF**




**PFD**




**PF**

## Assembly Method
Paneled ceiling mounting 600 x 600 mm.

## Selection table PFD/ PF

### PFD - supply

| Air flow ($m^3/h$) | LxH | 595x595 |
|---|---|---|
| | Ak [$m^2$] | 0.150 |
| **700** | Veff(m/s) | 1.3 |
| | X [m] | 1.8 |
| | NR [dB(A)] | 27.0 |
| | ΔPt [pa] | 7.0 |
| **1000** | Veff(m/s) | 1.9 |
| | X [m] | 2.6 |
| | NR [dB(A)] | 39.0 |
| | ΔPt [pa] | 15.0 |
| **1600** | Veff(m/s) | 3.0 |
| | X [m] | 3.1 |
| | NR [dB(A)] | 54.0 |
| | ΔPt [pa] | 36.0 |

### PF - exhaust

| Air flow ($m^3/h$) | LxH | 595x595 |
|---|---|---|
| | Ak [$m^2$] | 0.150 |
| **700** | Veff(m/s) | |
| | NR [dB(A)] | |
| | ΔPt [pa] | |
| **1000** | Veff(m/s) | 1.9 |
| | NR [dB(A)] | 20.0 |
| | ΔPt [pa] | 33.0 |
| **1600** | Veff(m/s) | 3.0 |
| | NR [dB(A)] | 36.0 |
| | ΔPt [pa] | 85.0 |

Note: The values for the supply mode from selection table are calculated for a 4 direction system diffuser and for an air velocity of Vk = 0.2 m/s.

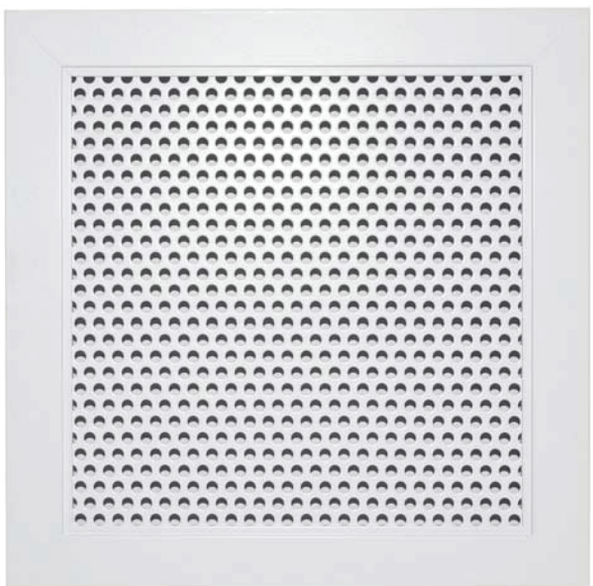
### Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

### Order codes

| **Type** | + | **ADP IZ/N** | / | **Dimensions** | / | **L x H** | / | **0** |
|---|---|---|---|---|---|---|---|---|
| **PFD/PF** | | Insulated/ Non-insulated adapter | | L X H<br>595 x 595 | | Other dimensions on request | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Perforated Diffuser PFF/PFDD



PFF

## Description
Rectangular ceiling diffuser for exhaust/ supply of the air flow provided with circular perforations and mounting frame PFF/PFDD. PFDD is provided with access door to ease access to the back of the diffuser.

## Application
Exhaust/ supply of the air flow.

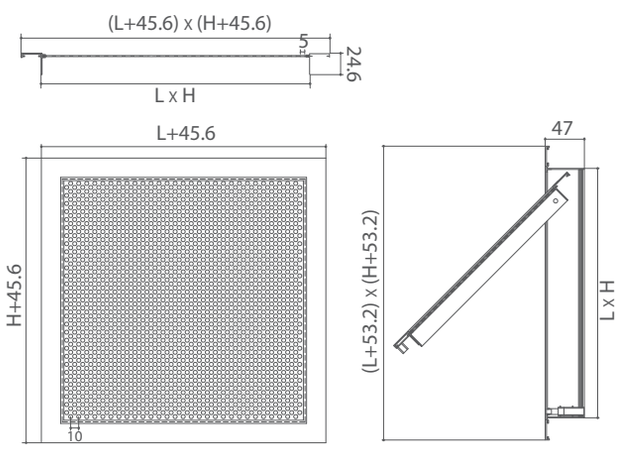
## Characteristics
The diffusers are executed with mounting holes on the frame, except those with dimensions 595 x 595 mm for paneled ceiling.
**PFDD** is provided with access door, in this case the mounting holes will not be executed on the frame, but on the inlet connection of the diffuser.
The access door is opened via a click type system (open/ close by pressing).
Circular perforations of 5 mm diameter.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.
An air filter FIL can be optionally mounted on the inlet connection of the diffuser .
The diffuser connects to the ventilation duct through the adapter.

## Accessories
Air filter **FIL**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Galvanized steel sheet painted in electrostatic field (white RAL 9016).
Other RAL colors available on request.

## Technical design

**PFF** **PFDD**




## Assembly Method
With screws (standard).
PFDD - screws on the inlet connection (optional).

## Selection table PFF/PFDD

### Supply

| Air flow m³/h | LxH | 300x300 | 400x400 | 500x500 | 595x595 |
|---|---|---|---|---|---|
| | Ak [m²] | 0.030 | 0.060 | 0.100 | 0.150 |
| 160 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 1.5<br>0.9<br>22.0<br>9.0 | | | |
| 200 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 1.8<br>1.1<br>31.0<br>14.0 | 0.9<br>0.8<br>< 20<br>4.0 | | |
| 300 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.8<br>1.7<br>44.0<br>31.0 | 1.4<br>1.2<br>25.0<br>8.0 | | |
| 400 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 3.7<br>2.1<br>53.0<br>53.0 | 1.9<br>1.6<br>35.0<br>15.0 | 1.1<br>1.3<br>20.0<br>5.0 | |
| 500 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 4.6<br>2.5<br>61.0<br>81.0 | 2.3<br>2.0<br>42.0<br>23.0 | 1.4<br>1.6<br>27.0<br>8.0 | |
| 700 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | | 3.2<br>2.2<br>52.0<br>45.0 | 1.9<br>2.2<br>39.0<br>16.0 | 1.3<br>1.8<br>27.0<br>7.0 |
| 1000 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | | 4.6<br>3.1<br>63.0<br>90.0 | 2.8<br>2.4<br>51.0<br>30.0 | 1.9<br>2.6<br>39.0<br>15.0 |
| 1600 | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | | | | 3.0<br>3.1<br>54.0<br>36.0 |

### Exhaust

| Air flow m³/h | LxH | 300x300 | 400x400 | 500x500 | 595x595 |
|---|---|---|---|---|---|
| | Ak [m²] | 0.030 | 0.060 | 0.100 | 0.150 |
| 160 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | 1.5<br><20<br>20.0 | | | |
| 200 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | 1.8<br><20<br>31.0 | | | |
| 300 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | 2.8<br>26.0<br>76.0 | | | |
| 400 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | 3.7<br>35.0<br>125.0 | 1.9<br><20<br>33.0 | | |
| 500 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | 4.6<br>42.0<br>185.0 | 2.3<br>23.0<br>47.0 | | |
| 700 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | | 3.2<br>34.0<br>95.0 | 1.9<br>20.0<br>37.0 | |
| 1000 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | | | 2.8<br>31.0<br>73.0 | 1.9<br>20.0<br>33.0 |
| 1600 | Veff(m/s)<br>NR [dB(A)]<br>ΔPt [pa] | | | | 3.0<br>36.0<br>85.0 |

NR < 35 | 35≤NR≤45 | NR>45

### Annotation

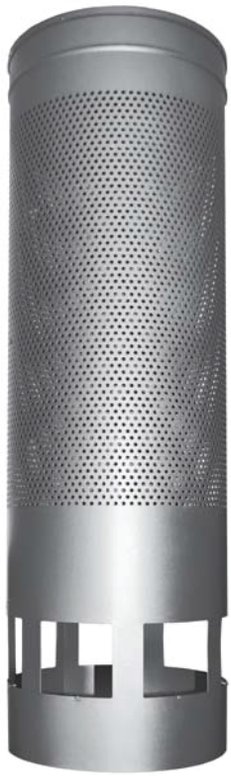
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* Vk = 0.2 m/s - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

### Order codes

**Type** + **FIL** + **OBD** + **ADP IZ/N** / **Dimensions** / **L x H** / **0**

- **Type**: **PFF / PFDD**
- **FIL**: Air filter
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: L X H — 300 x 300, 400 x 400, 500 x 500, 595 x 595
- **L x H**: Other dimensions on request
- **0**: **0** Painted white RAL 9016, **R** Other RAL paints

# Perforated Cylinder Diffuser PC



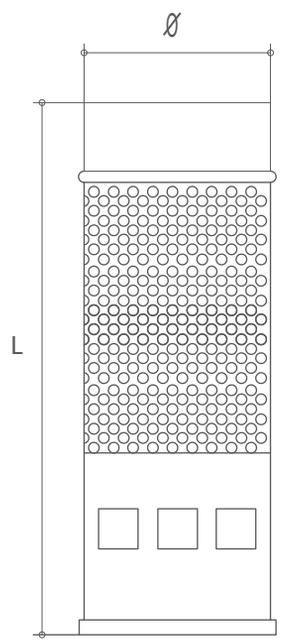
## Description
Perforated cylinder for use in industrial areas.

## Application
Air supply

## Characteristics
Available diameters: D250, D355, D450 and D560 mm.
Length of the cylinder: 825 mm.
Height for mounting is over 3.5 m.
Suitable for use for both heating and cooling.
Horizontal and vertical dispersion of air.
Can function as an air damper, which makes the air supply possible on both horizontal and vertical levels, depending on requirements (heating/ cooling).
The diffuser is provided with a internal adjustable disk manipulated by a chain drive.

## Materials and finishing
Perforated cylinder made of galvanized steel sheet
RAL colors available on request.

## Technical design



## Selection table PC

| Air flow ($m^3/h$) | Ø | 250 | 355 | 450 | 560 |
|---|---|---|---|---|---|
| **700** | NR [dB(A)] | 31.0 | | | |
| | X(m) | 2.2 | | | |
| | ΔPt [pa] | 12.0 | | | |
| **1000** | NR [dB(A)] | 42.0 | 26.0 | | |
| | X(m) | 3.5 | 2.6 | | |
| | ΔPt [pa] | 23.0 | 7.0 | | |
| **1500** | NR [dB(A)] | 55.0 | 38.0 | 21.0 | |
| | X(m) | 5.2 | 3.5 | 2.8 | |
| | ΔPt [pa] | 50.0 | 18.0 | 7.0 | |
| **2000** | NR [dB(A)] | | 47.0 | 31.0 | <25 |
| | X(m) | | 4.6 | 3.3 | 2.5 |
| | ΔPt [pa] | | 31.0 | 13.0 | 6.0 |
| **3000** | NR [dB(A)] | | >55 | 43.0 | 30.0 |
| | X(m) | | 7.0 | 5.2 | 4.2 |
| | ΔPt [pa] | | 70.0 | 28.0 | 13.0 |
| **4000** | NR [dB(A)] | | | 53.0 | 40.0 |
| | X(m) | | | 7.0 | 5.2 |
| | ΔPt [pa] | | | 50.0 | 27.0 |
| **6000** | NR [dB(A)] | | | | 53.0 |
| | X(m) | | | | 8.0 |
| | ΔPt [pa] | | | | 60.0 |

## Annotation

* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length

Note: The values in the selection table are calculated for an air jet discharge set in heating mode and for temperature deviation of ΔT= + 10 K.

## Order codes

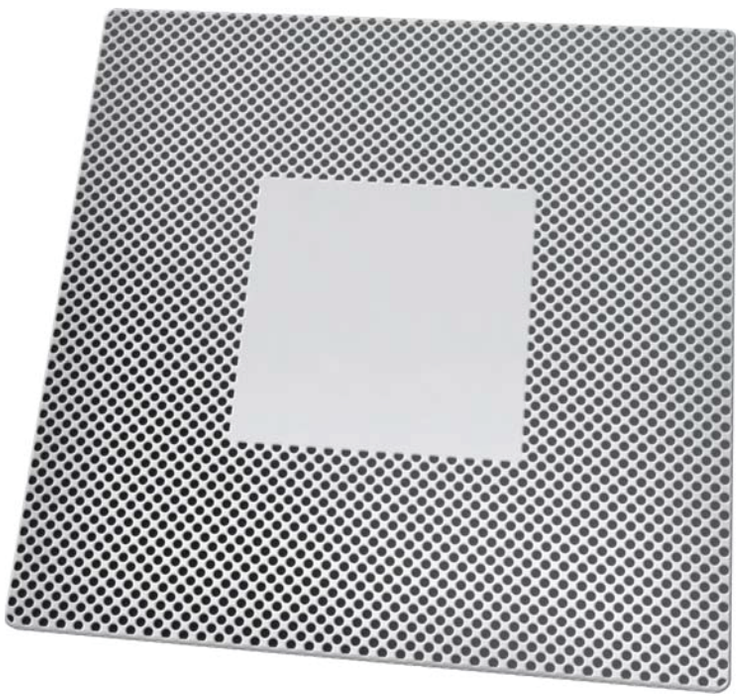
**Type** / **Ø** / **0**

**PC**

250
355
450
560

Natural steel material
**0** Painted white RAL 9016
**R** Other RAL paints

# Architectural Diffuser PA



## Description
Architectural diffuser with circular perforations, for paneled ceiling mounting.

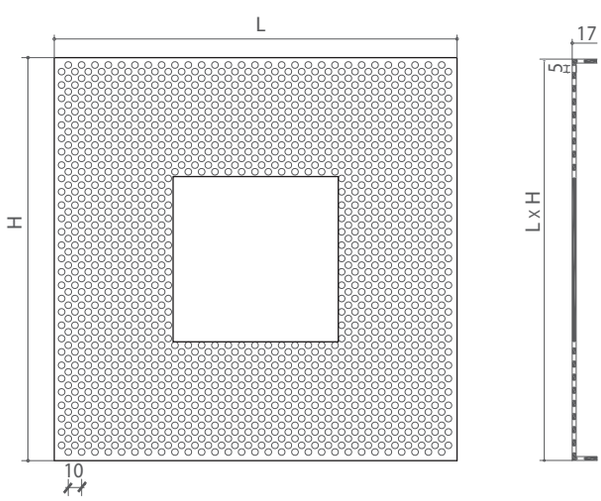
## Application
Supply/ exhaust ot the air flow.

## Characteristics
Diffuser is provided with circular perforation of Ø 5 mm placed on the front side.
Specially designed for paneled ceiling mounting - size 600 x 600 mm.
Connects to the air ducts through the adapter.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm

## Selection table PA

| Air flow ($m^3/h$) | L x H | 595x595 |
|---|---|---|
| **450** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.3<br><10<br>100 |
| **500** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.6<br>10.0<br>125.0 |
| **600** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 3.1<br>15.0<br>150.0 |
| **700** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 3.6<br>18.0<br>180.0 |
| **780** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br>20.0<br>200.0 |
| **890** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 4.6<br>25.0<br>250.0 |
| **1100** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 5.5<br>30.0<br>300.0 |
| **1300** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 6.8<br>35.0<br>380.0 |
| **1450** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 7.5<br>40.0<br>450.0 |

## Annotation

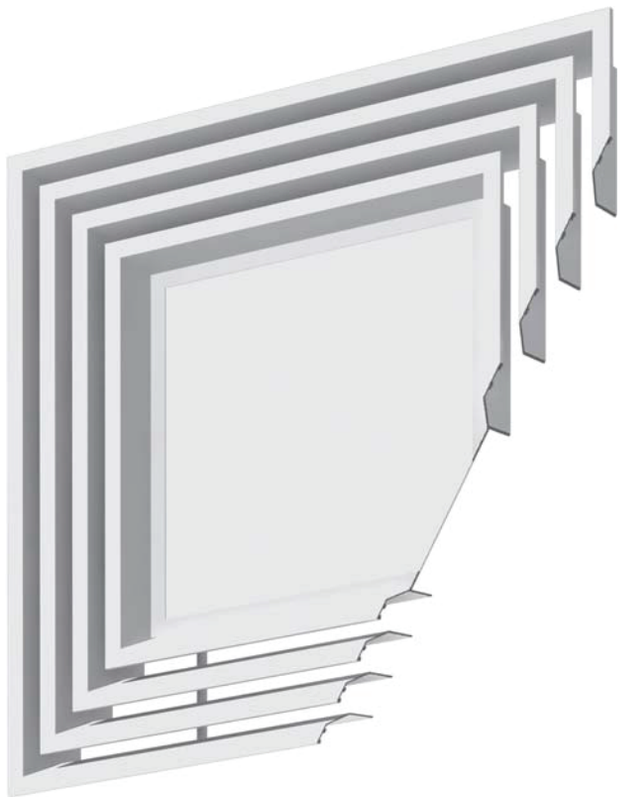
* X [m] - Air jet length at 0.2 m/s velocity
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **O**

- **Type**: **PA**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: L X H, 595 x 595
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

# Linear Ceiling Diffuser LCD



## Description
Rectangular diffuser, paneled ceiling type, with four direction air jet diffuser.

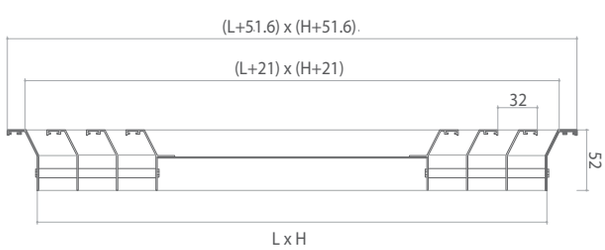
## Application
Designed for various air supply or exhaust systems.

## Characteristics
Preferred for paneled ceiling, 600 x 600 mm diffuser with 1, 2 or 3 peripheral slots.
The air flow can be adjusted by mounting an air control damper OBD on the diffuser inlet.

## Accessories
Air control damper with opposable blades **OBD**
Insulated/non-insulated adapter **ADP IZ/N**

## Materials and finishing
Extruded aluminum blades, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm (standard)

## Selection table LCD

| Air flow ($m^3/h$) | 595X595 | | | |
|---|---|---|---|---|
| | No. slots | 1 | 2 | 3 |
| | Ak [$m^2$] | 0.028 | 0.048 | 0.068 |
| **150** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 1.5<br>0.8<br><20<br>2.1 | | |
| **200** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.0<br>1.0<br>20.0<br>3.3 | 1.2<br>0.7<br><20<br>1.1 | |
| **300** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 3.0<br>1.4<br>28.0<br>7.3 | 1.7<br>1.1<br>22.0<br>2.5 | 1.2<br>0.9<br><20<br>1.2 |
| **400** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br>1.9<br>34.0<br>13.0 | 2.3<br>1.5<br>29.0<br>4.4 | 1.6<br>1.2<br>25.0<br>2.2 |
| **500** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 5.0<br>2.4<br>39.0<br>20.4 | 2.9<br>1.8<br>33.0<br>6.9 | 2.0<br>1.5<br>30<br>3.4 |
| **600** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 6.0<br>2.9<br>43.0<br>29.3 | 3.5<br>2.2<br>37.0<br>9.9 | 2.5<br>1.9<br>33.0<br>4.9 |
| **700** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 6.9<br>3.4<br>46.0<br>39.9 | 4.1<br>2.6<br>40.0<br>13.5 | 2.9<br>2.2<br>37.0<br>6.7 |
| **1000** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 9.9<br>4.8<br>54.0<br>81.5 | 5.8<br>3.7<br>48.0<br>27.6 | 4.1<br>3.1<br>44.0<br>13.7 |
| **1500** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | | 8.7<br>5.5<br>56.0<br>62.1 | 6.1<br>4.6<br>53.0<br>30.9 |

NR < 35 | 35≤NR≤45 | NR>45

## Annotation

* Ak [$m^2$] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Length airflow at speed Vk [m/s] = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

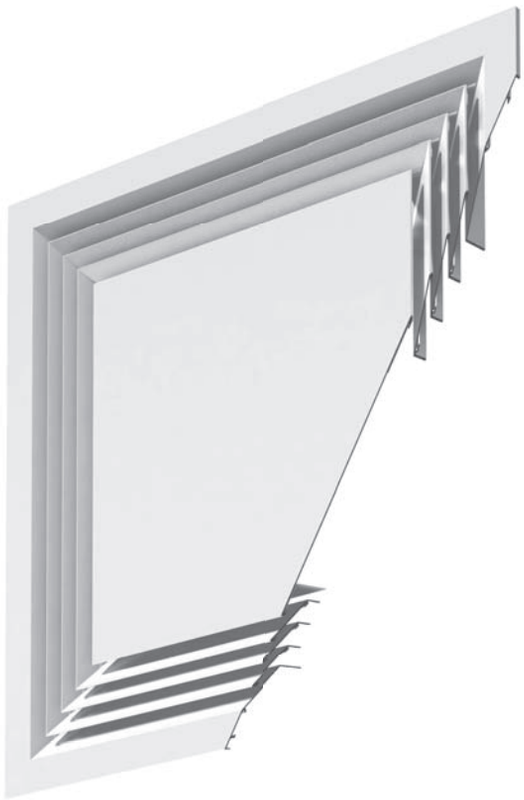
Note: The values in the selection table are calculated for 2.8 m high ceiling.

## Order codes

**Type - Number of slots** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **LCD**
- 1 / 2 / 3
- Insulated/Non-insulated adapter
- **L X H** (595 x 595)
- Other dimensions on request
- **0** Painted white RAL 9016
- **R** Other RAL paints

# Decorative Diffuser AC



## Description
Rectangular diffuser, paneled ceiling type, with four direction air jet diffuser.

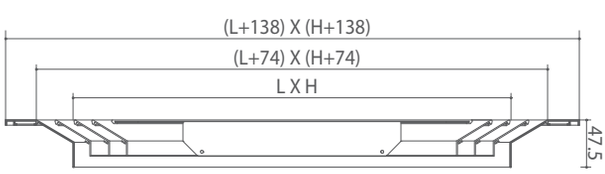
## Application
Supply or exhaust of the airflow.

## Characteristics
Preferred for paneled ceiling, 600 x 600 mm diffuser with 1, 2, 3 or 4 outlying slots around the square core.
Removable central core.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.
Uniform aspect with adjacent diffusers.

## Accessories
Air control damper with opposable blades **OBD.**
Insulated/non-insulated adaptor **ADP IZ/N.**

## Materials and finishing
Extruded shaped aluminum blades , electrostatic field painted white RAL 9016
Other RAL colors available on request.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes for diffuser connection (optional)

## Selection table AC

| Air flow ($m^3/h$) | No. slots | 595 X 595 | | | |
|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 |
| | Ak(mp) | 0.022 | 0.039 | 0.048 | 0.055 |
| **90** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 1.12<br>22.00<br>3.00<br>1.80 | | | |
| **125** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 1.56<br>28.00<br>6.00<br>2.10 | | | |
| **180** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 2.24<br>35.00<br>12.00<br>2.60 | | | |
| **250** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 3.11<br>42.00<br>24.00<br>3.60 | 1.79<br>25.00<br>17.00<br>2.40 | | |
| **320** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 3.99<br>47.00<br>40.00<br>4.80 | 2.30<br>34.00<br>29.00<br>3.00 | 1.84<br>20.00<br>8.00<br>2.40 | |
| **430** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt[pa]<br>X [m] | | 3.09<br>45.00<br>51.00<br>4.20 | 2.48<br>30.00<br>14.00<br>3.20 | 2.17<br>26.00<br>11.00<br>2.80 |
| **540** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | 3.88<br>55.00<br>80.00<br>5.20 | 3.11<br>38.00<br>21.00<br>4.00 | 2.73<br>33.00<br>17.00<br>3.60 |
| **650** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | 3.75<br>45.00<br>31.00<br>4.80 | 3.28<br>39.00<br>24.00<br>4.20 |
| **750** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | 4.32<br>52.00<br>42.00<br>5.60 | 3.79<br>44.00<br>33.00<br>5.00 |
| **850** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | | 4.29<br>49.00<br>43.00<br>5.60 |

## Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* NR [dB(A)] - Noise level without room attenuation
* ΔPt [pa] - Pressure loss inside diffuser
* X [m] - Airflow length
* Vk = 0.2 m/s - Speed at which airflow length was calculated "x" [m]

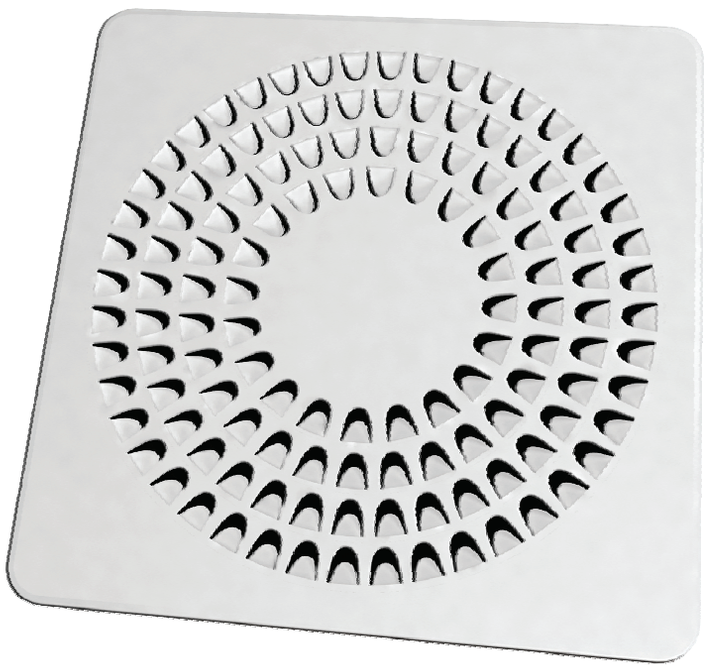
NR < 35 | 35≤NR≤45 | NR>45

## Order codes

**Type - Number of slots** + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **AC**
- 1, 2, 3
- Air control damper
- Insulated/non-insulated adapter
- **L X H** 595 x 595
- Other dimensions on request
- **0** Painted white RAL 9016
- **R** Other RAL paints…

# Ceiling Diffuser CD-CK



## Description
Square diffuser preferred for paneled ceiling mounting.

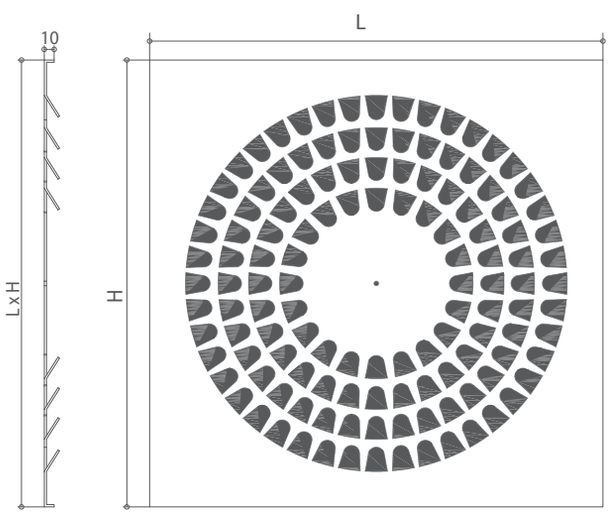
## Application
Supply/ exhaust of airflow.

## Characteristics
Specifically designed for paneled ceiling mounting - size 600 x 600 mm.
Keeps a uniform aspect with adjacent diffusers.
Connects to the air ducts through the adapter.
The diffuser is delivered with fitting gaskets.

## Accessories
Insulated/ Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016
Other RAL colors available on request

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm

## Selection table CD-CK

### Supply

| Air flow ($m^3/h$) | LxH | 595 X 595 |
|---|---|---|
| **480** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 5.7<br>5.0<br>7.0 |
| **540** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 6.3<br>10.0<br>8.0 |
| **610** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 7.2<br>20.0<br>10.0 |
| **720** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 8.5<br>25.0<br>15.0 |
| **890** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 11.0<br>32.0<br>22.0 |
| **1010** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 13.0<br>35.0<br>28.0 |
| **1200** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 14.0<br>40.0<br>40.0 |
| **1315** | X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 16.0<br>45.0<br>50.0 |

### Exhaust

| Air flow ($m^3/h$) | LxH | 595 X 595 |
|---|---|---|
| **480** | NR [dB(A)]<br>ΔPt [pa] | 15.0<br>7.5 |
| **540** | NR [dB(A)]<br>ΔPt [pa] | 22.0<br>9.0 |
| **610** | NR [dB(A)]<br>ΔPt [pa] | 26.0<br>11.0 |
| **720** | NR [dB(A)]<br>ΔPt [pa] | 32.0<br>17.0 |
| **890** | NR [dB(A)]<br>ΔPt [pa] | 37.0<br>24.0 |
| **1010** | NR [dB(A)]<br>ΔPt [pa] | 42.0<br>30.0 |
| **1200** | NR [dB(A)]<br>ΔPt [pa] | 45.0<br>40.0 |
| **1315** | NR [dB(A)]<br>ΔPt [pa] | 50.0<br>50.0 |

## Annotation

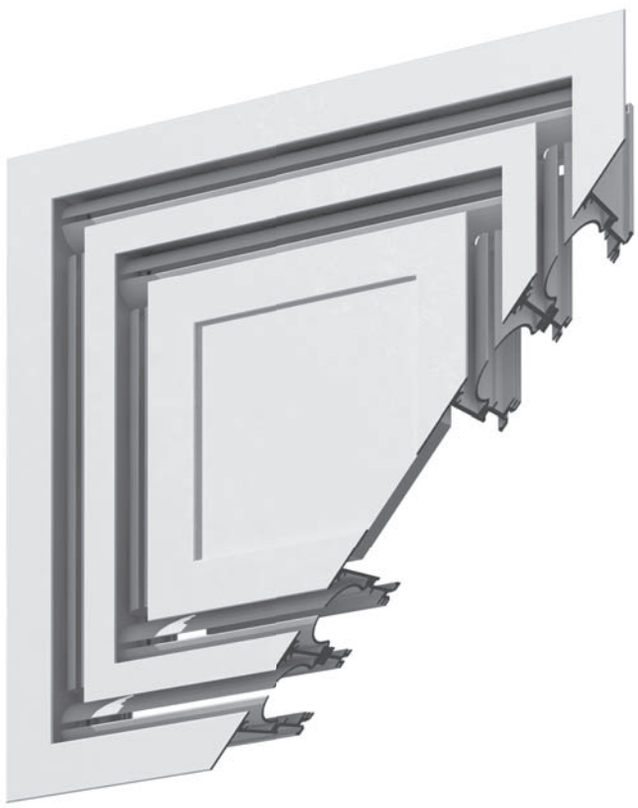
* X [m] - Length air jet at speed Vk = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **O**

- **CD-CK**
- Insulated/Non-insulated adapter
- **L X H** 595 x 595
- **0** Painted white RAL 9016
  **R** Other RAL paints

# 4 Direction Slot Diffuser SL



## Description
Air diffuser specially designed for paneled ceiling mounting.

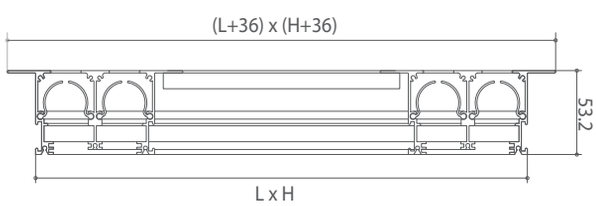
## Application
Supply/ exhaust of the airflow

## Characteristics
The diffuser is designed with two slots perimetrically located around a square center of plasterboard or other material, keeping a uniform aspect with the rest of the ceiling.
Inside each slot there are individually adjustable twin blades mounted, which allows adjustment of air flow passing through each slot.
The blades inside the slot can be painted differently from the diffuser, thus giving a decorative look.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Extruded aluminum blades, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm (standard)

## Selection table SL

| Air flow ($m^3/h$) | 595 x 595 | |
|---|---|---|
| | Ak (mp) | 0.076 |
| **86** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 4.00<br>3.00<br>---<br>4.02 |
| **176** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 5.00<br>4.00<br>18.00<br>7.94 |
| **219** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 8.00<br>7.00<br>30.00<br>22.16 |
| **262** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 9.00<br>7.00<br>34.00<br>31.00 |
| **305** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 10.00<br>8.00<br>38.00<br>43.00 |
| **352** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 10.00<br>9.00<br>41.00<br>56.00 |
| **395** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 11.00<br>9.00<br>44.00<br>72.00 |
| **485** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 12.00<br>10.00<br>47.00<br>73.00 |
| **552** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 12.50<br>10.50<br>48.00<br>73.50 |
| **625** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 12.80<br>11.00<br>50.00<br>74.00 |
| **850** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>ΔPt [pa] | 13.50<br>12.00<br>52.00<br>92.00 |

## Annotation

* Ak [m²] - Free surface of the diffuser
* X [m] - Length airflow
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **SL**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L X H** — 595 x 595
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

# Ceiling Diffuser FD

## Description
Aluminum rectangular diffuser with individual adjustable blades.

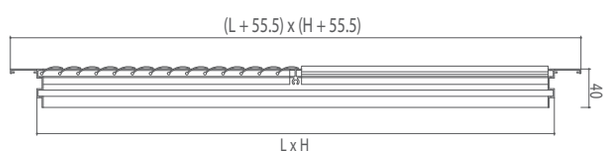
## Application
Designed for air exhaust from ventilation and air conditioning systems.

## Characteristics
The diffuser is equipped with adjustable blades arranged in four directions, causing rotation of the air with Coanda effect, obtaining a high rate of induction and reduced stratification.
On request the diffuser can be provided with 1, 2 or 3 discharge directions.
Recommended for rooms with high ceilings (eg. factories, airports, sports halls, etc.)
Minimum recommended height is 3.8 m.
Designed specifically for integration into the paneled ceiling box (dimensions 600 x 600 mm), the speaker is provided without mounting holes.
On request, the diffuser can be produced in custom sizes, in which case it is provided with mounting holes.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
With screws, visible on the diffuser's frame.

## Selection table FD

| Air flow ($m^3/h$) | Dimensions | 595x595 |
|---|---|---|
| | Ak [$m^2$] | 0.1819 |
| **960** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 1.5<br>3.0<br>23.0<br>13.0 |
| **1150** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 1.8<br>3.8<br>30.0<br>18.0 |
| **1380** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.1<br>4.5<br>35.0<br>25.0 |
| **1610** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.5<br>5.5<br>38.0<br>33.0 |
| **1860** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 2.8<br>6.2<br>42.0<br>47.0 |
| **2300** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 3.4<br>7.5<br>50.0<br>75.0 |
| **2700** | Veff(m/s)<br>X [m]<br>NR [dB(A)]<br>ΔPt [pa] | 4.1<br>8.0<br>55.0<br>100.0 |

## Annotation

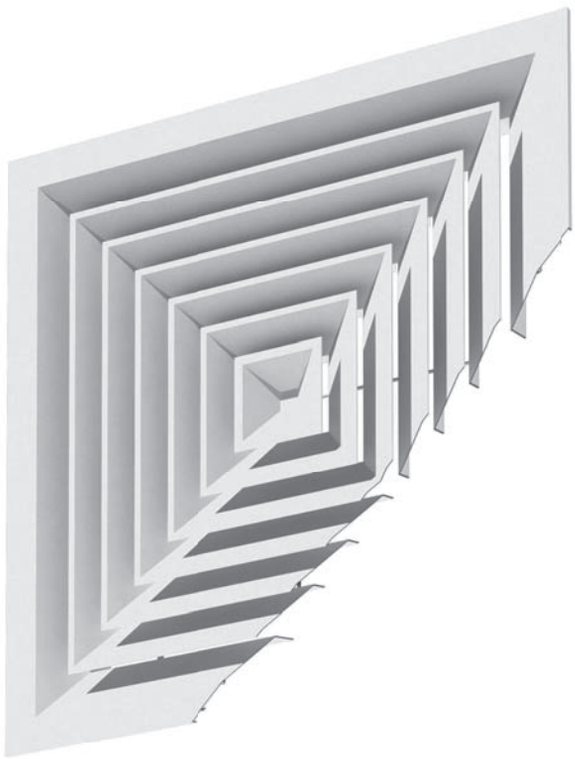
* Ak [$m^2$] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Length air jet at speed Vk = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Note: Table values are for maximum blade opening

## Order codes

| **Type** | + | **ADP IZ/N** | / | **Dimensions** | / | **L X H** | / | **O** |
|---|---|---|---|---|---|---|---|---|
| **FD** | | Insulated/ Non-insulated adapter | | **L X H**<br>595 x 595 | | Other dimensions on request | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Ceiling Diffuser CD



## Description
4 direction aluminum ceiling diffuser, for paneled ceiling mounting.

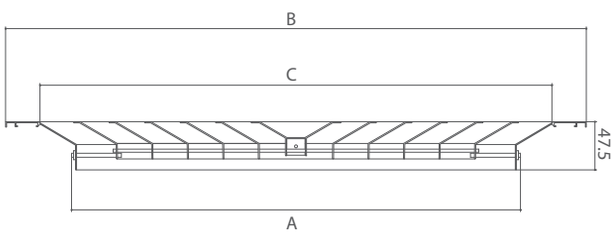
## Application
Supply or exhaust of the airflow

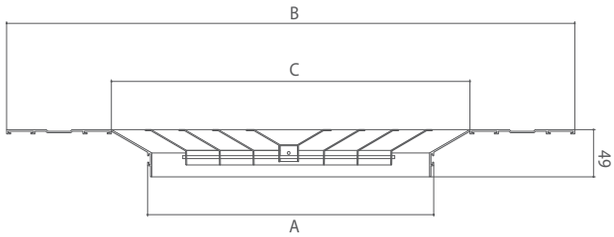
## Characteristics
Recommended for rooms with a maximum height of 4 meters .
Removable central cone for easy installation and cleaning.
The difference between the recommended supply air temperature and the room air temperature is ± 10 K.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

## Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

## Technical design
CD narrow/large frame




## Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes on diffuser connection (optional)

## Selection table CD

| Air flow ($m^3/h$) | LXH | 150x150 | 225x225 | 300x300 | 375x375 | 455x455 | 525x525 | 600x600 |
|---|---|---|---|---|---|---|---|---|
| | Ak [$m^2$] | 0.0109 | 0.0244 | 0.0435 | 0.0679 | 0.0978 | 0.1331 | 0.1739 |
| 100 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | 2.6<br>0,6<br>18.0<br>5.0 | | | | | | |
| 160 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | 4.1<br>1,0<br>29.0<br>12.0 | 1.8<br>0,7<br>13.0<br>2.0 | | | | | |
| 200 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | 5.1<br>1,2<br>35.0<br>18.0 | 2.3<br>0,8<br>18.0<br>4.0 | | | | | |
| 300 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | 7.7<br>1,9<br>45.0<br>41.0 | 3.4<br>1,3<br>28.0<br>8.0 | 1.9<br>0,9<br>16.0<br>3.0 | | | | |
| 400 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | 4.6<br>1,7<br>35.0<br>15.0 | 2.6<br>1,3<br>24.0<br>5.0 | 1.6<br>1,0<br>15.0<br>2.0 | | | |
| 500 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | 5.7<br>2,1<br>41.0<br>23.0 | 3.2<br>1,6<br>29.0<br>7.0 | 2.1<br>1,3<br>20.0<br>3.0 | | | |
| 600 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | 6.8<br>2,5<br>45.0<br>33.0 | 3.8<br>1,9<br>34.0<br>10.0 | 2.5<br>1,5<br>25.0<br>4.0 | 1.7<br>1,3<br>17.0<br>2.0 | | |
| 700 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | 8.0<br>2,9<br>49.0<br>44.0 | 4.5<br>2,2<br>37.0<br>14.0 | 2.9<br>1,8<br>28.0<br>6.0 | 2.0<br>1,5<br>21.0<br>3.0 | | |
| 800 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | 9.1<br>3,3<br>53.0<br>58.0 | 5.1<br>2,5<br>41.0<br>18.0 | 3.3<br>2,0<br>32.0<br>7.0 | 2.3<br>1,7<br>24.0<br>4.0 | 1.7<br>1,4<br>18.0<br>2.0 | |
| 900 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | 5.8<br>2,8<br>44.0<br>23.0 | 3.7<br>2,3<br>35.0<br>9.0 | 2.6<br>1,9<br>27.0<br>5.0 | 1.9<br>1,6<br>21.0<br>2.0 | |
| 1000 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | 6.4<br>3,1<br>46.0<br>29.0 | 4.1<br>2,5<br>37.0<br>12.0 | 2.8<br>2,1<br>30.0<br>6.0 | 2.1<br>1,8<br>23.0<br>3.0 | 1.6<br>1,6<br>18.0<br>2.0 |
| 1200 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | 7.7<br>3,8<br>51.0<br>41.0 | 4.9<br>3,0<br>42.0<br>17.0 | 3.4<br>2,5<br>34.0<br>8.0 | 2.5<br>2,1<br>28.0<br>4.0 | 1.9<br>1,9<br>23.0<br>3.0 |
| 1600 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | | 6.6<br>4,0<br>49.0<br>30.0 | 4.5<br>3,3<br>41.0<br>14.0 | 3.3<br>2,9<br>35.0<br>8.0 | 2.6<br>2,5<br>30.0<br>5.0 |
| 2000 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | | 8.2<br>5,0<br>54.0<br>47.0 | 5.7<br>4,2<br>47.0<br>23.0 | 4.2<br>3,6<br>41.0<br>12.0 | 3.2<br>3,1<br>35.0<br>7.0 |
| 3000 | Veff [m/s]<br>X ( m )<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 8.5<br>6,3<br>57.0<br>51.0 | 6.3<br>5,4<br>51.0<br>27.0 | 4.8<br>4,7<br>45.0<br>16.0 |

NR < 35 | 35≤NR≤45 | NR>45

## Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser
* Vk = 0.2 [m/s] - Speed at which airflow length was calculated "x" [m]

## Standard dimensions

| CD | | | |
|---|---|---|---|
| **neck / external size** | **A** | **B** | **C** |
| **150 / 290** | 150 x 150 | 290 x 290 | 230 x 230 |
| **150 / 595** | 150 x 150 | 595 x 595 | 235 x 235 |
| **225 / 365** | 225 x 225 | 365 x 365 | 305 x 305 |
| **225 / 595** | 225 x 225 | 595 x 595 | 315 x 315 |
| **300 / 440** | 300 x 300 | 440 x 440 | 380 x 380 |
| **300 / 595** | 300 x 300 | 595 x 595 | 385 x 385 |
| **375 / 515** | 375 x 375 | 515 x 515 | 455 x 455 |
| **375 / 595** | 375 x 375 | 595 x 595 | 465 x 465 |
| **455 / 595** | 455 x 455 | 595 x 595 | 535 x 535 |
| **480 / 620** | 480 x 480 | 620 x 620 | 560 x 560 |
| **525 / 665** | 525 x 525 | 665 x 665 | 605 x 605 |
| **600 / 740** | 600 x 600 | 740 x 740 | 680 x 680 |

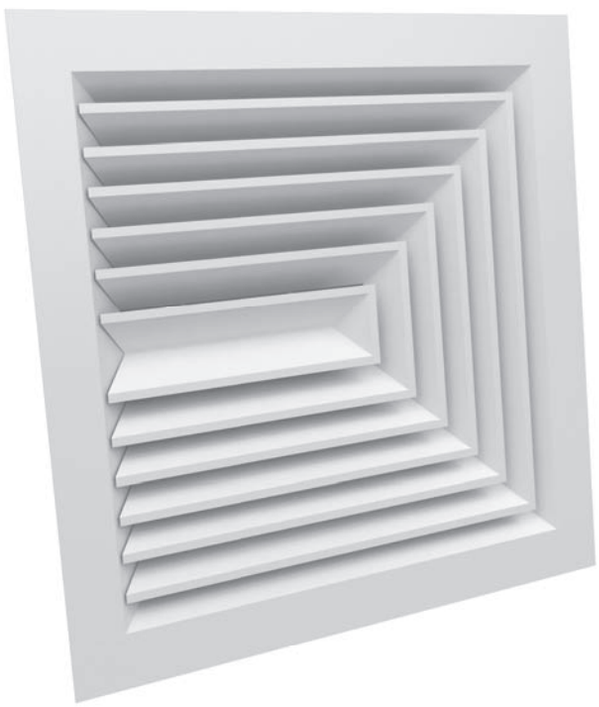
## Order codes

**Type** + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **CD**
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions** — **neck / external size**: 150 / 290; 150 / 595; 225 / 365; 225 / 595; 300 / 440; 300 / 595; 375 / 515; 375 / 595; 455 / 595; 480 / 620; 525 / 665; 600 / 740
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

# Ceiling Diffuser CD-S





CD-S3

### Description
1, 2 or 3 direction square diffuser preferred for paneled ceiling mounting.

### Application
Supply/ exhaust of airflow.

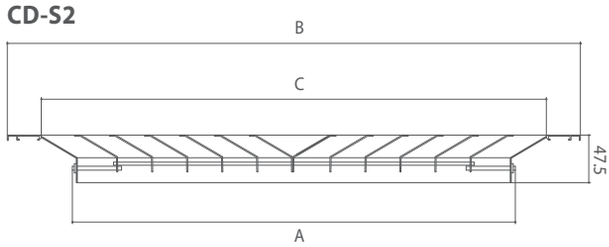
### Characteristics
Recommended for rooms of a maximum height of 4 meters .
Removable central cone for easy installation and cleaning.
Suitable for horizontal air supply.
The difference between the recommended supply air temperature and the room air temperature is ± 10 K.
The air flow can be adjusted by mounting a air control damper **OBD** on the diffuser inlet.

### Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

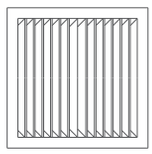
### Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request.
Natural anodised aluminum.

### Technical design

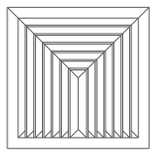
**CD-S2**



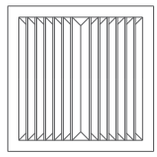

**CD-S1**



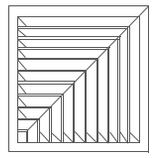
**CD-S3**



**CD-S2**



**CD-S2-90**



### Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes on diffuser connection (optional)

# Ceiling Diffuser CD-S

## Selection table CD-S

| Air flow ($m^3/h$) | LxH | 150 x 150 | | | 225 x 225 | | | 300 x 300 | | | 375 x 375 | | | 455 x 455 | | | 525 x 525 | | | 600 x 600 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Model | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 |
| | Ak($m^2$) | 0.0096 | 0.0093 | 0.0088 | 0.0215 | 0.0209 | 0.02 | 0.0383 | 0.0373 | 0.0355 | 0.0598 | 0.0582 | 0.0554 | 0.0863 | 0.0838 | 0.0798 | 0.1174 | 0.1141 | 0.1086 | 0.1534 | 0.149 | 0.1418 |
| 100 | Veff [m/s] | 2.9 | 3.0 | | | | | | | | | | | | | | | | | | | |
| | X (m) | 2.2 | 0.8 | | | | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | 22 | 22 | | | | | | | | | | | | | | | | | | | |
| | ΔPt(Pa) | 5.9 | 6.7 | | | | | | | | | | | | | | | | | | | |
| 160 | Veff [m/s] | 4.6 | 4.8 | 5.1 | | | | | | | | | | | | | | | | | | |
| | X (m) | 3.5 | 1.2 | 1.2 | | | | | | | | | | | | | | | | | | |
| | NR [dB(A)] | 33 | 33 | 29 | | | | | | | | | | | | | | | | | | |
| | ΔPt(Pa) | 15 | 17.1 | 22 | | | | | | | | | | | | | | | | | | |
| 200 | Veff [m/s] | 5.8 | 6.0 | 6.3 | 2.6 | 2.7 | 2.8 | | | | | | | | | | | | | | | |
| | X (m) | 4.4 | 1.5 | 1.5 | 2.9 | 1 | 1 | | | | | | | | | | | | | | | |
| | NR [dB(A)] | 39 | 39 | 34 | 22 | 22 | <20 | | | | | | | | | | | | | | | |
| | ΔPt(Pa) | 23.4 | 26.8 | 34 | 4.7 | 5.3 | 7 | | | | | | | | | | | | | | | |
| 300 | Veff [m/s] | 8.7 | 9.0 | 9.5 | 3.9 | 4.0 | 4.2 | 2.2 | 2.2 | | | | | | | | | | | | | |
| | X (m) | 6.6 | 2.3 | 2.3 | 4.4 | 1.5 | 1.5 | 3.3 | 1.1 | | | | | | | | | | | | | |
| | NR [dB(A)] | 49 | 49 | 44 | 32 | 32 | 28 | 20 | 21 | | | | | | | | | | | | | |
| | ΔPt(Pa) | 52.7 | 60.2 | 77 | 10.5 | 11.9 | 15 | 3.3 | 3.7 | | | | | | | | | | | | | |
| 400 | Veff [m/s] | | | | 5.2 | 5.3 | 5.6 | 2.9 | 3.0 | 3.1 | 1.9 | 1.9 | | | | | | | | | | |
| | X (m) | | | | 5.8 | 2 | 2 | 4.4 | 1.5 | 1.5 | 3.5 | 1.2 | | | | | | | | | | |
| | NR [dB(A)] | | | | 39 | 40 | 35 | 27 | 28 | 23 | <20 | <20 | | | | | | | | | | |
| | ΔPt(Pa) | | | | 18.7 | 21.2 | 27 | 5.9 | 6.7 | 9 | 2.4 | 2.7 | | | | | | | | | | |
| 600 | Veff [m/s] | | | | 7.8 | 8.0 | 8.3 | 4.4 | 4.5 | 4.7 | 2.8 | 2.9 | 3.0 | 1.9 | 2.0 | | | | | | | |
| | X (m) | | | | 8.8 | 3 | 3 | 6.6 | 2.3 | 2.3 | 5.3 | 1.8 | 1.8 | 4.4 | 1.5 | | | | | | | |
| | NR [dB(A)] | | | | 49 | 50 | 45 | 37 | 38 | 33 | 28 | 29 | 24 | 20 | 21 | | | | | | | |
| | ΔPt(Pa) | | | | 42.1 | 47.7 | 61 | 13.3 | 15 | 19 | 5.4 | 6.2 | 8 | 2.6 | 3 | | | | | | | |
| 800 | Veff [m/s] | | | | | | | 5.8 | 6.0 | 6.3 | 3.7 | 3.8 | 4.0 | 2.6 | 2.7 | 2.8 | 1.9 | 1.9 | | | | |
| | X (m) | | | | | | | 8.8 | 3 | 3 | 7 | 2.4 | 2.4 | 5.8 | 2 | 2 | 5 | 1.7 | | | | |
| | NR [dB(A)] | | | | | | | 44 | 45 | 40 | 35 | 36 | 31 | 27 | 28 | 24 | 21 | 22 | | | | |
| | ΔPt(Pa) | | | | | | | 23.6 | 26.6 | 34 | 9.7 | 10.9 | 14 | 4.6 | 5.3 | 7 | 2.5 | 2.8 | | | | |
| 1000 | Veff [m/s] | | | | | | | 7.3 | | | 4.6 | 4.8 | 5.0 | 3.2 | 3.3 | 3.5 | 2.4 | 2.4 | 2.6 | 1.8 | 1.9 | |
| | X (m) | | | | | | | 11 | | | 8.8 | 3 | 3 | 7.3 | 2.5 | 2.5 | 6.3 | 2.2 | 2.1 | 5.5 | 1.9 | |
| | NR [dB(A)] | | | | | | | 50 | | | 41 | 41 | 37 | 33 | 34 | 29 | 26 | 28 | 23 | 21 | 22 | |
| | ΔPt(Pa) | | | | | | | 36.8 | | | 15.1 | 17.1 | 22 | 7.3 | 8.2 | 11 | 3.9 | 4.4 | 6 | 2.3 | 2.6 | |
| 1200 | Veff [m/s] | | | | | | | | | | 5.6 | 5.7 | 6.0 | 3.9 | 4.0 | 4.2 | 2.8 | 2.9 | 3.1 | 2.2 | 2.2 | 2.4 |
| | X (m) | | | | | | | | | | 10.5 | 3.7 | 3.6 | 8.8 | 3 | 3 | 7.5 | 2.6 | 2.6 | 6.6 | 2.3 | 2.3 |
| | NR [dB(A)] | | | | | | | | | | 45 | 46 | 41 | 37 | 38 | 34 | 31 | 32 | 28 | 25 | 27 | 22 |
| | ΔPt(Pa) | | | | | | | | | | 21.7 | 24.6 | 32 | 10.4 | 11.9 | 15 | 5.6 | 6.4 | 8 | 3.3 | 3.8 | 5 |
| 1600 | Veff [m/s] | | | | | | | | | | 7.4 | | 8.0 | 5.1 | 5.3 | 5.6 | 3.8 | 3.9 | 4.1 | 2.9 | 3.0 | 3.1 |
| | X (m) | | | | | | | | | | 14 | | 4.8 | 11.7 | 4.1 | 4 | 10 | 3.5 | 3.4 | 8.8 | 3 | 3 |
| | NR [dB(A)] | | | | | | | | | | 52 | | 48 | 44 | 46 | 41 | 38 | 39 | 35 | 33 | 34 | 29 |
| | ΔPt(Pa) | | | | | | | | | | 38.7 | | 56 | 18.6 | 21.2 | 27 | 10 | 11.4 | 15 | 5.9 | 6.7 | 9 |
| 2000 | Veff [m/s] | | | | | | | | | | | | | | | | | 4.9 | 5.1 | | 3.7 | 3.9 |
| | X (m) | | | | | | | | | | | | | | | | | 4.3 | 4.3 | | 3.8 | 3.8 |
| | NR [dB(A)] | | | | | | | | | | | | | | | | | 45 | 40 | | 39 | 35 |
| | ΔPt(Pa) | | | | | | | | | | | | | | | | | 17.8 | 23 | | 10.4 | 13 |
| 3000 | Veff [m/s] | | | | | | | | | | | | | | | | | | 7.7 | | 5.6 | 5.9 |
| | X (m) | | | | | | | | | | | | | | | | | | 6.4 | | 5.7 | 5.6 |
| | NR [dB(A)] | | | | | | | | | | | | | | | | | | 50 | | 49 | 45 |
| | ΔPt(Pa) | | | | | | | | | | | | | | | | | | 51 | | 23.5 | 30 |

NR < 35 | 35≤NR≤45 | NR>45

## Annotation

- Ak [m²] - Free surface of the diffuser
- Veff[m/s] - Air velocity in diffuser
- X [m] - Air jet length
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss
- Vk = 0.25 [m/s] - Speed at which air jet length was calculated "x" [m]

## Standard dimensions

| CD-S 1,2,3 | | | |
|---|---|---|---|
| **neck / external size** | **A** | **B** | **C** |
| **150 / 290** | 150 x 150 | 290 x 290 | 230 x 230 |
| **150 / 595** | 150 x 150 | 595 x 595 | 235 x 235 |
| **225 / 365** | 225 x 225 | 365 x 365 | 305 x 305 |
| **225 / 595** | 225 x 225 | 595 x 595 | 315 x 315 |
| **300 / 440** | 300 x 300 | 440 x 440 | 380 x 380 |
| **300 / 595** | 300 x 300 | 595 x 595 | 385 x 385 |
| **375 / 515** | 375 x 375 | 515 x 515 | 455 x 455 |
| **375 / 595** | 375 x 375 | 595 x 595 | 465 x 465 |
| **455 / 595** | 455 x 455 | 595 x 595 | 535 x 535 |
| **480 / 620** | 480 x 480 | 620 x 620 | 560 x 560 |
| **525 / 665** | 525 x 525 | 665 x 665 | 605 x 605 |
| **600 / 740** | 600 x 600 | 740 x 740 | 680 x 680 |

## Order codes

| **Type** | – | **No. of directions** | + | **OBD** | + | **ADP IZ/N** | / | **Dimensions** | / | **L X H** | / | **O** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CD-S** | | 1, 2, 3 | | Air control damper | | Insulated/ Non-insulated adapter | | **neck / external**<br>150 / 290<br>150 / 595<br>225 / 365<br>225 / 595<br>300 / 440<br>300 / 595<br>375 / 515<br>375 / 595<br>455 / 595<br>480 / 620<br>525 / 665 | | Other dimensions on request | | **0** Painted white RAL 9016<br>**R** Other RAL paints<br>Natural anodised |

# Ceiling Diffuser With Double Deflection CD-DD

## Description
Aluminum rectangular diffuser designed for paneled ceiling mounting.

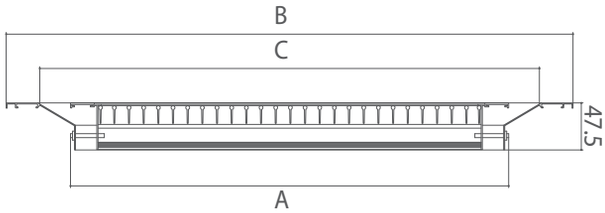
## Application
Suitable for supply and exhaust air systems.

## Characteristics
The diffuser is provided with a central core that consists of a **DD** -grille (double deflection) with two rows of individually adjustable blades.
The adjustable baldes allow airflow distribution to desired areas.
Removable central cone for easy installation and cleaning.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet connection.

## Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

## Technical design




## Assembly Methods
Paneled ceiling 600x600mm (standard)
Mounting holes on the diffuser inlet connection (optional)

## Selection table CD-DD

| Air flow m³/h | LXH | 150x150 | 225x225 | 300x300 | 375x375 | 455x455 | 480x480 | 525x525 | 600x600 |
|---|---|---|---|---|---|---|---|---|---|
| | Ak [m²] | 0.014 | 0.028 | 0.048 | 0.075 | 0.110 | 0.123 | 0.147 | 0.192 |
| **100** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | 2.0<br>3.0<br>20.0<br>3.0 | | | | | | | |
| **160** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | 3.2<br>3.7<br>23.0<br>7.0 | 1.6<br>2.4<br>20.0<br>2.0 | | | | | | |
| **200** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br>5.0<br>25.0<br>12.0 | 2.0<br>3.5<br>20.0<br>3.0 | | | | | | |
| **300** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | 6.0<br>7.5<br>30.0<br>30.0 | 3.0<br>5.3<br>25.0<br>7.0 | 1.7<br>2.5<br>20.0<br>3.0 | | | | | |
| **400** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | 4.0<br>7.0<br>30.0<br>12.0 | 2.3<br>4.0<br>23.0<br>5.0 | 1.5<br>2.0<br>15.0<br>2.0 | | | | |
| **500** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | 4.9<br>9.0<br>30.0<br>20.0 | 2.9<br>5.0<br>28.0<br>7.0 | 1.9<br>1.3<br>19.0<br>3.0 | | | | |
| **600** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | 5.9<br>9.0<br>35.0<br>33.0 | 3.4<br>7.0<br>33.0<br>9.0 | 2.2<br>4.0<br>25.0<br>4.0 | 1.5<br>3.0<br>17.0<br>2.0 | | | |
| **700** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | 6.9<br>12.0<br>40.0<br>40.0 | 4.0<br>8.0<br>35.0<br>15.0 | 2.6<br>7.5<br>30.0<br>5.0 | 1.8<br>5.0<br>21.0<br>3.0 | | | |
| **800** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | 7.9<br>18.0<br>45.0<br>50.0 | 4.6<br>11.0<br>40.0<br>20.0 | 3.0<br>8.0<br>31.0<br>7.0 | 2.0<br>6.0<br>23.0<br>4.0 | 1.8<br>3.0<br>20.0<br>3.0 | 1.5<br>2.0<br>17.0<br>2.0 | |
| **900** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | 5.2<br>12.0<br>43.0<br>25.0 | 3.3<br>9.0<br>34.0<br>15.0 | 2.3<br>7.0<br>26.0<br>5.0 | 2.0<br>4.0<br>23.0<br>3.0 | 1.7<br>2.5<br>20.0<br>2.0 | |
| **1000** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | 5.7<br>15.0<br>46.0<br>32.0 | 3.7<br>12.0<br>36.0<br>18.0 | 2.5<br>8.0<br>29.0<br>6.0 | 2.3<br>7.0<br>25.0<br>4.0 | 1.9<br>4.0<br>23.0<br>3.0 | 1.4<br>2.0<br>18.0<br>2.0 |
| **1200** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | 6.9<br>20.0<br>50.0<br>41.0 | 4.4<br>15.0<br>41.0<br>20.0 | 3.0<br>11.0<br>33.0<br>8.0 | 2.7<br>9.0<br>30.0<br>6.0 | 2.3<br>7.0<br>27.0<br>4.0 | 1.7<br>3.0<br>22.0<br>3.0 |
| **1600** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | | 5.9<br>17.0<br>48.0<br>29.0 | 4.0<br>13.0<br>40.0<br>14.0 | 3.6<br>12.0<br>37.0<br>10.0 | 3.0<br>10.0<br>33.0<br>8.0 | 2.3<br>7.0<br>29.0<br>5.0 |
| **2000** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | | 7.4<br>20.0<br>53.0<br>45.0 | 5.0<br>17.0<br>45.0<br>23.0 | 4.5<br>15.0<br>42.0<br>18.0 | 3.8<br>11.0<br>40.0<br>12.0 | 2.9<br>8.5<br>34.0<br>7.0 |
| **3000** | Veff [m/s]<br>X (m)<br>NR [dB(A)]<br>ΔPt [pa] | | | | | 7.6<br>20.0<br>55.0<br>50.0 | 6.8<br>18.0<br>53.0<br>40.0 | 5.7<br>15.0<br>50.0<br>25.0 | 4.3<br>12.0<br>44.0<br>16.0 |

## Annotation

- Ak [m²] - Free surface of the diffuser
- Veff[m/s] - Air velocity in diffuser
- X [m] - Air jet length
- ΔPt [pa] - Pressure loss

* NR [dB(A)] - Noise level without the room attenuation

## Standard dimensions

| CD - DD | | | |
|---|---|---|---|
| **neck/external size** | **A** | **B** | **C** |
| **150 / 290** | 150 x 150 | 290 x 290 | 230 x 230 |
| **150 / 595** | 150 x 150 | 595 x 595 | 235 x 235 |
| **225 / 365** | 225 x 225 | 365 x 365 | 305 x 305 |
| **225 / 595** | 225 x 225 | 595 x 595 | 315 x 315 |
| **300 / 440** | 300 x 300 | 440 x 440 | 380 x 380 |
| **300 / 595** | 300 x 300 | 595 x 595 | 385 x 385 |
| **375 / 515** | 375 x 375 | 515 x 515 | 455 x 455 |
| **375 / 595** | 375 x 375 | 595 x 595 | 465 x 465 |
| **455 / 595** | 455 x 455 | 595 x 595 | 535 x 535 |
| **480 / 620** | 480 x 480 | 620 x 620 | 560 x 560 |
| **525 / 665** | 525 x 525 | 665 x 665 | 605 x 605 |
| **600 / 740** | 600 x 600 | 740 x 740 | 680 x 680 |

## Order codes

**Type** + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **CD-DD**
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **neck/external size**
  - 150 / 290
  - 150 / 595
  - 225 / 365
  - 225 / 595
  - 300 / 440
  - 375 / 515
  - 375 / 595
  - 455 / 595
  - 480 / 620
  - 525 / 665
  - 600 / 740
- **L X H**: Other dimensions on request
- **O**:
  - **0** Painted white RAL 9016
  - **R** Other RAL paints
  - Natural anodised

# Linear Slot Diffuser SLOT

## Description
Linear aluminum diffuser with adjustable blades.

## Application
Supply/ exhaust of the airflow

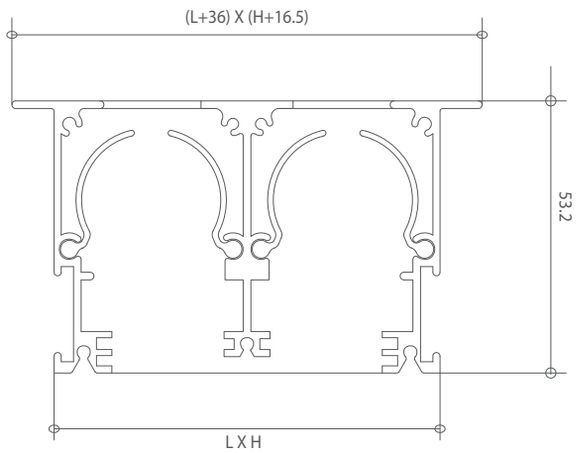
## Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
The recommended temperature difference between supplied air and indoor air is ± 10 K.
Long run diffusers are made of assembled modules (maximum 3 meters full length). In this case the product is delivered with joining elements between modules.
Blades are used for air flow control.
On request, the diffuser can be made perimetral.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**
Attachment spring **A**

## Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Anodised aluminium

## Assembly Methods
Attachment spring.

## Technical design




| No. of slots | H interior (height) slot = 19 mm | H interior (height) slot = 25 mm |
|---|---|---|
| 1 | 40 | 45 |
| 2 | 76 | 88 |
| 3 | 112 | 130 |
| 4 | 149 | 174 |
| 5 | 186 | 216 |
| 6 | 223 | 260 |
| 7 | 260 | 304 |
| 8 | 297 | 348 |

## Selection table SLOT 19 mm

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 1<br>0.019 |
|---|---|---|
| **Length = 1000 mm** | | |
| **33** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 1.00<br>1.00<br>-<br>0.98 |
| **67** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>4.02 |
| **106** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 4.00<br>3.00<br>15.00<br>9.22 |
| **140** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>21.00<br>16.18 |
| **173** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>27.00<br>25.40 |
| **206** | X [m] (Horizontal<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>5.00<br>31.00<br>36.38 |
| **240** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>5.00<br>35.00<br>49.52 |
| **279** | X [m] (Horizontal)<br>X[m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>5.00<br>38.00<br>64.72 |
| **312** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>5.00<br>41.00<br>82.18 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 2<br>0.038 |
|---|---|---|
| **Length = 1000 mm** | | |
| **67** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 1.00<br>1.00<br>-<br>0.98 |
| **140** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 3.00<br>3.00<br>-<br>4.02 |
| **206** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>18.00<br>9.22 |
| **279** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>5.00<br>24.00<br>16.18 |
| **346** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>6.00<br>30.00<br>25.40 |
| **413** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>6.00<br>34.00<br>36.38 |
| **485** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>7.00<br>38.00<br>49.52 |
| **552** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>7.00<br>41.00<br>64.72 |
| **625** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>8.00<br>44.00<br>82.18 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 3<br>0.057 |
|---|---|---|
| **Length = 1000 mm** | | |
| **106** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>0.98 |
| **206** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 4.00<br>3.00<br>-<br>4.02 |
| **312** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>5.00<br>19.00<br>9.22 |
| **413** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>6.00<br>26.00<br>16.18 |
| **519** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>7.00<br>32.00<br>25.40 |
| **625** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>8.00<br>36.00<br>36.38 |
| **725** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>40.00<br>49.52 |
| **831** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>43.00<br>64.72 |
| **932** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>10.00<br>46.00<br>82.18 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 4<br>0.076 |
|---|---|---|
| **Length = 1000 mm** | | |
| **140** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>0.98 |
| **279** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>-<br>4.02 |
| **413** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>6.00<br>21.00<br>9.22 |
| **552** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>7.00<br>27.00<br>16.18 |
| **692** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>8.00<br>33.00<br>25.40 |
| **831** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>37.00<br>36.38 |
| **971** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>10.00<br>41.00<br>49.52 |
| **1105** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>10.00<br>44.00<br>64.72 |
| **1244** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 15.00<br>11.00<br>47.00<br>82.18 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 5<br>0.095 |
|---|---|---|
| **Length = 1000 mm** | | |
| **173** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>0.98 |
| **346** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>-<br>4.02 |
| **519** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>6.00<br>22.00<br>9.22 |
| **692** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>8.00<br>28.00<br>16.18 |
| **865** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>34.00<br>25.40 |
| **1038** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>10.00<br>38.00<br>36.38 |
| **1211** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 15.00<br>11.00<br>42.00<br>49.52 |
| **1384** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 16.00<br>12.00<br>45.00<br>64.72 |
| **1557** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 17.00<br>12.00<br>48.00<br>82.18 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 6<br>0.114 |
|---|---|---|
| **Length = 1000 mm** | | |
| **206** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 3.00<br>2.00<br>-<br>0.98 |
| **413** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>5.00<br>-<br>4.02 |
| **625** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>7.00<br>22.00<br>9.22 |
| **831** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>29.00<br>16.18 |
| **1038** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>10.00<br>35.00<br>25.40 |
| **1244** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 15.00<br>11.00<br>39.00<br>36.38 |
| **1451** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 16.00<br>12.00<br>43.00<br>49.52 |
| **1663** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 17.00<br>13.00<br>46.00<br>64.72 |
| **1869** | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 18.00<br>14.00<br>49.00<br>82.18 |

## Selection table SLOT 25 mm

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 1<br>0.025 |
|---|---|---|
| Length = 1000 mm | | |
| 39 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 1.00<br>1.00<br>-<br>0.98 |
| 73 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>1.00<br>3.53 |
| 112 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 4.00<br>4.00<br>14.00<br>7.94 |
| 145 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>4.00<br>21.00<br>14.22 |
| 184 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>5.00<br>27.00<br>22.16 |
| 223 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>5.00<br>31.00<br>31.87 |
| 257 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>5.00<br>35.00<br>43.34 |
| 296 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>6.00<br>38.00<br>56.78 |
| 329 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>6.00<br>41.00<br>71.78 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 2<br>0.05 |
|---|---|---|
| Length = 1000 mm | | |
| 73 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 1.00<br>2.00<br>-<br>0.98 |
| 145 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 4.00<br>3.00<br>-<br>3.53 |
| 223 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>5.00<br>17.00<br>7.94 |
| 296 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>6.00<br>24.00<br>14.22 |
| 368 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>7.00<br>30.00<br>22.16 |
| 441 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>7.00<br>34.00<br>31.87 |
| 513 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>8.00<br>38.00<br>43.34 |
| 591 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>9.00<br>41.00<br>56.78 |
| 664 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>9.00<br>44.00<br>71.78 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 3<br>0.075 |
|---|---|---|
| Length= 1000 mm | | |
| 112 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>0.98 |
| 223 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 4.00<br>4.00<br>-<br>3.53 |
| 329 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>6.00<br>19.00<br>7.94 |
| 441 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>7.00<br>26.00<br>14.22 |
| 552 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>8.00<br>31.00<br>22.16 |
| 664 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>9.00<br>36.00<br>31.87 |
| 776 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>10.00<br>40.00<br>43.34 |
| 882 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>10.00<br>43.00<br>56.78 |
| 993 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>11.00<br>46.00<br>71.78 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 4<br>0.1 |
|---|---|---|
| Length = 1000 mm | | |
| 145 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>2.00<br>-<br>0.98 |
| 296 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>5.00<br>-<br>3.53 |
| 441 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 7.00<br>7.00<br>20.00<br>7.94 |
| 591 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 10.00<br>9.00<br>27.00<br>14.22 |
| 737 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>9.00<br>33.00<br>22.16 |
| 882 | X[m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>10.00<br>37.00<br>31.87 |
| 1032 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>11.00<br>41.00<br>43.34 |
| 1177 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 15.00<br>12.00<br>44.00<br>56.78 |
| 1328 | X [m] (Orizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 16.00<br>13.00<br>47.00<br>71.78 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 5<br>0.125 |
|---|---|---|
| Length = 1000 mm | | |
| 184 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 2.00<br>3.00<br>-<br>0.98 |
| 368 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 5.00<br>5.00<br>-<br>3.53 |
| 552 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 8.00<br>8.00<br>21.00<br>7.94 |
| 737 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 11.00<br>9.00<br>28.00<br>14.22 |
| 921 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 13.00<br>10.00<br>34.00<br>22.16 |
| 1105 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>12.00<br>38.00<br>31.87 |
| 1289 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 15.00<br>12.00<br>42.00<br>43.34 |
| 1473 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 16.00<br>13.00<br>45.00<br>56.78 |
| 1657 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 17.00<br>14.00<br>48.00<br>71.78 |

| Air flow ($m^3/h$) | No. of slots<br>Ak (mp) | 6<br>0.15 |
|---|---|---|
| Length = 1000 mm | | |
| 223 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 3.00<br>3.00<br>-<br>0.98 |
| 441 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 6.00<br>6.00<br>-<br>3.53 |
| 664 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 9.00<br>9.00<br>22.00<br>7.94 |
| 882 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 12.00<br>10.00<br>29.00<br>14.22 |
| 1105 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 14.00<br>12.00<br>34.00<br>22.16 |
| 1328 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 16.00<br>12.00<br>39.00<br>31.87 |
| 1546 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 17.00<br>14.00<br>43.00<br>43.34 |
| 1769 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 18.00<br>15.00<br>46.00<br>56.78 |
| 1986 | X [m] (Horizontal)<br>X [m] (Vertical)<br>NR [dB(A)]<br>△Pt [pa] | 19.00<br>16.00<br>48.00<br>71.78 |

## Annotation

* Ak [m²] - Free surface
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

**Type - No. of slots** + **ADP IZ/N** + **A** ( Attachment spring ) / **Dimensions** / **Blades color** / **O**

- **Type - No. of slots**
  - SLOT
  - 1…..8 slots
- **ADP IZ/N**
  - Insulated/Non-insulated adapter
- **A** ( Attachment spring )
  - No. of springs depending on length and no. of slots
- **Dimensions**
  - L (length) on request
- **Blades color**
  - Same as the frame
  - Black RAL 9005
- **O**
  - **0** Painted white RAL 9016
  - **R** Other RAL paints
  - Natural anodised

# Linear Diffuser With Nozzles BD1/BD2/BDV

BDV

## Description
Aluminum linear diffuser with individually maneuvarable nozzles. Recommended for use in traffic areas, classrooms, laboratories, etc.

## Application
Airflow supply.

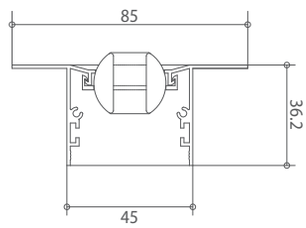
## Characteristics
Recommended for rooms between 2.40 and 4.00 m in height.
Long run diffusers are made of assembled modules (maximum 3 m full length). In this case the product is delivered with joining elements between modules.
The spherical design of the nozzles enables air flow redirection in the desired areas.
The spherical nozzles are available in two colors: white RAL 9016 and black RAL 9005.
Depending on the technical and architectural requirements, the diffuser is made with a single row of nozzles - **BD1** or two parallel rows of nozzles - **BD2/BDV**.

## Accessories
Attachment spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

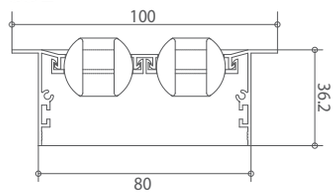
## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural/black anodised aluminum.
Polypropylene nozzles.

## Technical design

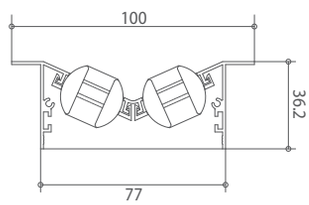
**BD1**




**BD2**




**BDV**




## Assembly Methods
Attachment spring.

## Selection table BD1, BD2, BDV

| Air flow m³/h | Model | BD1 | | BD2 | | BDV |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 1 | 2 | 3 |
| | Ak(m²) | 0.0034 | 0.0028 | 0.0067 | 0.0057 | 0.0063 |
| **30** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | -<br><20<br>11.0<br>2.45 | -<br><20<br>13.0<br>2.98 | | | |
| **40** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 1.5<br>21.0<br>19.0<br>3.27 | 1.5<br>25.0<br>23.0<br>3.97 | | | |
| **50** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 1.6<br>26.0<br>29.0<br>4.08 | 1.6<br>30.0<br>36.0<br>4.96 | 1.7<br><20<br>9.0<br>2.07 | 1.6<br><20<br>10.0<br>2.44 | -<br><20<br>9.0<br>2.20 |
| **60** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 1.8<br>31.0<br>41.0<br>4.90 | 1.7<br>35.0<br>52.0<br>5.95 | 1.9<br><20<br>12.0<br>2.49 | 1.7<br>21.0<br>14.0<br>2.92 | 1.8<br><20<br>13.0<br>2.65 |
| **70** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 2.0<br>35.0<br>55.0<br>5.72 | 1.9<br>39.0<br>71.0<br>6.94 | 2.1<br>23.0<br>16.0<br>2.90 | 1.8<br>25.0<br>19.0<br>3.41 | 1.9<br>23.0<br>17.0<br>3.09 |
| **80** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 2.3<br>38.0<br>71.0<br>6.54 | 2.1<br>42.0<br>93.0<br>7.94 | 2.3<br>26.0<br>21.0<br>3.32 | 1.9<br>29.0<br>25.0<br>3.90 | 2.0<br>26.0<br>22.0<br>3.53 |
| **90** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | 2.6<br>41.0<br>90.0<br>7.35 | 2.3<br>45.0<br>118.0<br>8.93 | 2.5<br>29.0<br>27.0<br>3.73 | 2.1<br>32.0<br>32.0<br>4.39 | 2.2<br>29.0<br>28.0<br>3.97 |
| **100** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | | | 2.7<br>32.0<br>33.0<br>4.15 | 2.2<br>34.0<br>39.0<br>4.87 | 2.4<br>31.0<br>34.0<br>4.41 |
| **110** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | | | 3.0<br>34.0<br>40.0<br>4.56 | 2.4<br>37.0<br>48.0<br>5.36 | 2.6<br>33.0<br>41.0<br>4.85 |
| **120** | X [m]<br>NR [dB(A)]<br>ΔPt [pa]<br>Veff [m/s] | | | 3.2<br>36.0<br>47.0<br>4.98 | 2.6<br>39.0<br>57.0<br>5.85 | 2.9<br>35.0<br>49.0<br>5.29 |

Note: The values of the selection table are given for a length of 1m.

### Annotation

- Ak [m²] - Free surface of the diffuser
- X [m] - Length airflow at speed Vk [m/s] = 0.3 m/s
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss
- Veff[m/s] - Air velocity in diffuser

COLUMN 1 - nozzles directed toward the residence area
COLUMN 2 - nozzles directed 50% toward the residence area and 50% toward the ceiling
COLUMN 3 - nozzles directed toward the ceiling

## Order codes

| **Type** | + | **A** (Attachment spring) | + | **ADP IZ/N** | / | **Dimensions** | / | **Nozzles color** | / | **O** |
|---|---|---|---|---|---|---|---|---|---|---|
| **BD1**<br>**BD2**<br>**BDV** | | No. of springs depending on the size of the diffuser | | Insulated/ Non-insulated adapter | | **L** (lenght) on request | | White RAL 9016<br>Black RAL 9005 | | **0** Painted white RAL 9016<br>**R** Other RAL paints<br>Black/natural anodised |

# Linear Disc Diffuser SP 2 / SP 3

SP3

### Description
Aluminium linear diffuser provided with aerodynamic shaped discs.

### Application
Airflow supply.

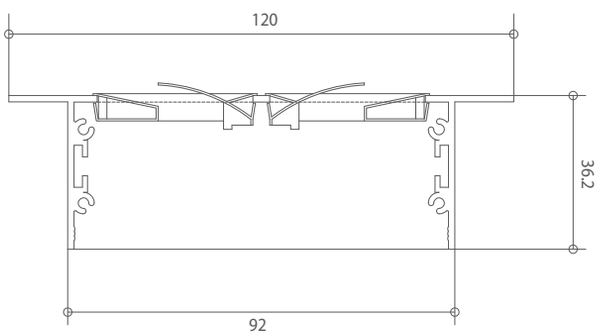
### Characteristics
Depending on the technical and architectural requirements, the diffuser is provided with 2 or 3 parallel rows of discs **SP2/ SP3**
Each disk can be rotated at 360 degrees, providing unlimited possibilities to diffuse the air, without affecting the air flow, pressure loss or noise level.
Discs are available in two colors: white RAL 9016 and black RAL 9005
Long run diffusers are made out of assembled modules (maximum 3 m full length). In this case the product is delivered with joining elements between modules.
Diffuser recommended for constant or variable air flow.

### Accessories
Attachment spring **A**
Insulated/ Non-insulated adapter **ADP IZ/N**

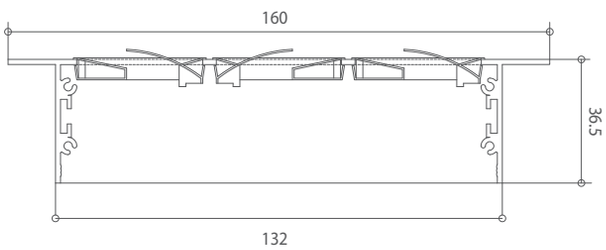
### Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural/ black anodised aluminum
Polypropylene discs.

### Technical design

**SP 2**




**SP 3**




### Assembly Method
Attachment spring

## Selection table SP 2, SP 3

SP2-900

| Air flow ($m^3/h$) | Length Vk[m/s] | 900 0.2 |
|---|---|---|
| **105** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 6.5<br>20.0<br>50.0 |
| **125** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 7.5<br>25.0<br>75.0 |
| **150** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 9.5<br>30.0<br>100.0 |
| **180** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 12.0<br>35.0<br>150.0 |
| **210** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 14.0<br>40.0<br>200.0 |

SP2-1500

| Air flow ($m^3/h$) | Length Vk[m/s] | 1500 0.2 |
|---|---|---|
| **190** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 8.0<br>20.0<br>40.0 |
| **240** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 10.0<br>27.0<br>75.0 |
| **270** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 12.0<br>30.0<br>90.0 |
| **320** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 14.0<br>35.0<br>130.0 |
| **380** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 17.0<br>40.0<br>180.0 |

SP3-900

| Air flow ($m^3/h$) | Length Vk[m/s] | 900 0.2 |
|---|---|---|
| **110** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 5.5<br>22<br>50 |
| **130** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 6.5<br>27<br>75 |
| **160** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 8<br>32<br>110 |
| **190** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 9.5<br>37<br>175 |
| **230** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 12<br>42<br>225 |

SP3-1500

| Air flow ($m^3/h$) | Length Vk[m/s] | 1500 0.2 |
|---|---|---|
| **270** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 9.5<br>20.0<br>40.0 |
| **340** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 12<br>27<br>70 |
| **400** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 14<br>32<br>85 |
| **470** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 16<br>37<br>120 |
| **520** | X [m]<br>NR [dB(A)]<br>△Pt [pa] | 18<br>38<br>150 |

## Annotation

* Vk [m/s] - Speed at which airflow length was calculated "x" [m]
* X [m] - Airflow length
* NR [dB(A)] - Noise level without room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

## Order codes

**Type** + **ADP IZ/N** + **A** (Attachment spring) / **Dimensions** / **Discs colour** / **O**

- **Type**: **SP2**, **SP3**
- **ADP IZ/N**: Insulated/non-insulated adapter
- **A** (Attachment spring): No of springs depending on grille dimensions
- **Dimensions**: L (length) on request
- **Discs colour**: White RAL 9016, Black RAL 9005
- **O**: **0** Painted white RAL 9016; **R** Painted RAL…; Natural/black anodised

# Swirl Diffuser SWR

## Description
Swirl jet diffuser with radial fixed blades **SWR**.

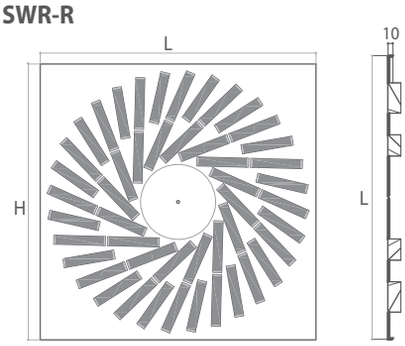
## Application
Supply/ exhaust of the air flow

## Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
Depending on the technical and architectural requirements, the diffuser is produced with round or rectangular front **R/C** .
Blade design, radial layout and swirl action, result in a high level induction rate.
The diffuser should be embedded into the ceiling to ensure a stable horizontal discharge of air.
The diffuser is supplied with seals, screw fastening and masking plastic lid.
Blades are available in two colors: white RAL 9016 and black RAL9005 .

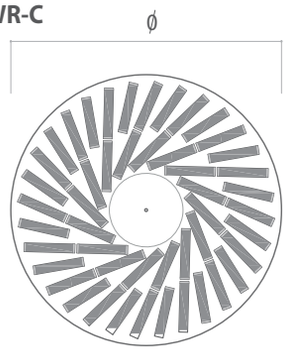
## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design
**SWR-R**




**SWR-C**




## Assembly Method
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm

# Swirl Diffuser SWR

## Selection table SWR

SWR - 8 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 300 x 300 / D300 | |
|---|---|---|---|
| | | 0.007 | |
| | | Exhaust | Supply |
| **100** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br><20<br>9.0 | 4.0<br><20<br>10.0 |
| **125** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.0<br><20<br>14.0 | 5.0<br><20<br>12.0 |
| **150** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 6.0<br>22.0<br>18.0 | 6.0<br><20<br>14.0 |
| **175** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 6.9<br>28.0<br>28.0 | 6.9<br><20<br>16.0 |
| **200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.9<br>30.0<br>30.0 | 7.9<br>22.0<br>17.0 |
| **250** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 9.9<br>38.0<br>52.0 | 9.9<br>26.0<br>22.0 |
| **300** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 11.9<br>35.0<br>35.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 15.9<br>42.0<br>60.0 |

SWR - 16 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 400 x 400/D400 | |
|---|---|---|---|
| | | 0.014 | |
| | | Exhaust | Supply |
| **200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br>15.0<br>10.0 | 4.0<br><20<br>5.0 |
| **250** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.0<br>25.0<br>15.0 | 5.0<br><20<br>6.0 |
| **300** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 6.0<br>30.0<br>25.0 | 6.0<br><20.0<br>10.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.9<br>38.0<br>40.0 | 7.9<br>26.0<br>15.0 |
| **500** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 9.9<br>46.0<br>60.0 | 9.9<br>33.0<br>25.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 11.9<br>48.0<br>75.0 | 11.9<br>40.0<br>36.0 |
| **700** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 13.9<br>45.0<br>50.0 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 15.9<br>50.0<br>65.0 |

SWR - 24 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 500 x 500 / D500 | |
|---|---|---|---|
| | | 0.021 | |
| | | Exhaust | Supply |
| **200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 2.6<br><25<br>7.0 | 2.6<br><20<br>3.0 |
| **300** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.0<br>28.0<br>13.0 | 4.0<br><20<br>4.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.3<br>35.0<br>25.0 | 5.3<br><20<br>7.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.9<br>45.0<br>52.0 | 7.9<br>30.0<br>17.0 |
| **700** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 9.3<br>50.0<br>70.0 | 9.3<br>33.0<br>20.0 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 10.6<br>55.0<br>80.0 | 10.6<br>42.0<br>32.0 |
| **1000** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 13.2<br>50.0<br>52.0 |
| **1200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 15.9<br>60.0<br>65.0 |

SWR - 24 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 595 x 595 / D595 | |
|---|---|---|---|
| | | 0.0295 | |
| | | Exhaust | Supply |
| **250** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 2.4<br><20<br>4.0 | 2.4<br><20<br>3.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.8<br>24.0<br>12.0 | 3.8<br><20<br>4.0 |
| **500** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.7<br>30.0<br>15.0 | 4.7<br><20<br>6.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.6<br>33.0<br>20.0 | 5.6<br><20<br>8.0 |
| **700** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 6.6<br>36.0<br>25.0 | 6.6<br>23.0<br>9.0 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.5<br>43.0<br>35.0 | 7.5<br>30.0<br>15.0 |
| **1000** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 9.4<br>35.0<br>20.0 |
| **1200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 11.3<br>40.0<br>30.0 |

## Selection table SWR

### SWR - 48 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 595 x 595 / D595 | |
|---|---|---|---|
| | | 0.0390 | |
| | | Exhaust | Supply |
| **350** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 2.5<br><20<br>6.0 | 2.5<br><20<br>3.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 2.8<br>23.0<br>12.0 | 2.8<br><20<br>4.0 |
| **500** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.6<br>27.0<br>14.0 | 3.6<br><20<br>5.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.3<br>30.0<br>17.0 | 4.3<br><20<br>6.0 |
| **700** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.0<br>33.0<br>20.0 | 5.0<br>23.0<br>7.5 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.7<br>36.0<br>28.0 | 5.7<br>25.0<br>8.0 |
| **1000** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.1<br>43.0<br>45.0 | 7.1<br>35.0<br>15.0 |
| **1400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 10.0<br>45.0<br>30.0 |

### SWR - 24 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 623 x 623 / D623 | |
|---|---|---|---|
| | | 0.0295 | |
| | | Exhaust | Supply |
| **300** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 2.8<br><20<br>6.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.8<br><20<br>4.0 | 3.8<br>27.0<br>12.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.6<br><20<br>8.0 | 5.6<br>33.0<br>18.0 |
| **700** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 6.6<br>25.0<br>10.0 | 6.6<br>35.0<br>23.0 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.5<br>30.0<br>15.0 | 7.5<br>42.0<br>33.0 |
| **1000** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 9.4<br>38.0<br>28.0 | 9.4<br>46.0<br>45.0 |
| **1200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 11.3<br>42.0<br>32.0 | 11.3<br>53.0<br>70.0 |
| **1400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 13.2<br>50.0<br>40.0 | |

### SWR - 54 slots

| Air flow ($m^3/h$) | Ak [$m^2$] | 623 x 623 / D623 | |
|---|---|---|---|
| | | 0.047 | |
| | | Exhaust | Supply |
| **300** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 1.8<br><20<br>3.0 |
| **400** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 2.4<br><20<br>6.0 |
| **600** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 3.5<br><20<br>3.0 | 3.5<br>27.0<br>13.0 |
| **800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 4.7<br>25.0<br>6.0 | 4.7<br>35.0<br>23.0 |
| **1000** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 5.9<br>30.0<br>8.5 | 5.9<br>43.0<br>40.0 |
| **1200** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 7.1<br>35.0<br>12.0 | 7.1<br>46.0<br>50.0 |
| **1500** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 8.9<br>42.0<br>20.0 | 8.9<br>52.0<br>73.0 |
| **1800** | Veff[m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 10.6<br>50.0<br>30.0 | |

## Annotation

- Ak [$m^2$] - Free surface
- Veff[m/s] - Air velocity inside the diffuser
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

## Order codes

| Type | - | R/C | + | ADP IZ/N | / | Dimensions - No. of slots | / | Blades color | / | O |
|---|---|---|---|---|---|---|---|---|---|---|
| **SWR** | | Rectangular / Circular | | Insulated/Non-insulated adapter | | 300 x 300 / D300 - 8<br>400 x 400 / D400 - 16<br>500 x 500 / D500 - 24<br>595 x 595 / D595 - 24<br>595 x 595 / D595 - 48<br>623 x 623 / D623 - 24<br>623 x 623 / D623 - 54 | | White RAL9016<br>Black RAL9005 | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Swirl Jet Diffuser STD/STD1

## Description
Swirl jet diffuser with radial fixed blades **STD/STD1**.

## Application
Supply/ exhaust of the air flow

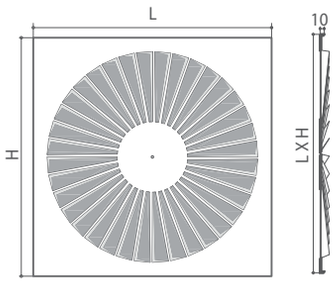
## Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
Depending on the technical and architectural requirements, the diffuser is produced with round or rectangular front **R/C.**
Blade design, radial layout and swirl action, result in a high level induction rate.
The diffuser should be embedded into the ceiling to ensure a stable horizontal discharge of air.
The diffuser is supplied with seals, screw fastening and masking plastic lid.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

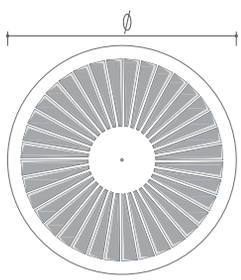
## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design

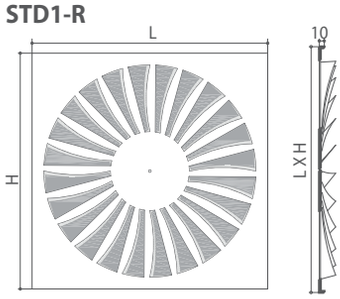
**STD-R**



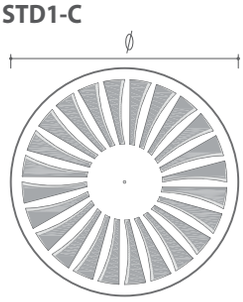

**STD-C**




**STD1-R**




**STD1-C**




## Assembly method
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm.

## Selection table STD/STD1

### STD-R / STD-C

| Air flow ($m^3/h$) | 595 x 595 / D595 | |
|---|---|---|
| | Ak [$m^2$] | 0.0367 |
| 300 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 2.27<br>1.80<br>22.00<br>22.00 |
| 450 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 3.41<br>3.30<br>25.00<br>30.00 |
| 500 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 3.78<br>3.90<br>35.00<br>33.00 |
| 600 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 4.54<br>4.80<br>45.00<br>38.00 |
| 700 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 5.30<br>5.70<br>64.00<br>43.00 |
| 800 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 6.06<br>6.10<br>79.00<br>49.00 |

### STD1-R / STD1-C

| Air flow ($m^3/h$) | 595 x 595 /D595 | |
|---|---|---|
| | Ak [$m^2$] | 0.058 |
| 300 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 1.44<br>2.60<br>-<br>8.00 |
| 450 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 2.16<br>3.80<br>15.00<br>18.00 |
| 500 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 2.39<br>4.20<br>18.00<br>20.00 |
| 600 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 2.87<br>5.00<br>26.10<br>26.00 |
| 700 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 3.35<br>5.70<br>35.00<br>32.00 |
| 800 | Veff[m/s]<br>X [m]<br>NR [dB(A)]<br>△Pt [pa] | 3.83<br>6.70<br>44.70<br>36.00 |

## Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

| Type | - | R/C | + | ADP IZ/N | / | O |
|---|---|---|---|---|---|---|
| **STD**<br>**STD1** | | Rectangular \ Circular | | Insulated/ Non-insulated adapter | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Swirl Jet Diffuser SS

## Description
Rectangular swirl jet diffuser suitable for paneled ceiling mounting.

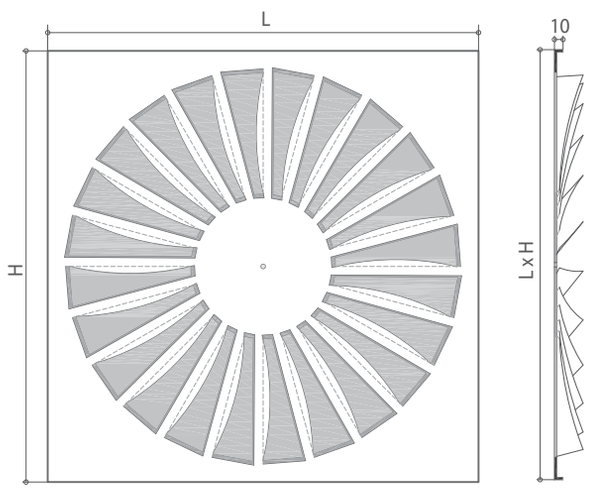
## Application
Supply /exhaust of the air flow

## Characteristics
Recommended for rooms between 2.60 to 4.00 m. in height
The design, radial layout and vortex action of the blades leads to a high level of induction rate.
The recommended temperature difference between supplied air and indoor air is ± 10 K.
The blades that guide the air jet are plastic white RAL 9016 or black RAL 9005.
The diffuser is supplied with gasket seals, fastening screw and plastic cover for concealment.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene blades.

## Technical design




## Assembly Methods
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm on the mounting traverse of the adapter.

## Selection table SS

| Air flow ($m^3/h$) | Dimensions | 595x595 | |
|---|---|---|---|
| | Ak [$m^2$] | 0.0440 | |
| | Vk[m/s] | 0.3 | 0.5 |
| **360** | Veff [m/s] | 2.27 | 2.27 |
| | X [m] | 1.9 | - |
| | NR [dB(A)] | <20 | <20 |
| | △Pt [pa] | 8.0 | 8.0 |
| **450** | Veff [m/s] | 2.84 | 2.84 |
| | X [m] | 2.3 | 1.4 |
| | NR [dB(A)] | 22.0 | 22.0 |
| | △Pt [pa] | 12.0 | 12.0 |
| **540** | Veff [m/s] | 3.41 | 3.41 |
| | X [m] | 2.8 | 1.7 |
| | NR [dB(A)] | 30.0 | 30.0 |
| | △Pt [pa] | 17.0 | 17.0 |
| **648** | Veff [m/s] | 4.09 | 4.09 |
| | X [m] | 3.3 | 2.0 |
| | NR [dB(A)] | 36.0 | 36.0 |
| | △Pt [pa] | 24.0 | 24.0 |
| **720** | Veff [m/s] | 4.55 | 4.55 |
| | X [m] | 3.7 | 2.2 |
| | NR [dB(A)] | 39.0 | 39.0 |
| | △Pt [pa] | 30.0 | 30.0 |
| **900** | Veff [m/s] | 5.68 | 5.68 |
| | X [m] | 4.6 | 2.8 |
| | NR [dB(A)] | 46.0 | 46.0 |
| | △Pt [pa] | 47.0 | 47.0 |

## Annotation

* Ak [$m^2$] - Free surface
* Vk - Velocity at which the air jet length was calculated "x" [m]
* Veff[m/s] - Air velocity inside diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

| Type | + | ADP IZ/N | / | Dimensions | / | Control blades | / | 0 |
|---|---|---|---|---|---|---|---|---|
| **SS** | | Insulated/ Non-insulated adapter | | L x H<br>595*595 | | White RAL 9016<br>Black RAL 9005 | | **0** Painted white RAL<br>**R** Other RAL paints |

# Swirl Jet Diffuser VS

## Description
Swirl jet diffuser with fixed and radial arranged blades.

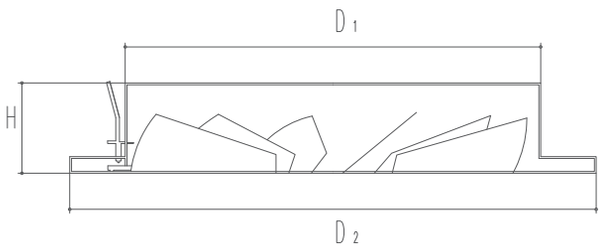
## Application
Supply/ exhaust of the air.

## Characteristics
Recommended for rooms of 2.60 to 4.00 m in height.
The design, radial layout and vortex action of the blades leads to a high level of induction rate.
Can be used in systems with both constant or variable air volumes.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016
Other RAL colors available on request

## Technical design




| Ø | D1 | H | D2 |
|---|---|---|---|
| 125 | 123 | 50 | 158 |
| 160 | 158 | 53 | 197 |
| 200 | 198 | 53 | 241 |
| 250 | 248 | 50 | 295 |
| 315 | 313 | 53 | 364 |
| 400 | 398 | 53 | 450 |

## Selection table VS

| Air flow (m³/h) | Ø Ak [m²] | 125 0.0025 |
|---|---|---|
| 35 | Veff(m/s) | 3.9 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 8.2 |
| 43 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 13.5 |
| 50 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 18.0 |
| 61 | Veff(m/s) | 6.8 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 28.0 |
| 76 | Veff(m/s) | 8.4 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 39.0 |
| 96 | Veff(m/s) | 10.7 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 60.0 |

| Air flow (m³/h) | Ø Ak [m²] | 160 0.004 |
|---|---|---|
| 43 | Veff(m/s) | 3.0 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 7.5 |
| 52 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 9.0 |
| 65 | Veff(m/s) | 4.5 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 15.0 |
| 81 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 23.0 |
| 100 | Veff(m/s) | 6.9 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 35.0 |
| 120 | Veff(m/s) | 8.3 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 50.0 |

| Air flow (m³/h) | Ø Ak [m²] | 200 0.007 |
|---|---|---|
| 90 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 12.0 |
| 120 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 18.0 |
| 140 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 25.0 |
| 160 | Veff(m/s) | 6.3 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 35.0 |
| 185 | Veff(m/s) | 7.3 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 48.0 |
| 230 | Veff(m/s) | 9.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 70.0 |

| Air flow (m³/h) | Ø Ak [m²] | 250 0.011 |
|---|---|---|
| 190 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 15.0 |
| 240 | Veff(m/s) | 6.1 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 20.0 |
| 280 | Veff(m/s) | 7.1 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 30.0 |
| 320 | Veff(m/s) | 8.1 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 40.0 |
| 410 | Veff(m/s) | 10.4 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 58.0 |
| 520 | Veff(m/s) | 13.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 60.0 |

## Selection table VS

| Air flow (m³/h) | Ø | 315 |
|---|---|---|
| | Ak [m²] | 0.021 |
| 240 | Veff(m/s) | 3.2 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 9.0 |
| 270 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 13.5 |
| 320 | Veff(m/s) | 4.2 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 18.0 |
| 380 | Veff(m/s) | 5.0 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 25.0 |
| 470 | Veff(m/s) | 6.2 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 35.0 |
| 530 | Veff(m/s) | 7.0 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 48.0 |

| Air flow (m³/h) | Ø | 400 |
|---|---|---|
| | Ak [m²] | 0.028 |
| 290 | Veff(m/s) | 2.9 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 10.0 |
| 330 | Veff(m/s) | 3.3 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 15.0 |
| 350 | Veff(m/s) | 3.5 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 20.0 |
| 500 | Veff(m/s) | 5.0 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 30.0 |
| 600 | Veff(m/s) | 6.0 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 42.0 |
| 720 | Veff(m/s) | 7.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 65.0 |

## Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

| Type | + | ADP IZ/N | / | Ø | / | 0 |
|---|---|---|---|---|---|---|
| **VS** | | Insulated/ Non-insulated adapter | | 125<br>160<br>200<br>250<br>315<br>400 | | **0** Painted white RAL<br>**R** Other RAL paints |

# Swirl Jet Diffuser VSR

## Description
Swirl jet diffuser with adjustable and radial arranged blades.

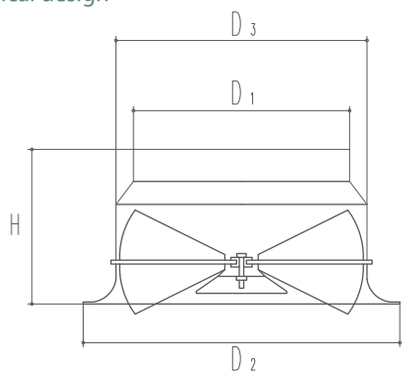
## Application
Supply/ exhaust of the air.

## Characteristics
Recommended for rooms over 3 m in height.
Circular and helical design promotes airflow swirling diffusion obtaining a high level of induction rate and reduces air stratification.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Aluminum diffuser electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




| D1 | D2 | D3 | H |
|---|---|---|---|
| 250 | 400 | 300 | 205 |
| 315 | 500 | 400 | 230 |
| 400 | 615 | 465 | 270 |
| 500 | 780 | 565 | 320 |
| 630 | 935 | 665 | 390 |

# Swirl Jet Diffuser VSR

Selection table VSR

| Air flow ($m^3/h$) | Ø | 250 |
|---|---|---|
| | Ak [$m^2$] | 0.050 |
| 250 | Veff(m/s) | 1.4 |
| | NR [dB(A)] | <20 |
| | X(m) | 1.2 |
| | △Pt [pa] | 10.0 |
| 300 | Veff(m/s) | 1.7 |
| | NR [dB(A)] | <20 |
| | X(m) | 1.5 |
| | △Pt [pa] | 15.0 |
| 350 | Veff(m/s) | 1.9 |
| | NR [dB(A)] | 22.0 |
| | X(m) | 1.6 |
| | △Pt [pa] | 20.0 |
| 400 | Veff(m/s) | 2.2 |
| | NR [dB(A)] | 26.0 |
| | X(m) | 1.8 |
| | △Pt [pa] | 28.0 |
| 500 | Veff(m/s) | 2.8 |
| | NR [dB(A)] | 33.0 |
| | X(m) | 2.8 |
| | △Pt [pa] | 45.0 |
| 600 | Veff(m/s) | 3.3 |
| | NR [dB(A)] | 38.0 |
| | X(m) | 3.6 |
| | △Pt [pa] | 70.0 |
| 700 | Veff(m/s) | 3.9 |
| | NR [dB(A)] | 44.0 |
| | X(m) | 4.6 |
| | △Pt [pa] | 91.0 |

| Air flow ($m^3/h$) | Ø | 315 |
|---|---|---|
| | Ak [$m^2$] | 0.078 |
| 500 | Veff(m/s) | 1.8 |
| | NR [dB(A)] | <20 |
| | X(m) | 1.6 |
| | △Pt [pa] | 17.0 |
| 600 | Veff(m/s) | 2.1 |
| | NR [dB(A)] | 24.0 |
| | X(m) | 2.0 |
| | △Pt [pa] | 23.0 |
| 700 | Veff(m/s) | 2.5 |
| | NR [dB(A)] | 27.0 |
| | X(m) | 2.6 |
| | △Pt [pa] | 31.0 |
| 800 | Veff(m/s) | 2.8 |
| | NR [dB(A)] | 33.0 |
| | X(m) | 3.0 |
| | △Pt [pa] | 41.0 |
| 900 | Veff(m/s) | 3.2 |
| | NR [dB(A)] | 38.0 |
| | X(m) | 3.8 |
| | △Pt [pa] | 51.0 |
| 1000 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 43.0 |
| | X(m) | 4.2 |
| | △Pt [pa] | 69.0 |
| 1100 | Veff(m/s) | 3.9 |
| | NR [dB(A)] | 46.0 |
| | X(m) | 5.0 |
| | △Pt [pa] | 77.0 |

| Air flow ($m^3/h$) | Ø | 400 |
|---|---|---|
| | Ak [$m^2$] | 0.125 |
| 900 | Veff(m/s) | 2 |
| | NR [dB(A)] | <20 |
| | X(m) | 1.8 |
| | △Pt [pa] | 13.0 |
| 1000 | Veff(m/s) | 2.2 |
| | NR [dB(A)] | 23.0 |
| | X(m) | 2.1 |
| | △Pt [pa] | 17.0 |
| 1100 | Veff(m/s) | 2.4 |
| | NR [dB(A)] | 26.0 |
| | X(m) | 2.5 |
| | △Pt [pa] | 21.0 |
| 1200 | Veff(m/s) | 2.7 |
| | NR [dB(A)] | 28.0 |
| | X(m) | 3.0 |
| | △Pt [pa] | 25.0 |
| 1400 | Veff(m/s) | 3.1 |
| | NR [dB(A)] | 32.0 |
| | X(m) | 3.8 |
| | △Pt [pa] | 33.0 |
| 1500 | Veff(m/s) | 3.3 |
| | NR [dB(A)] | 35.0 |
| | X(m) | 4.0 |
| | △Pt [pa] | 38.0 |
| 1600 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 38.0 |
| | X(m) | 4.5 |
| | △Pt [pa] | 44.0 |

## Selection table VSR

| Air flow ($m^3/h$) | Ø / Ak [$m^2$] | 500 / 0.195 |
|---|---|---|
| 1200 | Veff(m/s) | 1.7 |
| | NR [dB(A)] | 30.0 |
| | X(m) | 1.8 |
| | △Pt [pa] | 17.0 |
| 1400 | Veff(m/s) | 2.0 |
| | NR [dB(A)] | 33.0 |
| | X(m) | 2.5 |
| | △Pt [pa] | 22.0 |
| 1600 | Veff(m/s) | 2.3 |
| | NR [dB(A)] | 36.0 |
| | X(m) | 3.0 |
| | △Pt [pa] | 28.0 |
| 1800 | Veff(m/s) | 2.6 |
| | NR [dB(A)] | 38.0 |
| | X(m) | 3.5 |
| | △Pt [pa] | 35.0 |
| 2000 | Veff(m/s) | 2.8 |
| | NR [dB(A)] | 41.5 |
| | X(m) | 4.0 |
| | △Pt [pa] | 42.0 |
| 2250 | Veff(m/s) | 3.2 |
| | NR [dB(A)] | 44.0 |
| | X(m) | 4.5 |
| | △Pt [pa] | 51.0 |
| 2500 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 47.0 |
| | X(m) | 5.2 |
| | △Pt [pa] | 70.0 |

| Air flow ($m^3/h$) | Ø / Ak [$m^2$] | 630 / 0.310 |
|---|---|---|
| 2100 | Veff(m/s) | 1.9 |
| | NR [dB(A)] | 25.0 |
| | X(m) | 3.5 |
| | △Pt [pa] | 18.0 |
| 2500 | Veff(m/s) | 2.2 |
| | NR [dB(A)] | 28.0 |
| | X(m) | 3.8 |
| | △Pt [pa] | 31.0 |
| 3000 | Veff(m/s) | 2.7 |
| | NR [dB(A)] | 35.0 |
| | X(m) | 4.8 |
| | △Pt [pa] | 38.0 |
| 3500 | Veff(m/s) | 3.1 |
| | NR [dB(A)] | 40.0 |
| | X(m) | 5.0 |
| | △Pt [pa] | 53.0 |
| 4000 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 44.0 |
| | X(m) | 5.5 |
| | △Pt [pa] | 72.0 |
| 4500 | Veff(m/s) | 4.0 |
| | NR [dB(A)] | 47.0 |
| | X(m) | 6.0 |
| | △Pt [pa] | 86.0 |
| 5000 | Veff(m/s) | 4.5 |
| | NR [dB(A)] | 50.0 |
| | X(m) | 7.0 |
| | △Pt [pa] | 110.0 |

### Note:
The values from selection table are calculated for 45° blades inclination and 5 degrees temperature deviation.

### Annotation
* Ak [$m^2$] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation
* X [m] - Air jet length

### Order codes

| Type | + | ADP IZ/N | / | Ø | / | 0 |
|---|---|---|---|---|---|---|
| VSR | | Insulated/ Non-insulated adapter | | 250<br>315<br>400<br>500<br>630 | | **0** Painted white RAL<br>**R** Other RAL paints |

# Access Grille AD

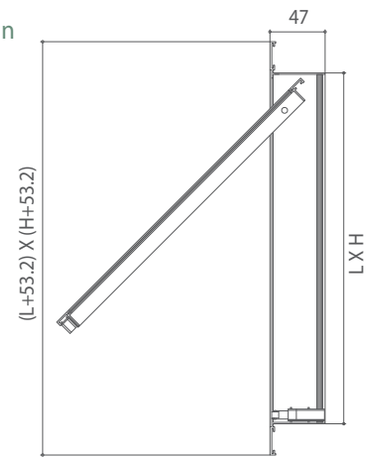
## Description
Aluminum access door is used in areas where regular access is required for installation works maintenance.

## Application
To access the desired maintenance area.

## Characteristics
Suitable for wall or ceiling mounting.
The access door is provided with countersunk holes for mounting the visible frame .
Size 595 x 595 mm (for paneled ceiling) is provided without mounting holes.
Opening/closing the door is done by pushing a click type system.

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Pickled steel sheet.
Other RAL colors available on request.

## Technical design




## Assembly Methods
Mounting countersunk holes on the visible frame (standard)
Mounting holes on the connection frame inlet (optional)

## Order codes

| Type | / | Dimensions | / | 0 |
|---|---|---|---|---|
| AD | | On request | | 0 Painted white RAL 9016<br>R Other RAL paints |

# Access Door AG

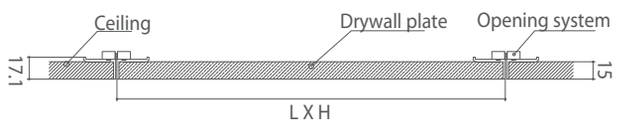
## Description
Access doors are used in areas where regular access is required for work maintenance and installations.

## Application
To access the desired maintenance area.

## Characteristics
Access door is made of an aluminum profile with frontal drywall plate.
Opening/closing of the roof is carried out via a click type system.
Perfect incorporation with the wall or ceiling.

## Materials and finishing
Natural aluminum frame
Gypsum plate

## Technical design




## Order codes

| Type | / | Dimensions | / | L x H |
|---|---|---|---|---|
| AG | | L X H<br>300 x 300<br>400 x 400<br>500 x 500<br>600 x 600 | | Other dimensions on request |

# Air Duct With Nozzles JR

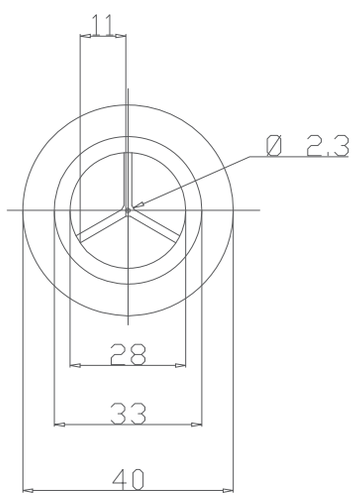
## Description
Air duct with nozzles.

## Application
Air supply

## Characteristics
Available lenghts: 0.5m, 1.0m, 1.5m and 2.0m.
The nozzles are 32 mm in diameter, available in two colors: white RAL 9016 and black RAL 9005
The ideal temperature difference between supplied air and room air is between 2 and 4K and the velocity between 2.5 and 4.5 m/s.
Uniform and permanent air distribution.
Low pressure loss.

## Materials and finishing
Galvanized steel sheet
Other RAL colors available on request.

## Technical design



## Selection table JR

| Q (m³/h) | Length = 2000 (mm) | | |
|---|---|---|---|
| | D (mm) | NR [dB(A)] | ΔPt [pa] |
| 128 | 160 | 27 | 5.0 |
| | 200 | 26 | 5.0 |
| | 250 | 26 | 5.0 |
| | 315 | 26 | 5.0 |
| No. of rows 1 | 400 | 26 | 5.0 |
| | 500 | 26 | 5.0 |
| 249 | 160 | 34 | 7.0 |
| | 200 | 30 | 7.0 |
| | 250 | 29 | 7.0 |
| | 315 | 29 | 7.0 |
| No. of rows 2 | 400 | 29 | 7.0 |
| | 500 | 29 | 7.0 |
| 337 | 160 | 41 | 9.0 |
| | 200 | 32 | 9.0 |
| | 250 | 29 | 9.0 |
| | 315 | 29 | 9.0 |
| No. of rows 3 | 400 | 29 | 9.0 |
| | 500 | 29 | 9.0 |
| 413 | 160 | >35 | 10.0 |
| | 200 | 35 | 10.0 |
| | 250 | 30 | 10.0 |
| | 315 | 29 | 10.0 |
| No. of rowsi 4 | 400 | 29 | 10.0 |
| | 500 | 29 | 10.0 |
| 483 | 160 | >35 | 11.0 |
| | 200 | >35 | 11.0 |
| | 250 | 31 | 11.0 |
| | 315 | 29 | 11.0 |
| No. of rows 5 | 400 | 29 | 11.0 |
| | 500 | 29 | 11.0 |
| 559 | 160 | >35 | 12.0 |
| | 200 | >35 | 12.0 |
| | 250 | 33 | 12.0 |
| | 315 | 29 | 12.0 |
| No. of rows 6 | 400 | 29 | 12.0 |
| | 500 | 29 | 12.0 |

| Q (m³/h) | Length = 3000 (mm) | | |
|---|---|---|---|
| | D (mm) | NR [dB(A)] | ΔPt [pa] |
| 199 | 160 | 31 | 6.0 |
| | 200 | 29 | 6.0 |
| | 250 | 29 | 6.0 |
| | 315 | 29 | 6.0 |
| No. of rows 1 | 400 | 29 | 6.0 |
| | 500 | 29 | 6.0 |
| 381 | 160 | - | 8.0 |
| | 200 | 35 | 8.0 |
| | 250 | 32 | 8.0 |
| | 315 | 31 | 8.0 |
| No. of rows 2 | 400 | 31 | 8.0 |
| | 500 | 31 | 8.0 |
| 518 | 160 | - | 10.0 |
| | 200 | 40 | 10.0 |
| | 250 | 44 | 10.0 |
| | 315 | 31 | 10.0 |
| No. of rows 3 | 400 | 31 | 10.0 |
| | 500 | 31 | 10.0 |
| 637 | 160 | - | 11.0 |
| | 200 | - | 11.0 |
| | 250 | 36 | 11.0 |
| | 315 | 32 | 11.0 |
| No. of rows 4 | 400 | 31 | 11.0 |
| | 500 | 31 | 11.0 |
| 746 | 160 | - | 12.0 |
| | 200 | - | 12.0 |
| | 250 | 39 | 12.0 |
| | 315 | 32 | 12.0 |
| No. of rows 5 | 400 | 31 | 12.0 |
| | 500 | 31 | 12.0 |
| 859 | 160 | - | 13.0 |
| | 200 | - | 13.0 |
| | 250 | - | 13.0 |
| | 315 | 33 | 13.0 |
| No. of rows 6 | 400 | 31 | 13.0 |
| | 500 | 31 | 13.0 |

| Q (m³/h) | Length= 4000 (mm) | | |
|---|---|---|---|
| | D (mm) | NR [dB(A)] | ΔPt [pa] |
| 270 | 160 | 36 | 9.0 |
| | 200 | 31 | 9.0 |
| | 250 | 30 | 9.0 |
| | 315 | 30 | 9.0 |
| No. of rows 1 | 400 | 30 | 9.0 |
| | 500 | 30 | 9.0 |
| 512 | 160 | - | 11.0 |
| | 200 | 40 | 11.0 |
| | 250 | 34 | 11.0 |
| | 315 | 33 | 11.0 |
| No. of rows 2 | 400 | 32 | 11.0 |
| | 500 | 32 | 11.0 |
| 698 | 160 | - | 16.0 |
| | 200 | - | 16.0 |
| | 250 | 38 | 16.0 |
| | 315 | 38 | 16.0 |
| No. of rows 3 | 400 | 32 | 16.0 |
| | 500 | 32 | 16.0 |
| 860 | 160 | - | 20.0 |
| | 200 | - | 20.0 |
| | 250 | 42 | 20.0 |
| | 315 | 33 | 20.0 |
| No. of rows 4 | 400 | 32 | 20.0 |
| | 500 | 32 | 20.0 |
| 1008 | 160 | - | 23.0 |
| | 200 | - | 23.0 |
| | 250 | - | 23.0 |
| | 315 | 36 | 23.0 |
| No. of rows 5 | 400 | 33 | 23.0 |
| | 500 | 32 | 23.0 |
| 1159 | 160 | - | 25.0 |
| | 200 | - | 25.0 |
| | 250 | - | 25.0 |
| | 315 | 38 | 25.0 |
| No. of rows 6 | 400 | 33 | 25.0 |
| | 500 | 32 | 25.0 |

## Note

Described values are for:

* Velocity in sitting area <0.2 m/s (measured at 1.7 meters when mounting)
* Assembly height - 3m (coefficient value F1=1)
* Temperature difference - 3K-COOLING (coefficient value F2=1)

THE CALCULATED AIR FLOW WILL BE ADJUSTED BY MULTIPLYING WITH F1 AND F2 COEFFICIENTS, DEPENDING ON APPLICATION Qeff= Q *F1*F2.

| Correction factor for temperature | |
|---|---|
| Temperature difference (K) | Correction coefficient F2 |
| 0 | 1.055 |
| 1 | 1.044 |
| 2 | 1.026 |
| 3 | 1 |
| 4 | 0.967 |
| 5 | 0.925 |
| 6 | 0.874 |

| Correction factor for the mounting height | |
|---|---|
| Mounting height (m) | Correction coefficient F1 |
| 2.6 | 0.856 |
| 2.8 | 0.932 |
| 3 | 1 |
| 3.2 | 1.065 |
| 3.4 | 1.125 |
| 3.6 | 1.183 |
| 3.8 | 1.238 |
| 4 | 1.3 |

## Annotation

* Dn (mm) - Duct diameter
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

## Order codes

| Type | + | Ø | + | No. of rows | / | L | / | Nozzles color | / | 0 |
|---|---|---|---|---|---|---|---|---|---|---|
| JR | | 160<br>200<br>250<br>315<br>400<br>500 | | 1<br>2<br>3<br>4<br>5<br>6 | | 500<br>1000<br>1500<br>2000 | | WHITE RAL 9016<br>BLACK RAL 9005 | | Natural<br>**0** Painted white RAL 9016<br>**R** Other RAL paints |

# Jet Diffuser JD

## Description
Diffuser with long range air supply capacity.
Ideal for use in traffic areas, theaters, discos, museums, concert halls, etc.

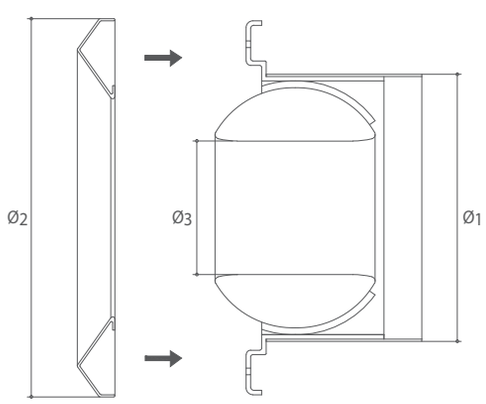
## Application
Air supply

## Characteristics
Low noise level.
Spherical nozzle bordered by a ring with mounting holes, allows airflow direction in the desired area.
Decorative ring for masked mounting.
Circular inlet connection for mounting to air ducts.

## Accessories
Insulated/ Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Aluminum nozzle and decorative ring, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene ring (mounting)
Galvanized steel sheet connection (for air ducts)

## Technical design




| Ø1 | Ø2 | Ø3 |
|---|---|---|
| 125 | 170 | 60 |
| 160 | 210 | 80 |
| 200 | 255 | 105 |
| 250 | 300 | 135 |
| 315 | 380 | 185 |
| 400 | 470 | 230 |
| 500 | 595 | 285 |

## Assembly Methods
Standard mounting, with screws.

## Selection table JD

| Air flow (m³/h) | Ø | 125 | 160 | 200 | 250 | 315 | 400 |
|---|---|---|---|---|---|---|---|
| | Ak(m²) | 0.0031 | 0.005 | 0.0085 | 0.0135 | 0.0225 | 0.0385 |
| 100 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 8.96<br>20<br>48<br>5 | 5.56<br><20<br>20<br>4 | | | | |
| 150 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 13.44<br>32<br>100<br>7 | 8.33<br>20<br>40<br>5.50 | 4.90<br><20<br>8.00<br>4.30 | | | |
| 200 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | 17.92<br>40<br>180<br>8.5 | 11.11<br>28<br>60<br>7 | 6.54<br><20<br>25<br>5.5 | | | |
| 300 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | 16.67<br>40<br>150<br>8.5 | 9.80<br>28<br>60<br>6.3 | 6.17<br><20<br>22<br>5.2 | | |
| 500 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | 16.34<br>45<br>140<br>9 | 10.29<br>37<br>60<br>8 | 6.17<br>20<br>25<br>6 | |
| 700 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | | 14.40<br>40<br>100<br>10 | 8.64<br>28<br>48<br>8 | 5.05<br><20<br>12<br>6.5 |
| 1000 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | | 20.58<br>52<br>230<br>13 | 12.35<br>40<br>95<br>12 | 7.22<br>25<br>28<br>10 |
| 1500 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | | | 18.52<br>52<br>200<br>18 | 10.82<br>35<br>62<br>14 |
| 2000 | Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa]<br>X [m] | | | | | 24.7<br>>50<br>320<br>20 | 14.4<br>45<br>120<br>18 |

## Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Note: The values are for ΔT = + 10 K.

## Order codes

**Type** + **ADP IZ/N** / **Dimensions** / **O**

- **Type**: **JD**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: 125, 160, 200, 250, 315, 400, 500
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

# Jet Diffuser with Nozzles JB

## Description
Aluminum linear diffuser with individually steerable nozzles. Recommended for use in traffic areas, cinemas, museums, theaters, etc.

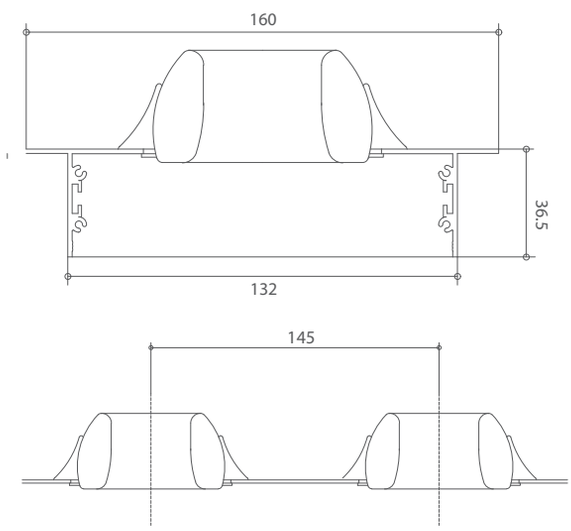
## Application
Various air supply systems

## Characteristics
The diffuser is produced, depending on technical and architectural requirements, with one or two rows of nozzles.
Nozzles have a diameter of 40 mm for air passage and are available in two colors: white RAL 9016 and black RAL 9005.
Due to the spherical design of the nozzle, the air flow can be directed into desired areas.
Allows unlimited airflow direction settings.

## Accessories
Attachment spring **A**.
Insulated/Non-insulated adapter **ADP IZ/N**.

## Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Polypropylene nozzles

## Technical design




| Number of nozzles | Standard height |
|---|---|
| 2 | 1row of nozzles = 132 mm<br>2 rows of nozzles = 294 mm |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |

## Assembly Methods
Attachment spring.

## Selection table JB

| Air flow m³/h | Number of nozzles | 3 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| | Ak(m²) | 0.0038 | 0.0063 | 0.0076 | 0.01 | 0.0126 |
| 120 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 9.5 5.7 2.8<br>8.77<br>< 20<br>43.0 | | | | |
| 150 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 11.8 7.1 3.5<br>10.96<br>22.0<br>68.0 | | | | |
| 200 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 15.8 9.5 4.7<br>14.62<br>29.0<br>121.0 | 12.2 7.3 3.7<br>8.82<br>< 20<br>43.0 | 11.2 6.7 3.3<br>7.31<br>< 20<br>30.0 | | |
| 250 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 19.7 11.8 5.9<br>18.27<br>35.0<br>189.0 | 15.3 9.2 4.6<br>11.02<br>24.0<br>68.0 | 13.9 8.4 4.2<br>9.14<br>< 20<br>47.0 | 12.1 7.2 3.6<br>6.94<br>< 20<br>27.0 | |
| 300 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 23.7 14.2 7.1<br>21.93<br>40.0<br>272.0 | 18.3 11.0 5.5<br>13.23<br>29.0<br>98.0 | 16.7 10.0 5.0<br>10.96<br>25.0<br>68.0 | 14.5 8.7 4.3<br>8.33<br>< 20<br>38.0 | |
| 350 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | 27.6 16.6 8.3<br>25.58<br>44.0<br>370.0 | 21.4 12.8 6.4<br>15.43<br>33.0<br>133.0 | 19.5 11.7 5.9<br>12.79<br>29.0<br>92.0 | 16.9 10.1 5.1<br>9.72<br>22.0<br>52.0 | 15.1 9.1 4.5<br>7.72<br>< 20<br>33.0 |
| 400 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | 24.4 14.7 7.3<br>17.64<br>36.0<br>174.0 | 22.3 13.4 6.7<br>14.62<br>32.0<br>121.0 | 19.3 11.6 5.8<br>11.11<br>26.0<br>68.0 | 17.3 10.4 5.2<br>8.82<br>21.0<br>43.0 |
| 500 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | > 30 18.3 9.2<br>22.05<br>42.0<br>272.0 | 27.9 16.7 8.4<br>18.27<br>38.0<br>189.0 | 24.1 14.5 7.2<br>13.89<br>32.0<br>106.0 | 21.6 13.0 6.5<br>11.02<br>27.0<br>68.0 |
| 600 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | > 30 22.0 11.0<br>26.46<br>47.0<br>391.0 | > 30 20.1 10.0<br>21.93<br>43.0<br>272.0 | 29.0 17.4 8.7<br>16.67<br>37.0<br>153.0 | 25.9 15.6 7.8<br>13.23<br>32.0<br>98.0 |
| 700 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | > 30 23.4 11.7<br>25.58<br>47.0<br>370.0 | > 30 20.3 10.1<br>19.44<br>41.0<br>208.0 | > 30 18.1 9.1<br>15.43<br>36.0<br>133.0 |
| 800 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | > 30 23.2 11.6<br>22.22<br>44.0<br>272.0 | > 30 20.7 10.4<br>17.64<br>39.0<br>174.0 |
| 1000 | $X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m]<br>Veff [m/s]<br>NR [dB(A)]<br>ΔPt [pa] | | | | | > 30 25.9 13.0<br>22.05<br>45.0<br>272.0 |

## Correction table for more rows

| Number of rows | 1 | 2 |
|---|---|---|
| FNR | 0 | 3 |
| Fx | 1 | 1.4 |

The values "X" will be corrected by multiplying with the coefficients from the table and "NR" through addition.

## Annotation

*Ak [m²] - Free surface of the diffuser
*Veff[m/s] - Air velocity in diffuser
*NR [dB(A)] - Noise level without the room attenuation
*ΔPt [pa] - Pressure loss

X0.3 - Air jet length at speed Vk = 0.3 m/s
X0.5 - Air jet length at speed Vk = 0.5 m/s
X1.0 - Air jet length at speed Vk = 1.0 m/s

## Order codes

**Type** - **Number of rows** + **ADP IZ/N** / **Number of nozzles** / **Color of nozzles** / **O**

- **JB**
- 1, 2
- Insulated/Non-insulated adapter
- On request
- White RAL 9016, Black RAL 9005
- **0** Painted white RAL 9016, **R** Other RAL paints

# Circular Valve EV

## Description
Circular steel valve with adjustable central cone for exhaust of the air, provided with mounting frame **EV**.

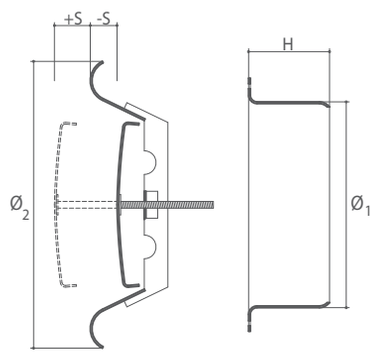
## Application
Exhaust the airflow

## Characteristics
Valve is provided with gasket seal and mounting frame
The adjustable center cone allows adjustment of air flow
The mounting frame is provided with holes for fixing

## Materials and finishing
Steel electrostatic field painted in RAL9016 white
Other RAL colors available on request.
Mounting frame made of galvanized metal sheet

## Assembly Methods
Screws on the mounting frame.

## Technical design




| Tip | Ø1 | Ø2 | H |
|---|---|---|---|
| EV | 100 | 140 | 45 |
| | 125 | 165 | 47 |
| | 150 | 200 | 47 |
| | 160 | 210 | 50 |
| | 200 | 250 | 50 |

## Selection table EV

| Air flow | Dimension (mm) | 100 | | | | | 125 | | | | | | | 160 | | | | | | 200 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ($m^3$/h) | S(mm) | -14 | -9 | -5 | 0 | 5 | -15 | -12 | -9 | -6 | -3 | 0 | 5 | -18 | -15 | -10 | -5 | 0 | 6 | -23 | -20 | -15 | -10 | -5 | 0 | 10 | 20 |
| **20** | NR [dB(A)] | 24.0 | <25 | | | | <25 | <25 | | | | | | | | | | | | | | | | | | | |
| | △ Pt [pa] | 200.0 | 25.0 | | | | 23.0 | 12.0 | | | | | | | | | | | | | | | | | | | |
| **30** | NR [dB(A)] | | <25 | <25 | | | <25 | <25 | <25 | | | | | | | | | | | | | | | | | | |
| | △ Pt [pa] | | 60.0 | 20.0 | | | 50.0 | 25.0 | 15.0 | | | | | | | | | | | | | | | | | | |
| **50** | NR [dB(A)] | | 40.0 | <25 | <25 | <25 | 30.0 | <25 | <25 | <25 | <25 | <25 | | <30 | <25 | <30 | | | | <30 | <25 | <30 | | | | | |
| | △ Pt [pa] | | 150.0 | 60.0 | 25.0 | 12.0 | 145.0 | 70.0 | 40.0 | 25.0 | 12.0 | 10.0 | | 70.0 | 30.0 | 12.0 | | | | 50.0 | 28.0 | 12.0 | | | | | |
| **70** | NR [dB(A)] | | >40 | 35.0 | <30 | <30 | 40.0 | 30.0 | <25 | <25 | <25 | <25 | <25 | 35.0 | <30 | <30 | <30 | | | 30.0 | <30 | <30 | <30 | | | | |
| | △ Pt [pa] | | 300.0 | 110.0 | 50.0 | 28.0 | 260.0 | 110.0 | 70.0 | 50.0 | 30.0 | 23.0 | 12.0 | 125.0 | 70.0 | 28.0 | 15.0 | | | 100.0 | 52.0 | 22.0 | 12.0 | | | | |
| **100** | NR [dB(A)] | | | >40 | 37.0 | 30.0 | | 42.0 | 37.0 | 30.0 | <35 | <30 | <25 | >40 | 35.0 | <30 | <30 | <30 | | 37.0 | 33.0 | <30 | <30 | <30 | | | |
| | △ Pt [pa] | | | 300.0 | 100.0 | 55.0 | | 220.0 | 150.0 | 90.0 | 65.0 | 40.0 | 28.0 | 280.0 | 125.0 | 52.0 | 30.0 | 18.0 | | 175.0 | 110.0 | 40.0 | 25.0 | 15.0 | | | |
| **200** | NR [dB(A)] | | | | | >40 | | | | | 50.0 | 45.0 | 40.0 | | | >40 | 38.0 | 33.0 | 28.0 | | | 40.0 | 35.0 | 30.0 | <30 | <30 | <30 |
| | △ Pt [pa] | | | | | 200.0 | | | | | 250.0 | 170.0 | 110.0 | | | 270.0 | 140.0 | 70.0 | 40.0 | | | 150.0 | 90.0 | 55.0 | 20.0 | 20.0 | 10.0 |
| **250** | NR [dB(A)] | | | | | | | | | | | | | | | | >40 | 40.0 | 34.0 | | | >40 | 40.0 | 36.0 | 31.0 | <30 | <30 |
| | △ Pt [pa] | | | | | | | | | | | | | | | | 200.0 | 120.0 | 70.0 | | | 250.0 | 140.0 | 80.0 | 52.0 | 30.0 | 15.0 |
| **300** | NR [dB(A)] | | | | | | | | | | | | | | | | >40 | >40 | 40.0 | | | | >40 | 41.0 | 36.0 | 40.0 | <30 |
| | △ Pt [pa] | | | | | | | | | | | | | | | | 300.0 | 200.0 | 110.0 | | | | 200.0 | 130.0 | 75.0 | 45.0 | 25.0 |
| **500** | NR [dB(A)] | | | | | | | | | | | | | | | | | | | | | | | | >40 | 42.0 | 37.0 |
| | △ Pt [pa] | | | | | | | | | | | | | | | | | | | | | | | | 200.0 | 110.0 | 55.0 |
| **700** | NR [dB(A)] | | | | | | | | | | | | | | | | | | | | | | | | | >40 | >40 |
| | △ Pt [pa] | | | | | | | | | | | | | | | | | | | | | | | | | 210.0 | 110.0 |

### Annotation

* S [mm] - Disk opening
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

| Type | / | Diameter | / | O |
|---|---|---|---|---|
| **EV** | | 100<br>125<br>150<br>160<br>200 | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Circular Valve SV

## Description
Circular steel valve with adjustable central cone for supply of the air, provided with mounting frame **SV**.

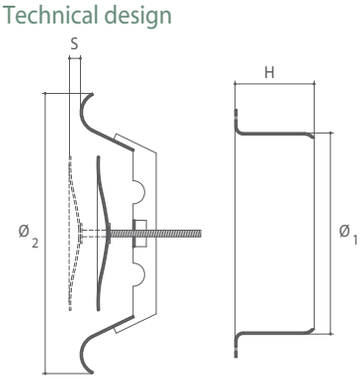
## Application
Supply the airflow

## Characteristics
Valve is provided with gasket seal and mounting frame
The adjustable center cone allows adjustment of air flow
The mounting frame is provided with holes for fixing.

## Materials and finishing
Steel electrostatic field painted in RAL9016 white
Other RAL colors available on request.
Mounting frame made of galvanized metal sheet

## Assembly Methods
Screws on the mounting frame

## Technical design




| Tip | Ø1 | Ø2 | H |
|---|---|---|---|
| SV | 100 | 140 | 48 |
| | 125 | 165 | 48 |
| | 150 | 200 | 44 |
| | 160 | 210 | 50 |
| | 200 | 250 | 45 |

## Selection table SV

| Air flow ($m^3/h$) | Dimension (mm) | 100 | | | 125 | | | | 160 | | | | | 200 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S(mm) | 4 | 9 | 12 | 3 | 7 | 12 | 15 | 4 | 9 | 12 | 15 | 20 | 9 | 12 | 15 | 20 |
| 20 | X [m] | 0.65 | | | 0.6 | | | | | | | | | | | | |
| | NR [dB(A)] | <10 | | | <25 | | | | | | | | | | | | |
| | △Pt [pa] | 15.0 | | | 20.0 | | | | | | | | | | | | |
| 30 | X [m] | 1.0 | | | 0.8 | | | | | | | | | | | | |
| | NR [dB(A)] | <10 | | | <25 | | | | | | | | | | | | |
| | △Pt [pa] | 28.0 | | | 37.0 | | | | | | | | | | | | |
| 50 | X [m] | 2.0 | 0.8 | 0.8 | 1.5 | 1.3 | | | 1.4 | | | | | | | | |
| | NR [dB(A)] | 32.0 | <30 | <25 | 30.0 | <25 | | | <25 | | | | | | | | |
| | △Pt [pa] | 83.0 | 20.0 | 5.0 | 105.0 | 25.0 | | | 38.0 | | | | | | | | |
| 70 | X [m] | 2.8 | 1.7 | 1.3 | 2.2 | 1.7 | 1.1 | | 2.0 | 1.3 | | | | | | | |
| | NR [dB(A)] | >35 | 30.0 | <30 | >35 | <25 | <25 | | 25.0 | <25 | | | | | | | |
| | △Pt [pa] | 130.0 | 32.0 | 20.0 | 190.0 | 40.0 | 15.0 | | 70.0 | 15.0 | | | | | | | |
| 100 | X [m] | | 2.5 | 1.8 | | 2.5 | 1.6 | 1.2 | 3.0 | 1.9 | 1.5 | 1.4 | | 2.2 | 1.5 | | |
| | NR [dB(A)] | | >35 | 40.0 | | 30.0 | <25 | <25 | 32.0 | <25 | <25 | <25 | | <25 | <25 | | |
| | △Pt [pa] | | 75.0 | 40.0 | | 80.0 | 30.0 | 22.0 | 145.0 | 35.0 | 20.0 | 13.0 | | 20.0 | 12.0 | | |
| 150 | X [m] | | >3 | >3 | | >3 | 2.5 | 2.0 | | 3.0 | 2.2 | 2.0 | 1.5 | 3.1 | 2.2 | 1.8 | 1.2 |
| | NR [dB(A)] | | >35 | >40 | | >35 | 37.0 | 35.0 | | 33.0 | <25 | <25 | <25 | <25 | 23.0 | <25 | <25 |
| | △Pt [pa] | | 160.0 | 100.0 | | 180.0 | 68.0 | 48.0 | | 72.0 | 43.0 | 28.0 | 18.0 | 43.0 | 25.0 | 15.0 | 10.0 |
| 200 | X [m] | | >3 | >3 | | >3 | >3 | 2.9 | | >4 | 3.5 | 2.9 | 2.2 | 4.7 | 3.1 | 2.6 | 2.2 |
| | NR [dB(A)] | | >35 | >40 | | >35 | >35 | 40.0 | | 40.0 | 35.0 | 30.0 | <30 | 35.0 | 30.0 | <25 | <25 |
| | △Pt [pa] | | 280.0 | 155.0 | | 300.0 | 120.0 | 70.0 | | 125.0 | 73.0 | 48.0 | 30.0 | 70.0 | 48.0 | 27.0 | 18.0 |
| 250 | X [m] | | | | | | | | | >4 | >4 | 3.9 | 3.0 | >5 | 4.0 | 3.6 | 3.0 |
| | NR [dB(A)] | | | | | | | | | >40 | 40.0 | 35.0 | 33.0 | 40.0 | 37.0 | 30.0 | <25 |
| | △Pt [pa] | | | | | | | | | 195.0 | 125.0 | 68.0 | 45.0 | 100.0 | 78.0 | 40.0 | 25.0 |
| 300 | X [m] | | | | | | | | | | | >4 | 3.8 | >5 | 4.8 | 4.2 | 3.5 |
| | NR [dB(A)] | | | | | | | | | | | 40.0 | 38.0 | >40 | 45.0 | 36.0 | 30.0 |
| | △Pt [pa] | | | | | | | | | | | 100.0 | 65.0 | 150.0 | 120.0 | 60.0 | 35.0 |
| 500 | X [m] | | | | | | | | | | | | | | | >5 | >5 |
| | NR [dB(A)] | | | | | | | | | | | | | | | >40 | 42.0 |
| | △Pt [pa] | | | | | | | | | | | | | | | 180.0 | 100.0 |
| 700 | X [m] | | | | | | | | | | | | | | | | >5 |
| | NR [dB(A)] | | | | | | | | | | | | | | | | >40 |
| | △Pt [pa] | | | | | | | | | | | | | | | | 160.0 |

## Annotation

* S [mm] - Disk opening
* X [m] - Air jet length at the velocity Vk = 0.2 m/s
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

| Type | / | Diameter | / | O |
|---|---|---|---|---|
| **SV** | | 100<br>125<br>150<br>160<br>200 | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Circular Valve VN/VLB/VLA

## Description
Stainless steel circular valve for exhaust/supply of the air flow.

## Application
Supply/ exhaust of the airflow

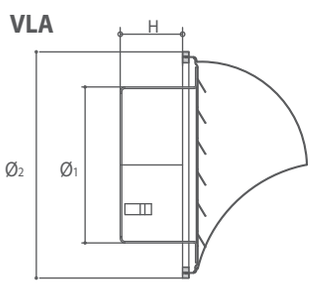
## Characteristics
VN - Circular valve with adjustable central cone for air flow adjustments.
- Mounting frame with mounting holes and clips provided
VLA - Exterior valve with blades, 1.5 x 1.5 mm mesh and sealing gasket.
- Clips on the connector and mounting holes on the outer frame
VLB - Exterior valve with 10 x 10 mm mesh and sealing gasket.
- Clips on the connector and mounting holes on the outer frame.

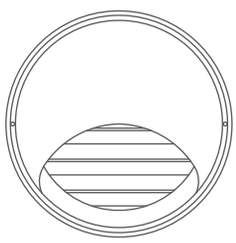
## Materials and finishing
Stainless steel, natural finish.

## Assembly Methods
Mounting with screw and/or using the clamp located at the connection/framework.

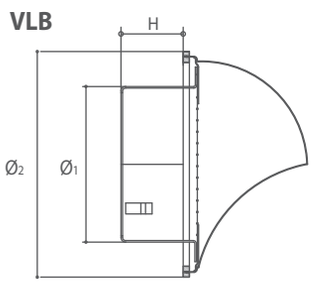
## Technical design

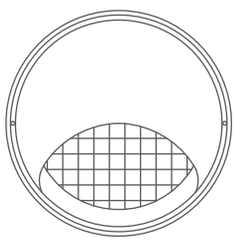
**VLA**



**VLB**



| Type | Ø1 | Ø2 | H |
|---|---|---|---|
| VN | 100 | 140 | 50 |
| | 125 | 170 | 50 |
| | 150 | 218 | 63 |
| | 160 | 218 | 63 |
| | 200 | 298 | 80 |
| VLA | 100 | 147 | 50 |
| | 125 | 187 | 50 |
| | 150 | 210 | 61 |
| | 160 | 210 | 64 |
| | 200 | 270 | 73 |
| VLB | 100 | 147 | 50 |
| | 125 | 187 | 50 |
| | 150 | 210 | 61 |
| | 160 | 210 | 64 |
| | 200 | 270 | 73 |

## Selection table VN, VLA, VLB

| VN | | | | | | |
|---|---|---|---|---|---|---|
| Air flow ($m^3/h$) | Dimensions (mm) | 100 | 125 | 150 | 160 | 200 |
| 30 | X [m] | 0.21 | | | | |
| | ΔPt [pa] | 14.00 | | | | |
| 40 | X [m] | 0.42 | 0.32 | | | |
| | ΔPt [pa] | 27.00 | 12.00 | | | |
| 60 | X [m] | 0.75 | 0.45 | 0.30 | 0.20 | |
| | ΔPt [pa] | 50.00 | 25.00 | 3.00 | 2.00 | |
| 80 | X [m] | 1.10 | 0.70 | 0.50 | 0.43 | |
| | ΔPt [pa] | 95.00 | 55.00 | 1.00 | 9.00 | |
| 130 | X [m] | | 1.50 | 1.00 | 0.95 | 1.2 |
| | ΔPt [pa] | | 125.00 | 25.00 | 25.00 | 45.00 |
| 190 | X [m] | | | 1.80 | 1.70 | 1.7 |
| | ΔPt [pa] | | | 56.00 | 50.00 | 60.00 |
| 220 | X [m] | | | | | 2.10 |
| | ΔPt [pa] | | | | | 120.00 |

| VLA | | | | |
|---|---|---|---|---|
| Air flow ($m^3/h$) | Dimensions (mm) | 100 | 125 | 150 |
| 30 | Veff[m/s] | 1.00 | 0.80 | |
| | ΔPt [pa] | 6.50 | 5.00 | |
| 60 | Veff[m/s] | 2.00 | 1.40 | 1.00 |
| | ΔPt [pa] | 23.00 | 18.00 | 7.00 |
| 80 | Veff[m/s] | 3.00 | 1.90 | 1.30 |
| | ΔPt [pa] | 45.00 | 27.00 | 13.00 |
| 110 | Veff[m/s] | 4.00 | 2.50 | 1.80 |
| | ΔPt [pa] | 90.00 | 47.00 | 23.00 |
| 140 | Veff[m/s] | 5.00 | 3.20 | 2.20 |
| | ΔPt [pa] | 145.00 | 72.00 | 34.00 |
| 170 | Veff[m/s] | | 3.90 | 2.70 |
| | ΔPt [pa] | | 125.00 | 50.00 |
| 180 | Veff[m/s] | | 4.00 | 2.90 |
| | ΔPt [pa] | | 137.00 | 58.00 |
| 190 | Veff[m/s] | | 4.20 | 3.00 |
| | ΔPt [pa] | | 150.00 | 67.00 |
| 220 | Veff[m/s] | | 5.00 | 3.50 |
| | ΔPt [pa] | | 210.00 | 85.00 |
| 250 | Veff[m/s] | | 5.70 | 4.00 |
| | ΔPt [pa] | | 240.00 | 105.00 |
| 270 | Veff[m/s] | | 6.00 | 4.30 |
| | ΔPt [pa] | | 270.00 | 115.00 |
| 320 | Veff[m/s] | | | 5.00 |
| | ΔPt [pa] | | | 170.00 |
| 380 | Veff[m/s] | | | 6.00 |
| | ΔPt [pa] | | | 250.00 |

| VLB | | | | |
|---|---|---|---|---|
| Air flow ($m^3/h$) | Dimensions (mm) | 100 | 125 | 150 |
| 30 | Veff[m/s] | 1.00 | 0.80 | |
| | ΔPt [pa] | 4.5 | 5.00 | |
| 60 | Veff[m/s] | 2.00 | 1.40 | 1.00 |
| | ΔPt [pa] | 15.00 | 4.00 | 3.00 |
| 80 | Veff[m/s] | 3.00 | 1.90 | 1.30 |
| | ΔPt [pa] | 29.00 | 8.50 | 5.00 |
| 110 | Veff[m/s] | 4.00 | 2.50 | 1.70 |
| | ΔPt [pa] | 55.00 | 15.00 | 16.00 |
| 140 | Veff[m/s] | 5.00 | 3.20 | 2.20 |
| | ΔPt [pa] | 90.00 | 25.00 | 24.00 |
| 170 | Veff[m/s] | 6.00 | 3.90 | 2.70 |
| | ΔPt [pa] | 135.00 | 37.00 | 37.00 |
| 180 | Veff[m/s] | | 4.00 | 2.90 |
| | ΔPt [pa] | | 40.00 | 43.00 |
| 190 | Veff[m/s] | | 4.20 | 3.00 |
| | ΔPt [pa] | | 45.00 | 48.00 |
| 220 | Veff[m/s] | | 5.00 | 3.50 |
| | ΔPt [pa] | | 65.00 | 63.00 |
| 250 | Veff[m/s] | | 5.70 | 4.00 |
| | ΔPt [pa] | | 70.00 | 73.00 |
| 270 | Veff[m/s] | | 6.00 | 4.30 |
| | ΔPt [pa] | | 85.00 | 80.00 |
| 320 | Veff[m/s] | | | 5.00 |
| | ΔPt [pa] | | | 125.00 |
| 380 | Veff[m/s] | | | 6.00 |
| | ΔPt [pa] | | | 180.00 |

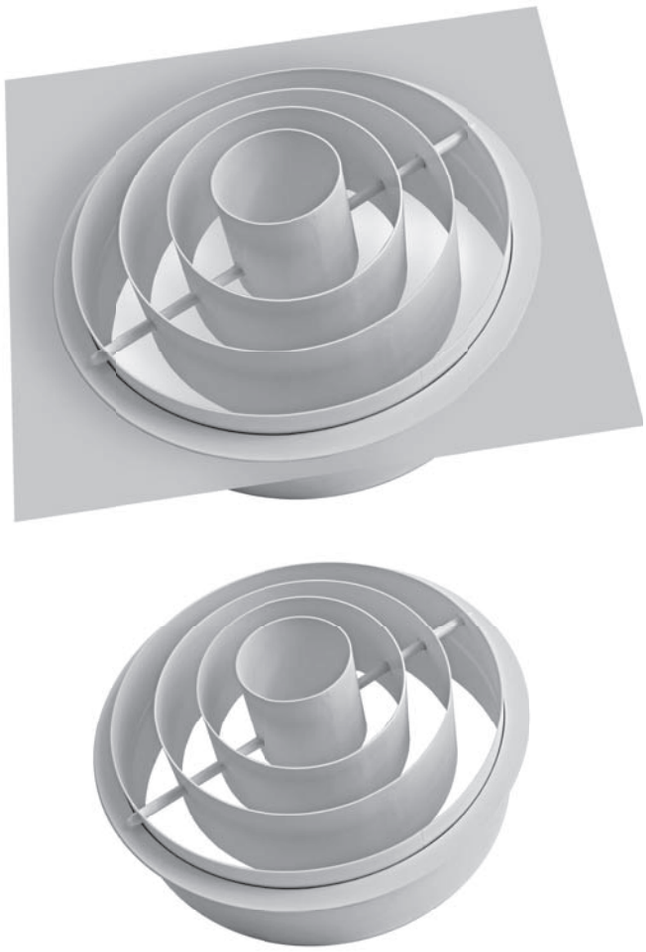
## Annotation

* Veff[m/s] - Air velocity in valve
* ΔPt [pa] - Pressure loss
* X [m] - Length air jet at speed Vk [m/s] = 0.5 m/s

## Order codes

| Type | / | Dimensions |
|---|---|---|
| | | Ø |
| VN<br>VLA<br>VLB | | 100<br>125<br>150<br>160<br>200 |

# Special Jet Diffuser SJD



## Description
Diffuser with long range air supply capacity. Ideal for use in traffic areas, theaters, discos, museums, concert halls, etc.

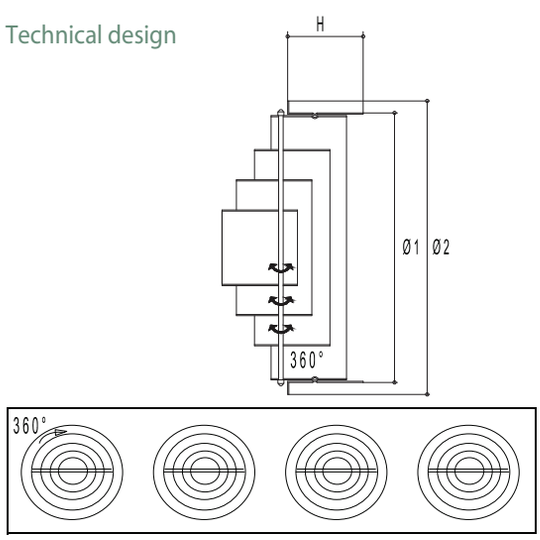
## Application
Air supply

## Characteristics
Concentric nozzles that allow airflow direction in the desired areas.
Rectangular or circular air duct mounting.
Mounting holes on flange.
Horizontal and vertical 360° rotation.
Produced in an assembly of 1, 2, 3, 4 pieces.
Manually adjustable.

## Accessories
Insulated/ Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Aluminum nozzles and galvanized steel sheet connection, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

## Technical design




| Ø1 | Ø2 | H |
|---|---|---|
| 200 | 240 | 100 |
| 250 | 290 | |
| 300 | 340 | |
| 350 | 390 | |

## Assembly Methods
Standard assembly with screws.

## Selection table SJD

| Air flow | Ø | 200 | | | | 250 | | | | 300 | | | | 350 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ($m^3$/h) | Elements no. | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| **342** | NR [dB(A)] | — | | | | | | | | | | | | | | | |
| | △Pt [pa] | 15.0 | | | | | | | | | | | | | | | |
| | X [m] | 8 | | | | | | | | | | | | | | | |
| **504** | NR [dB(A)] | 25 | | | | | | | | | | | | | | | |
| | △Pt [pa] | 35.0 | | | | | | | | | | | | | | | |
| | X [m] | 9 | | | | | | | | | | | | | | | |
| **684** | NR [dB(A)] | 31 | — | | | | | | | | | | | | | | |
| | △Pt [pa] | 60 | 15 | | | | | | | | | | | | | | |
| | X [m] | 12 | 8 | | | | | | | | | | | | | | |
| **828** | NR [dB(A)] | 39 | 22 | | | — | | | | — | | | | | | | |
| | △Pt [pa] | 90 | 25 | | | 30 | | | | 15 | | | | | | | |
| | X [m] | 15 | 9 | | | 12 | | | | 11 | | | | | | | |
| **1008** | NR [dB(A)] | 43 | 23 | | | 29 | | | | — | | | | | | | |
| | △Pt [pa] | 125 | 35 | | | 45 | | | | 25 | | | | | | | |
| | X [m] | 18 | 11 | | | 15 | | | | 14 | | | | | | | |
| **1512** | NR [dB(A)] | | 40 | 29 | 22 | 41 | — | | | 34 | 22 | | | | | | |
| | △Pt [pa] | | 75 | 35 | 20 | 95 | 30 | | | 50 | 15 | | | | | | |
| | X [m] | | 15 | 12 | 9 | 21 | 12 | | | 18 | 11 | | | | | | |
| **2052** | NR [dB(A)] | | 45 | 37 | 31 | 48 | 32 | — | | 42 | 24 | — | | 35 | — | | |
| | △Pt [pa] | | 125 | 60 | 35 | 190 | 50 | 20 | | 90 | 25 | 10 | | 45 | 10 | | |
| | X [m] | | 21 | 15 | 12 | 28 | 15 | 12 | | 24 | 14 | 11 | | 20 | 12 | | |
| **3060** | NR [dB(A)] | | | 48 | 43 | | 46 | 31 | — | 52 | 38 | 25 | | 46 | 29 | — | |
| | △Pt [pa] | | | 125 | 75 | | 95 | 45 | 30 | 200 | 50 | 25 | | 115 | 25 | 19 | |
| | X [m] | | | 23 | 17 | | 24 | 17 | 14 | 37 | 21 | 15 | | 30 | 18 | 12 | |
| **3744** | NR [dB(A)] | | | 53 | 47 | | 52 | 40 | 31 | | 44 | 31 | 27 | 52 | 38 | 24 | |
| | △Pt [pa] | | | 190 | 105 | | 145 | 65 | 40 | | 75 | 35 | 20 | 150 | 40 | 20 | |
| | X [m] | | | 28 | 21 | | 29 | 21 | 17 | | 26 | 18 | 14 | 37 | 21 | 15 | |
| **4428** | NR [dB(A)] | | | | 52 | | 57 | 45 | 38 | | 47 | 39 | 30 | 57 | 43 | 29 | — |
| | △Pt [pa] | | | | 150 | | 200 | 90 | 50 | | 105 | 50 | 25 | 200 | 55 | 25 | 15 |
| | X [m] | | | | 24 | | 34 | 24 | 18 | | 29 | 21 | 17 | 43 | 26 | 18 | 15 |
| **5940** | NR [dB(A)] | | | | 58 | | | 54 | 46 | | 55 | 46 | 39 | | 49 | 40 | 32 |
| | △Pt [pa] | | | | 275 | | | 155 | 90 | | 190 | 80 | 40 | | 100 | 45 | 25 |
| | X [m] | | | | 32 | | | 32 | 24 | | 37 | 28 | 21 | | 34 | 24 | 18 |
| **7632** | NR [dB(A)] | | | | | | | 61 | 53 | | | 51 | 48 | | 55 | 47 | 40 |
| | △Pt [pa] | | | | | | | 250 | 150 | | | 140 | 80 | | 155 | 60 | 40 |
| | X [m] | | | | | | | 40 | 32 | | | 35 | 28 | | 40 | 32 | 24 |
| **10260** | NR [dB(A)] | | | | | | | | 62 | | | 57 | 54 | | 65 | 53 | 49 |
| | △Pt [pa] | | | | | | | | 250 | | | 250 | 145 | | 250 | 125 | 70 |
| | X [m] | | | | | | | | 43 | | | 49 | 37 | | 58 | 41 | 32 |
| **13608** | NR [dB(A)] | | | | | | | | | | | | 61 | | | 63 | 56 |
| | △Pt [pa] | | | | | | | | | | | | 250 | | | 210 | 125 |
| | X [m] | | | | | | | | | | | | 55 | | | 55 | 43 |

## Annotation

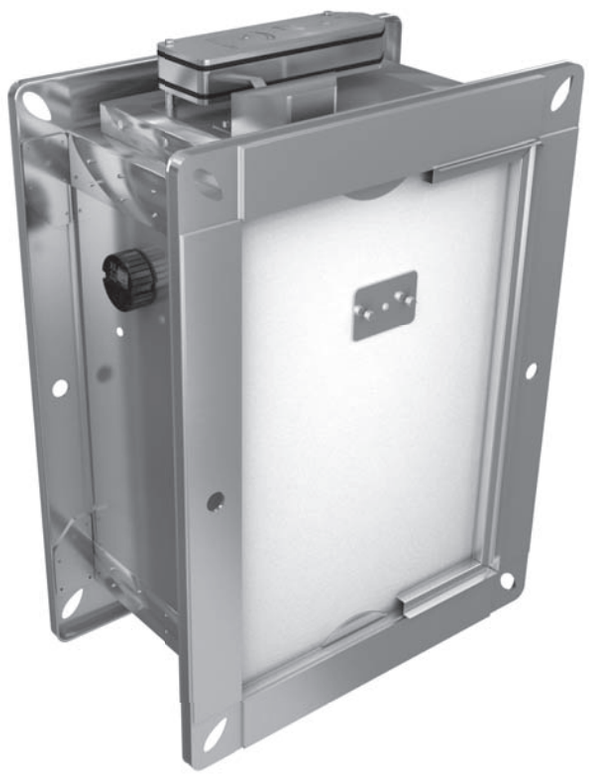
* X [m] - Air flow length for Vk=0.25m/s speed
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level for 8dB room attenuation

**Note:** The values in the table are for ΔT = + 10 K.

## Order codes

| Type | / | Dimensions | / | O |
|---|---|---|---|---|
| **SJD** | | 200<br>250<br>300<br>350 | | **0** Painted white RAL 9016<br>**R** Other RAL paints |

# Rectangular Fire Damper SFD



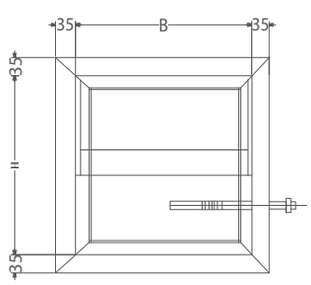
## Description
Rectangular fireproof damper designed for automatic isolation of ventilation and air conditioning systems compartments.
Suitable for mounting into concrete walls and ceilings.

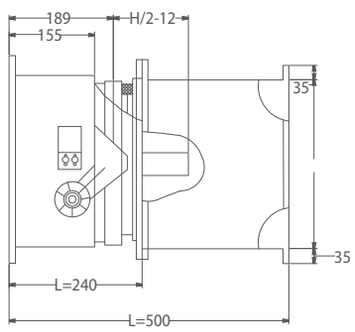
## Application
For closing and sealing the ventilation ducts in case of fire.

## Characteristics
Drive system with fusible element.
Automatic closing fire damper fusible element is determined to be active at temperatures> 72 ° C.
Various classes of fire resistance (RF60, RF90, RF120).
Optional contact signal for start/ end of the cycle.
Alternatively, an electric actuator can be used.

## Materials and finishing
Galvanized sheet steel case
Special insulated damper blade

## Technical design



## Selection table SFD

| Veff (m/s) | H (mm) / B=400 (mm) | 200 | 250 | 300 | 350 | 400 | 450 | 500 | 600 | 700 | 800 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2.0 | NR [db(A)] | 33.00 | 31.00 | 28.00 | 27.00 | 26.00 | 25.00 | 25.50 | 25.00 | 24.50 | 24.00 |
| | ΔPt[pa] | 18.00 | 8.50 | 6.00 | 4.50 | 3.80 | 3.10 | 2.80 | 2.20 | 1.90 | 1.70 |
| 2.5 | NR [db(A)] | 38.00 | 34.00 | 33.00 | 32.50 | 31.50 | 31.00 | 30.50 | 30.00 | 29.50 | 29.00 |
| | ΔPt[pa] | 28.00 | 14.00 | 9.20 | 7.00 | 5.80 | 4.80 | 4.20 | 3.50 | 2.90 | 2.70 |
| 3.0 | NR [db(A)] | 42.50 | 40.00 | 38.00 | 36.80 | 36.00 | 35.80 | 35.50 | 35.00 | 34.50 | 34.00 |
| | ΔPt[pa] | 40.00 | 19.00 | 14.00 | 10.00 | 8.00 | 7.00 | 6.00 | 5.00 | 4.30 | 3.80 |
| 5.0 | NR [db(A)] | 53.00 | 51.80 | 50.70 | 49.00 | 48.00 | 47.80 | 47.20 | 46.80 | 46.70 | 46.50 |
| | ΔPt[pa] | 110.00 | 55.00 | 38.00 | 28.00 | 22.00 | 19.50 | 17.00 | 14.00 | 12.00 | 11.00 |
| 7.0 | NR [db(A)] | 61.00 | 58.00 | 57.20 | 56.70 | 56.50 | 56.10 | 55.90 | 55.70 | 55.20 | 55.00 |
| | ΔPt[pa] | 150.00 | 110.00 | 70.00 | 55.00 | 45.00 | 38.00 | 32.00 | 28.00 | 22.00 | 20.00 |

| Aerodynamic and acoustic corrections based on other lengths of multispot B | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B(mm) | 200 | 250 | 300 | 350 | 400/450 | 500 | 600 | 700/800 | 900 | 1000/1100 | 1200-1400 | 1500 |
| ΔPt | 1.7 | 1.5 | 1.3 | 1.1 | 1.1 | 0.9 | 0.8 | 0.7 | 0.6 | 0.6 | 0.6 | 0.5 |
| NR | -1 | -1 | 0 | 0 | 0 | 0 | 1 | 2 | 2 | 3 | 4 | 5 |

ΔPt [pa] = Correction coefficient x Δ Pt (B=400 mm)
NR[dB(A)] = Correction coefficient + NR (B=400 mm)

## Annotation

* Veff[m/s] - Air velocity in damper
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

| Type | / | Dimensions | |
|---|---|---|---|
| | | B | H |
| SFD | | 200 | 200 |
| | | 250 | 250 |
| | | 300 | 300 |
| | | 350 | 350 |
| | | 400 | 400 |
| | | 450 | 450 |
| | | 500 | 500 |
| | | 600 | 600 |
| | | 700 | 700 |
| | | 800 | 800 |
| | | 900 | |
| | | 1000 | |
| | | 1100 | |
| | | 1200 | |
| | | 1300 | |
| | | 1400 | |
| | | 1500 | |

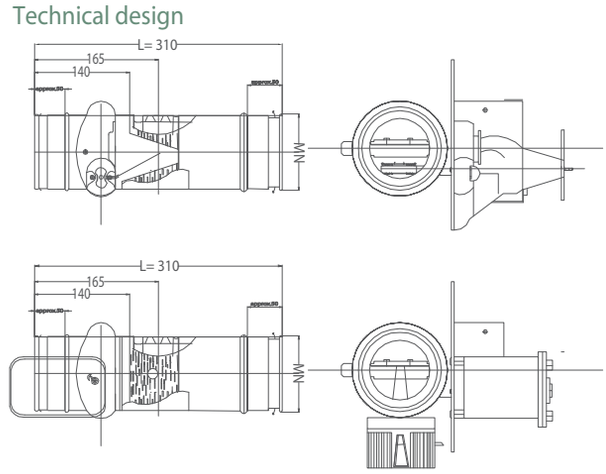
# Round Fire Damper RFD

## Description
Round fireproof damper designed for automatic isolation of ventilation and air conditioning systems compartments. Suitable for mounting into concrete walls and ceilings.

## Application
For closing and sealing the ventilation ducts in case of fire.

## Characteristics
Drive system with fusible element.
Automatic closing fire damper fusible element is determined to be active at temperatures> 72 ° C.
Various classes of fire resistance (RF60, RF90, RF120).
Optional contact signal for start/ end of the cycle.
Alternatively, an electric actuator can be used.

## Materials and finishing
Galvanized sheet steel case
Special insulated damper blade

## Technical design


## Selection table RFD

| Veff (m/s) | Diameter (mm) | 100 | 125 | 160 | 200 |
|---|---|---|---|---|---|
| **4** | NR [dB(A)]<br>ΔPt [pa] | 34.00<br>18.20 | 28.00<br>10.60 | 25.00<br>8.60 | 20.00<br>5.80 |
| **6** | NR [dB(A)]<br>ΔPt [pa] | 41.00<br>41.00 | 38.00<br>23.80 | 34.00<br>19.40 | 30.00<br>13.00 |
| **8** | NR [dB(A)]<br>ΔPt [pa] | 48.00<br>73.00 | 45.00<br>42.20 | 42.00<br>34.60 | 38.00<br>23.00 |
| **10** | NR [dB(A)]<br>ΔPt [pa] | 53.00<br>114.00 | 51.00<br>66.00 | 47.00<br>34.60 | 45.00<br>36.00 |
| **12** | NR [dB(A)]<br>ΔPt [pa] | 57.00<br>164.20 | 55.00<br>95.00 | 52.00<br>77.80 | 50.00<br>51.80 |
| **15** | NR [dB(A)]<br>ΔPt [pa] | 63.00<br>256.50 | 60.00<br>148.50 | 58.00<br>121.50 | 55.00<br>81.00 |

## Annotation

* Veff[m/s] - Air velocity in damper
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

**Type** / **Dimensions**

**RFD**

100
125
160
200

# Air Flow Control Damper VCD

## Description
Aluminum rectangular air flow control damper with opposable blades.

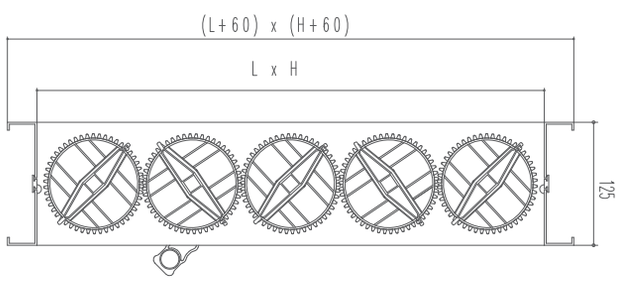
## Application
Air flow adjustment in ventilation systems.

## Characteristics
The standard control damper is provided with a drive lever and blocking system, an electric drive (actuator) can be delivered on request.
Actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.)
Control dampers can be made, on request, without drive system - in this case it will be considered the diameter of the blade's drive shaft for coupling the desired actuator. (eg. Control damper without drive lever, prepared for actuator)
Drive lever / actuator operates the opening / closing of the blades through 12 X 12mm and 150mm length shaft .
Aerodynamic blades are a main characteristic of the control damper.
The control damper is provided with gasket seals on both the blades and the frame.
Blade is 100mm height and cannot be adjusted, having the following two cases:
* Case 1: 200 x 200 mm - the control damper consists of two adjustable blades and the final measurements will become 200 x 212 mm [W (H + 12 mm)].
* Case 2: 200 x 150 mm - the control damper consists of one adjustable blade and the difference will be covered with a fixed blade (final measurements - 200 x 150 mm).

## Accessories
Drive actuator

## Materials and finishing
Natural aluminum material for frame and blades
Polypropylene sprocket wheels

## Technical design




## Assembly Methods
Screws on the mounting flange.

## Selection table VCD

| Air flow | Size [LxH] | 800x500 | | 1000x500 | | 1500x500 | | 800x800 | | 1000x800 | | 1200x800 | | 1500x800 | | 1200x1200 | | 1500x1200 | | 1800x1200 | | 2000x1200 | | 2200x1200 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ($m^3$/h) | α [°] | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 | 0 | 45 |
| 1000 | Veff [m/s] | 1.3 | 3.7 | | | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 1.3 | 8.8 | | | | | | | | | | | | | | | | | | | | | | |
| 1500 | Veff [m/s] | 1.9 | 5.5 | | | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 2.5 | 19.4 | | | | | | | | | | | | | | | | | | | | | | |
| 2000 | Veff [m/s] | 2.5 | 7.4 | | | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 4.2 | 34.3 | | | | | | | | | | | | | | | | | | | | | | |
| 2500 | Veff [m/s] | 3.1 | 9.2 | | | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 6.5 | 53.5 | | | | | | | | | | | | | | | | | | | | | | |
| 3000 | Veff [m/s] | 3.8 | 11.1 | 1.9 | 5.5 | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 9.2 | 76.8 | 2.5 | 19.4 | | | | | | | | | | | | | | | | | | | | |
| 3500 | Veff [m/s] | 4.4 | 12.9 | 2.2 | 6.5 | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 12.4 | 104.5 | 3.3 | 26.3 | | | | | | | | | | | | | | | | | | | | |
| 4000 | Veff [m/s] | 5.0 | 14.8 | 2.5 | 7.4 | | | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 16.1 | 136.4 | 4.2 | 34.3 | | | | | | | | | | | | | | | | | | | | |
| 4500 | Veff [m/s] | 5.7 | 16.6 | 2.8 | 8.3 | 1.9 | 5.5 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 20.3 | 172.5 | 5.3 | 43.4 | 2.5 | 19.4 | | | | | | | | | | | | | | | | | | |
| 5000 | Veff [m/s] | 6.3 | 18.4 | 3.1 | 9.2 | 2.1 | 6.2 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | 25.0 | 212.9 | 6.5 | 53.5 | 3.0 | 23.9 | | | | | | | | | | | | | | | | | | |
| 5500 | Veff [m/s] | | | 3.5 | 10.1 | 2.3 | 6.8 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | 7.8 | 64.6 | 3.6 | 28.9 | | | | | | | | | | | | | | | | | | |
| 6000 | Veff [m/s] | | | 3.8 | 11.1 | 2.5 | 7.4 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | 9.2 | 76.8 | 4.2 | 34.3 | | | | | | | | | | | | | | | | | | |
| 6500 | Veff [m/s] | | | 4.1 | 12.0 | 2.7 | 8.0 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | 10.7 | 90.1 | 4.9 | 40.2 | | | | | | | | | | | | | | | | | | |
| 7000 | Veff [m/s] | | | 4.4 | 12.9 | 2.9 | 8.6 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | 12.4 | 104.5 | 5.7 | 46.6 | | | | | | | | | | | | | | | | | | |
| 7500 | Veff [m/s] | | | | | 3.1 | 9.2 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | | | 6.5 | 53.5 | | | | | | | | | | | | | | | | | | |
| 8000 | Veff [m/s] | | | | | 3.4 | 9.8 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | | | 7.3 | 60.8 | | | | | | | | | | | | | | | | | | |
| 8500 | Veff [m/s] | | | | | 3.6 | 10.5 | | | | | | | | | | | | | | | | | | |
| | △Pt [pa] | | | | | 8.2 | 68.6 | | | | | | | | | | | | | | | | | | |
| 10000 | Veff [m/s] | | | | | | | 5.0 | 14.6 | 4.0 | 11.7 | 3.3 | 9.7 | 2.7 | 7.8 | 2.2 | 6.5 | | | | | | | | |
| | △Pt [pa] | | | | | | | 15.7 | 132.8 | 1.0 | 85.1 | 0.7 | 59.2 | 4.7 | 38.0 | 3.4 | 26.8 | | | | | | | | |
| 11000 | Veff [m/s] | | | | | | | 5.5 | 16.0 | 4.4 | 12.8 | 3.6 | 10.7 | 2.9 | 8.5 | 2.4 | 7.2 | | | | | | | | |
| | △Pt [pa] | | | | | | | 18.9 | 160.6 | 12.2 | 102.9 | 8.6 | 71.5 | 5.6 | 45.9 | 4.0 | 32.3 | | | | | | | | |
| 15000 | Veff [m/s] | | | | | | | | | 6.0 | 17.5 | 5.0 | 14.6 | 4.0 | 11.7 | 3.3 | 9.8 | 2.7 | 7.8 | 2.2 | 6.5 | 2.0 | 5.9 | | |
| | △Pt [pa] | | | | | | | | | 22.4 | 191.1 | 15.7 | 132.8 | 10.1 | 85.1 | 7.2 | 59.9 | 4.7 | 38.4 | 3.4 | 26.8 | 2.8 | 21.7 | | |
| 16000 | Veff [m/s] | | | | | | | | | 6.4 | 18.6 | 5.3 | 15.5 | 4.2 | 12.4 | 3.6 | 10.4 | 2.8 | 8.3 | 2.4 | 6.9 | 2.1 | 6.3 | 1.9 | 5.6 |
| | △Pt [pa] | | | | | | | | | 25.5 | 217.3 | 17.8 | 151.1 | 11.5 | 96.8 | 8.2 | 68.1 | 5.3 | 43.7 | 3.8 | 30.4 | 3.1 | 24.7 | 2.5 | 20.1 |
| 17000 | Veff [m/s] | | | | | | | | | 6.8 | 19.8 | 5.6 | 16.5 | 4.5 | 13.2 | 3.8 | 11.1 | 3.0 | 8.9 | 2.5 | 7.4 | 2.3 | 6.6 | 2.0 | 6.0 |
| | △Pt [pa] | | | | | | | | | 28.7 | 245.3 | 20.0 | 170.4 | 12.9 | 109.2 | 9.2 | 76.8 | 6.0 | 49.3 | 4.2 | 34.3 | 3.5 | 27.8 | 2.9 | 22.5 |
| 18000 | Veff [m/s] | | | | | | | | | | | 6.0 | 17.5 | 4.8 | 14.0 | 4.0 | 11.7 | 3.2 | 9.4 | 2.7 | 7.8 | 2.4 | 7.0 | 2.1 | 6.3 |
| | △Pt [pa] | | | | | | | | | | | 22.4 | 191.1 | 14.5 | 122.4 | 10.3 | 86.1 | 6.7 | 55.2 | 4.7 | 38.4 | 3.9 | 31.2 | 3.2 | 25.4 |
| 20000 | Veff [m/s] | | | | | | | | | | | 6.6 | 19.4 | 5.3 | 15.5 | 4.4 | 13.0 | 3.6 | 10.4 | 3.0 | 8.7 | 2.7 | 7.8 | 2.4 | 6.2 |
| | △Pt [pa] | | | | | | | | | | | 27.6 | 235.8 | 17.8 | 151.0 | 12.6 | 106.2 | 8.2 | 68.1 | 5.8 | 47.4 | 4.7 | 38.4 | 3.8 | 31.1 |
| 30000 | Veff [m/s] | | | | | | | | | | | | | | | 6.7 | 19.5 | 5.3 | 15.6 | 4.4 | 13.0 | 4.0 | 11.7 | 3.6 | 10.5 |
| | △Pt [pa] | | | | | | | | | | | | | | | 28.0 | 238.5 | 18.0 | 152.8 | 12.6 | 106.2 | 10.3 | 86.1 | 8.4 | 69.8 |
| 35000 | Veff [m/s] | | | | | | | | | | | | | | | | | 6.2 | 18.2 | 5.2 | 15.2 | 4.7 | 13.7 | 4.2 | 12.3 |
| | △Pt [pa] | | | | | | | | | | | | | | | | | 24.4 | 207.8 | 17.0 | 144.4 | 13.8 | 117.0 | 11.2 | 94.8 |
| 40000 | Veff [m/s] | | | | | | | | | | | | | | | | | | | 5.9 | 17.4 | 5.3 | 15.6 | 4.8 | 14.1 |
| | △Pt [pa] | | | | | | | | | | | | | | | | | | | 22.1 | 188.5 | 18.0 | 152.8 | 14.7 | 123.9 |
| 45000 | Veff [m/s] | | | | | | | | | | | | | | | | | | | 6.7 | 19.5 | 6.0 | 17.6 | 5.4 | 15.8 |
| | △Pt [pa] | | | | | | | | | | | | | | | | | | | 28.0 | 238.5 | 22.7 | 193.3 | 18.4 | 156.7 |

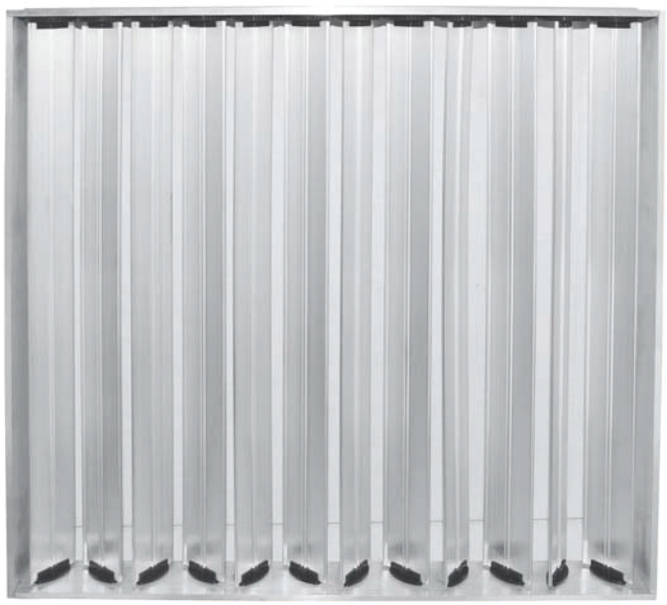
## Annotation

* α [°] - Closing angle of the blades
* Veff[m/s] - Air velocity through the damper
* ΔPt [pa] - Pressure loss
* *Veff [m/s] = Air flow [mc/h] / 3600[s] / Ak[$m^2$]

## Order codes

| Type | / | S | / | L x H |
|---|---|---|---|---|
| VCD | | Actuator | | On request |

# Opposable Blades Damper OBD



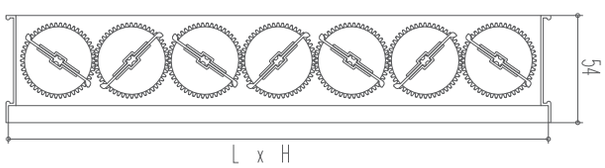
## Description
Air control register is mounted on connection of the grille and it is used to adjust the air flow.

## Application
Air flow adjustment in ventilation systems

## Characteristics
Built with aluminium opposable blades and sprocket wheels.
Manual operation.

## Materials and finishing
Natural aluminium blades and frame
Polypropylene sprocket wheels

## Technical design




## Assembly Methods
With pop rivets on the inlet connection of the grille.

## Order codes

| Type | / | Dimensions |
|---|---|---|
| OBD | | On request |

# Non Return Circular Damper RSD

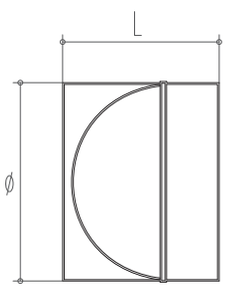
## Description
Non return circular damper for mounting between two circular sections of tubing.

## Application
It is used to prevent the return of the air when the fan stops.

## Characteristics
The damper blades will open when exposed to excess pressure.
The damper is provided with a return spring , which determines closing of the blades when pressure drops.
By closing the blades return of the air is prevented when the fan stops.

## Materials and finishing
Galvanized steel cylinder
Aluminium blades
Natural galvanized finish

## Technical design




| Dimensions | L |
|---|---|
| 100 | 100 |
| 125 | 100 |
| 150 | 150 |
| 160 | 150 |
| 200 | 150 |
| 250 | 200 |
| 315 | 200 |
| 400 | 250 |

## Order codes

| Type | / | Ø |
|---|---|---|
| RSD | | 100 200<br>125 250<br>150 315<br>160 400 |

# Circular Air Control Damper RD

## Description
Circular air control damper with driving system.

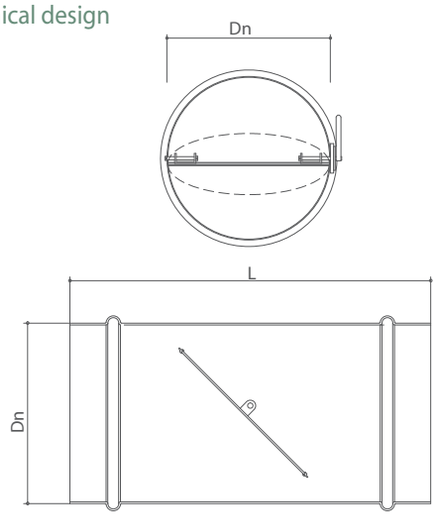
## Application
To adjust the air flow in ventilation/air conditioning ducts systems.

## Characteristics
The standard damper is produced with manual driving device (lever), unless electric drive (actuator) is requested.
The actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.).
Actuator operates the opening/ closing of the blade through Ø8 mm shaft.
The damper is provided with a sealing gasket.

## Accessories
Electric drive (actuator).

## Materials and finishing
Galvanized steel sheet

## Technical design




| Ø | L | Dn |
|---|---|---|
| 100 | 200 | 95 |
| 125 | 200 | 120 |
| 160 | 200 | 155 |
| 200 | 220 | 195 |
| 250 | 270 | 245 |
| 315 | 335 | 310 |
| 400 | 420 | 395 |

## Selection table RD

| Ø [mm] | Ak [mp] |
|---|---|
| 100 | 0.0077 |
| 125 | 0.0121 |
| 160 | 0.0198 |
| 200 | 0.0311 |
| 250 | 0.0487 |
| 315 | 0.0774 |
| 400 | 0.1250 |

| Correction factor values | | | | |
|---|---|---|---|---|
| opening angle(˚) | 0 | 15 | 30 | 45 |
| Correction coeff. | 1 | 1,8 | 8 | 20 |

ΔPt Corrected = ΔPt * Correction coeff.

| Ø (mm) | Air flow m³/h | 100 | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 | 1500 | 2000 | 3000 | 4000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 100 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | 18<br>7<br>3.6 | 33<br>25<br>7.2 | 46<br>55<br>10.8 | | | | | | | | | | | |
| 125 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | 18<br>6<br>4.6 | 30<br>13<br>6.9 | 35<br>20<br>9.2 | | | | | | | | | | |
| 160 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | | 8<br>4<br>4.2 | 20<br>6<br>5.6 | 27<br>8<br>7.0 | 30<br>14<br>8.4 | | | | | | | | |
| 200 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | | | 8<br>2<br>3.6 | 10<br>4<br>4.5 | 20<br>5<br>5.4 | 24<br>6.5<br>6.3 | 27<br>8<br>7.1 | 33<br>12<br>8.0 | | | | | |
| 250 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | | | | | 4<br>1<br>3.4 | 6<br>1.8<br>4.0 | 8<br>2.5<br>4.6 | 18<br>4<br>5.1 | 21<br>4.5<br>5.7 | 30<br>8<br>8.6 | 40<br>18<br>11.4 | | |
| 315 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | | | | | | | | | | 25<br>2.5<br>5.4 | 33<br>4.5<br>7.2 | 44<br>9<br>10.8 | |
| 400 | NR [dB(A)]<br>ΔPt [pa]<br>Veff[m/s] | | | | | | | | | | | | | 37<br>3<br>6.7 | 45<br>55<br>8.9 |

## Annotation

* Ak [m²] - Free surface
* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity through the damper

## Order codes

| Type | + | S | / | Ø |
|---|---|---|---|---|
| RD | | Actuator | | On request |

# Air Volume Control Damper IRIS

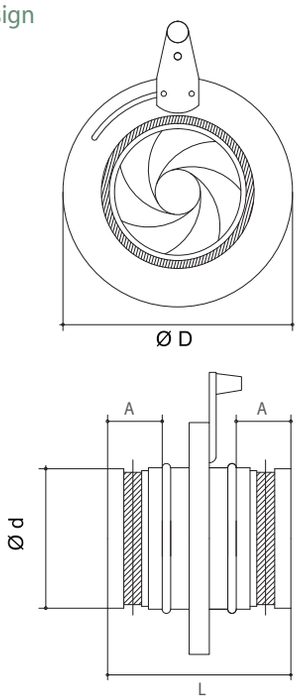
## Description
Iris diaphragm damper allows adjustment of the air flow by measuring pressure at two points. .

## Application
Adjustment of the air flow in ventilation and air conditioning systems.

## Characteristics
The damper is provided with an iris diaphragm, pressure measuring points, operating lever and fitting gaskets.
Various levels of blade opening.
The iris is adjusted by operating lever.
The damper is delivered along with a selection chart (pressure loss/ air flow).

## Materials and finishing
Galvanized steel sheet
Plastic adjustment device

## Technical design



## Selection table IRIS

| IRIS | | | |
|---|---|---|---|
| Ø d | Ø D | L | A |
| 100 | 165 | 110 | 30 |
| 160 | 230 | 110 | 30 |
| 200 | 285 | 110 | 30 |
| 250 | 335 | 135 | 35 |
| 315 | 410 | 135 | 35 |

| Diameter | Air flow | |
|---|---|---|
| | min ($m^3/h$) 35 dB(A) | max ($m^3/h$) 55 dB(A) |
| 100 | 40 | 270 |
| 160 | 60 | 400 |
| 200 | 120 | 600 |
| 250 | 270 | 1100 |
| 315 | 400 | 2500 |

## Order codes

| Type | / | Diameter |
|---|---|---|
| IRIS | | 100<br>160<br>200<br>250<br>315 |

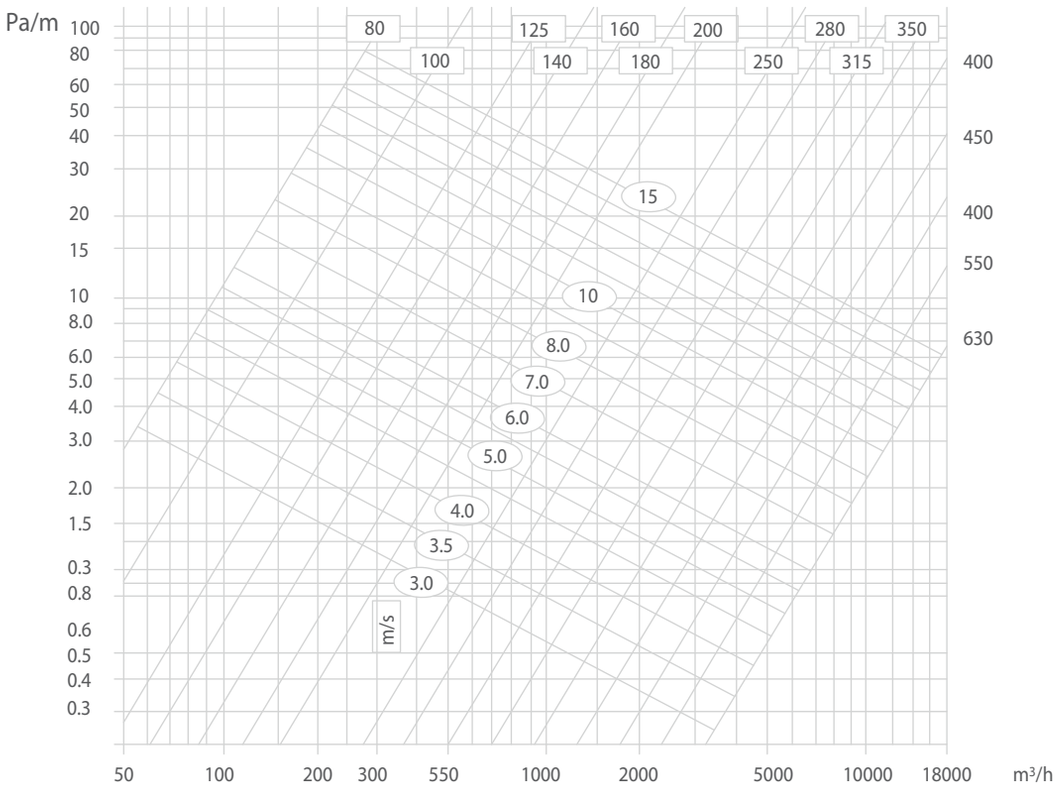
# Flexible Ducts

## Description
Aluminum flexible duct used in ventilation and air conditioning installations.

## Application
Supply/ exhaust of the air.

## Characteristics
Flexible tubing is available both non-insulated and insulated.
The standard length is 10 m/pc.
Fiberglass insulation.
The temperature range is - 30°C ÷ 140°C.
Maximum velocity is 20 m/s.
Not suitable for air transport with high concentration of acids and bases or for discharge of combustion gases.

## Pressure loss chart



# Adapter ADP

Description
Adapter (plenum) made of galvanized sheet, used as an intermediary element between the grille and ventilation ducts.

Characteristics
Adapter can be insulated or non-insulated **ADP IZ/N**
It is made with one or more connections for ventilation ducts.
The connector (socket) is positioned opposite to the grille (standard) or side (optional)
On request adapter is provided with mounting flange

# Special Parts

Description
Parts made of galvanized sheet metal for ventilation ducts, type: "T", "Y", sockets, bent pipes, etc.

Characteristics
Can be isolated on request
Dimensions on request

# Accessories

## Air Filter FIL

Description
Air filter for grille's inlet connection.

Characteristics
The air filter consists of filter material and "U" shaped aluminum frame
Class 2 polyester filter material, with high retention level and thickness of 12 mm.

## Mounting Spring

Description
Mounting spring is used for wall or ceiling mounting of the grille.

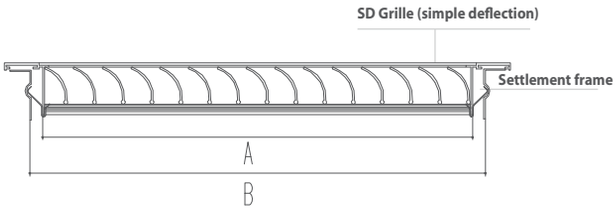
Characteristics
Optional accessory used as a way of mounting.
Not visible after mounting
Spring length is 105 mm.

# Settlement Frame For Mounting CC

## Description
Settlement frame made of galvanized steel sheet used for wall or ceiling mounting with clip mechanism.

## Characteristics
Optional accessory used as a way of mounting.
The mounting of the grille with settlement frame is not visible.
It is mounted in the hole and allows quick detachment of the grille.
The grille is attached to the settlement frame through the clips mounted on the grille's inlet.

## Technical design




A - interior size of the grille
B - size of the settlement frame

# Mounting Clips

## Description
Clip made of steel sheet for mounting the grille in frame.

Ram air flow

פתרונות דקורטיביים לפתחי מיזוג אוויר

רח' המורג 40
מושב עולש | מיקוד 42855
טל: 09-8940950 | פקס 09-8985156
www.ramaf.co.il

מקבוצת ה"תותח"

RAM
Airflow

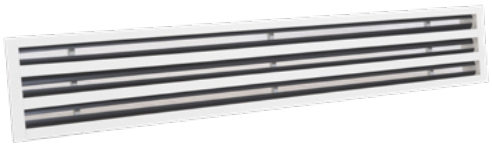
# מפזרי אוויר מעוצבים

**פתרונות מיזוג אוויר מעוצבים**

קטלוג מוצרים

# סלוטים

סלוט צר - 2 חריצים



סלוט צר - 3 חריצים

## סלוט צר

| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | | | | |
|---|---|---|---|---|---|---|---|
| | | עומק | רוחב | רוחב עם ברגים | שוליים מבחוץ | שוליים מבפנים | אורך מירבי ללא חיבור |
| סלוט צר חריץ 1 | 1 מ"א | 25 מ"מ | 48 | 52 | 20 מ"מ | 7.5 | 3.80 מטר |
| סלוט צר 2 חריצים | 1 מ"א | 25 מ"מ | 95 | 99 | 20 מ"מ | 0 | 6 מ'- 9016 . 3.8 מ' -צבעים אחרים |
| סלוט צר 3 חריצים | 1 מ"א | 25 מ"מ | 142 | 146 | 20 מ"מ | 0 | 6 מטר |
| סלוט צר 4 חריצים | 1 מ"א | 25 מ"מ | 189 | 193 | 20 מ"מ | 0 | 6 מטר |
| סלוט צר צוקל 2 חריצים | 1 מ"א | 25 מ"מ | | | 0 מ"מ | 0 | 6 מטר |
| סלוט צר 2 ח' 95*495 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*795 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*995 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*1495 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*1995 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*2495 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |
| סלוט צר 2 ח' 95*2995 | יחידה | 25 מ"מ | | | 20 מ"מ | 9.5 | - |

*ישנה אופצייה לסלוט 5 חריצים

## סלוט

| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | אורך מירבי לחתיכה | |
|---|---|---|---|---|---|
| | | עומק | רוחב אפשרי | שוליים 18 מבחוץ | שוליים 27 מבחוץ |
| סלוט חריץ 1 | 1 מ"א | 50 מ"מ | 40-62 | 3.8 מטר | 3.2 מטר |
| סלוט 2 חריצים | 1 מ"א | 50 מ"מ | 73-116 | 9016 -6 מ'. אחר-3.2 מ' | 3.2 מטר |
| סלוט 3 חריצים | 1 מ"א | 50 מ"מ | 117-170 | 3.8 מטר | 3.2 מטר |
| סלוט 4 חריצים | 1 מ"א | 50 מ"מ | 171-224 | 3.8 מטר | 3.2 מטר |
| סלוט 5 חריצים | 1 מ"א | 50 מ"מ | 225-278 | 3.8 מטר | 3.2 מטר |
| סלוט 6 חריצים | 1 מ"א | 50 מ"מ | 201-332 | 3.8 מטר | 3.2 מטר |
| סלוט 7 חריצים | 1 מ"א | 50 מ"מ | 332-387 | 3.8 מטר | 3.2 מטר |
| סלוט 8 חריצים | 1 מ"א | 50 מ"מ | 387-441 | 3.8 מטר | 3.2 מטר |
| סלוט 2 ח' 95*495 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*795 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*995 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*1495 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*1995 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*2495 | יחידה | 50 מ"מ | | | - |
| סלוט 2 ח' 95*2995 | יחידה | 50 מ"מ | | | - |


החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

# דקור פרמיום - במידות מיוחדות

## שוליים 12 ס"מ

| תיאור מוצר | מינימום לחישוב | אורך מירבי ביחידה אחת | | |
|---|---|---|---|---|
| | | עומק | 9016 | אחר |
| דקור פרמיום גובה 95 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום גובה 145 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום גובה 195 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום לבן על רקע שחור גובה 95 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום לבן על רקע שחור גובה 145 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום לבן על רקע שחור גובה 195 | 1 מ"א | 27 מ"מ | 5 מטר | 5 מטר |
| דקור פרמיום עד גובה 200 מ"מ | 0.1 מ"ר | 27 מ"מ | 6 מטר | 5 מטר |
| דקור פרמיום מעל גובה 200 מ"מ | 0.1 מ"ר | 27 מ"מ | 6 מטר | 5 מטר |
| דקור פרמיום עד גובה 200 מ"מ לבן על רקע שחור | 0.1 מ"ר | 27 מ"מ | 6 מטר | 5 מטר |
| דקור פרמיום מעל גובה 200 מ"מ לבן על רקע שחור | 0.1 מ"ר | 27 מ"מ | 6 מטר | 5 מטר |

# דקור

| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | | |
|---|---|---|---|---|---|
| | | עומק | ללא שוליים | שוליים 18 מ"מ | שוליים 27 / שוליים 10 |
| דקור עד גובה 200 מ"מ | 0.1 מ"ר | 20 | | 3.8 מטר | 3.2 מטר |
| דקור מעל גובה 200 מ"מ | 0.1 מ"ר | 20 | | 3.8 מטר | 3.2 מטר |
| דקור שדרות שחורות | 0.1 מ"ר | 20 | | 3.8 מטר | 3.2 מטר |
| דקור נפתח על ציר + פילטר | 0.2 מ"ר | 40 | | 3.2 מטר | |
| דקור עם משקוף עיוור | 0.1 מ"ר | 49 | 3.2 מטר | | |
| להבי הטיה אחוריים | 0.1 מ"ר | 16 | | | |
| דקור צוקל עד 200 מ"מ | 0.1 מ"ר | 30 | 3 מטר | | |
| דקור צוקל מעל 200 מ"מ | 0.1 מ"ר | 30 | 3 מטר | | |
| סטריפליין חד/דו כיווני | 0.1 מ"ר | 24 | 3.2 מטר | | |

# דקור במידות קבועות במלאי מיידי

| תיאור מוצר |
|---|
| דקור 95*500 |
| דקור 95*600 |
| דקור 95*800 |
| דקור 95*995 |
| דקור 95*1495 |
| דקור 95*1995 |
| דקור 95*2495 |
| דקור 95*2995 |
| דקור 95*4000 |
| דקור 200*400 |
| דקור 200*800 |
| דקור 150*400 |
| דקור 150*600 |
| דקור פרמיום 150*1000 |
| דקור פרמיום 150*1500 |
| דקור פרמיום 150*2000 |
| דקור פרמיום 150*2500 |
| דקור פרמיום 150*3000 |
| דקור פרמיום 200*1000 |
| דקור פרמיום 200*1500 |
| דקור פרמיום 200*2000 |
| דקור פרמיום 200*2500 |
| דקור פרמיום 200*3000 |

דקור פרימיום

דקור

דקור שוליים 0.7

החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

# סלוט צוקל

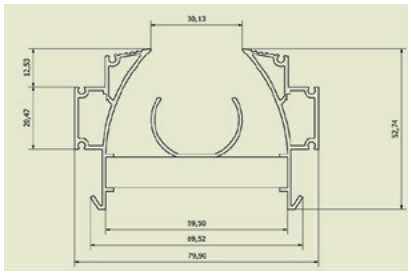
| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | | | | |
|---|---|---|---|---|---|---|---|
| | | עומק | רוחב | רוחב עם ברגים | שוליים מבחוץ | שוליים מבפנים | אורך מירבי ללא חיבור |
| סלוט צוקל צר חריץ 1 | 1 מ"א | 25 מ"מ | 48 | | | | 3.80 מטר |
| סלוט צוקל צר 2 חריצים | 1 מ"א | 25 מ"מ | 95 | | | | 3.80 מטר |
| סלוט צוקל צר 3 חריצים | 1 מ"א | 25 מ"מ | 142 | | | | 3.80 מטר |

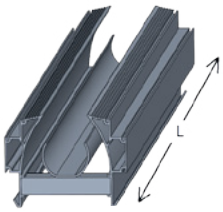
סלוט צוקל

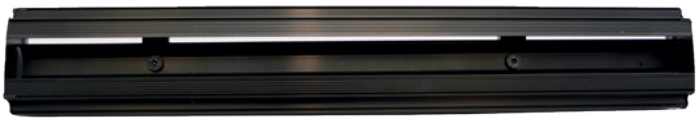
# שחר

| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | אורך מירבי לחתיכה |
|---|---|---|---|---|
| | | עומק | רוחב אפשרי | שוליים 18 מבחוץ |
| שחר חריץ אחד | 1 מ"א | 53 מ"מ | | 6 מטר |
| שחר שני חריצים | 1 מ"א | 53 מ"מ | | 3.2 מטר |
| שחר MDF | 1 מ"א | 46 מ"מ | | 2.4 מטר |
| תוספת לחיתוך 45 מעלות | יחידה | | | |
| קופסאת ניפוח כולל בידוד | 1 מ"א | | | |
| קופסאת ניפוח ללא בידוד | 1 מ"א | | | |

*תוספת חיתוך גרונג - 100₪







תריס אספקה קווים-שחר

החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

# מפזר אוויר תקרתי מרובע
# סלוט - SL

**דגמים:**

- ✔ תקרתי חריץ 1
- ✔ תקרתי 2 חריצים

**שניתן לקבל בשני אופנים**
חריץ 1 ו - 2 חריצים.

סלוט מרובע

*ניתן לקבל את הליבה עם אלומיניום, תקרה אקוסטית, תאורה.

## סלוט צר

| תיאור מוצר | מינימום לחישוב | נתונים טכניים | | | |
|---|---|---|---|---|---|
| | | עומק | רוחב | שוליים מבחוץ | שוליים מבפנים |
| סלוט צר חריץ 1 מרובע 600*600 כולל סגירה | יחידה | 25 מ"מ | 20 מ"מ | 20 מ"מ | 7.5 |
| סלוט צר חריץ 1 מרובע 600*600 ללא סגירה | יחידה | 25 מ"מ | 20 מ"מ | 20 מ"מ | 7.5 |
| סלוט צר 2 ח' מרובע 600*600 כולל סגירה | יחידה | 25 מ"מ | 20 מ"מ | 20 מ"מ | 7.5 |
| סלוט צר 2 ח' מרובע 600*600 ללא סגירה | יחידה | 25 מ"מ | 20 מ"מ | 20 מ"מ | 7.5 |


החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

# תריסי אספקה

| תיאור מוצר | מינימום לחישוב | שוליים |
|---|---|---|
| תריס אספקה 150*300 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 200*300 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 300*300 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 150*400 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 200*400 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 150*500 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 200*500 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 150*600 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 200*600 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה 200*800 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| אוויר חוזר משולב אספקה 150*600 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| אוויר חוזר משולב אספקה 200*600 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| אוויר חוזר משולב אספקה 200*800 להב אחורי שחור | 1 יחידה | 27 מ"מ |
| תריס אספקה | 0.1 מ"ר | 27 מ"מ |
| תריס אספקה להב אחורי שחורי | 0.1 מ"ר | 27 מ"מ |
| אספקה לתעלה עגולה | 0.1 מ"ר | 27 מ"מ |
| תריס אספקה 220*220 | 1 יחידה | 27 מ"מ |
| תריס אספקה 300*150 | 1 יחידה | 27 מ"מ |
| תריס אספקה 300*200 | 1 יחידה | 27 מ"מ |
| תריס אספקה 300*300 | 1 יחידה | 27 מ"מ |
| תריס אספקה 400*150 | 1 יחידה | 27 מ"מ |
| תריס אספקה 400*200 | 1 יחידה | 27 מ"מ |
| תריס אספקה 500*150 | 1 יחידה | 27 מ"מ |
| תריס אספקה 500*200 | 1 יחידה | 27 מ"מ |
| תריס אספקה 600*150 | 1 יחידה | 27 מ"מ |
| תריס אספקה 600*200 | 1 יחידה | 27 מ"מ |
| תריס אספקה 800*200 | 1 יחידה | 27 מ"מ |
| תריס אספקה 1500*100 | 1 יחידה | 27 מ"מ |
| תריס אספקה 2000*100 | 1 יחידה | 27 מ"מ |
| אויר חוזר משולב עם אספקה | 0.1 מ"ר | 27 מ"מ |

תריס אספקה שדרות שחורות

אספקה משולב

תריסי אספקה


החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

# עגולים

## תקרות נמוכות

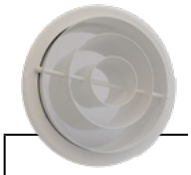
| תיאור מוצר | דגם |
|---|---|
| תקרות נמוכות 6" | RCDL150 |
| תקרות נמוכות 8" | RCDL200 |
| תקרות נמוכות 10" | RCDL250 |
| תקרות נמוכות 12" | RCDL300 |
| תקרות נמוכות 14" | RCDL350 |
| תקרות נמוכות 16" | RCDL400 |
| תקרות נמוכות 18" | RCDL450 |
| תקרות נמוכות 20" | RCDL500 |

## ג'ט - JET

| תיאור מוצר | דגם |
|---|---|
| ג'ט 6" | JET150 |
| ג'ט 8" | JET200 |
| ג'ט 10" | JET250 |
| ג'ט 12" | JET300 |
| ג'ט 14" | JET350 |
| ג'ט 16" | JET400 |

## תקרות גבוהות

| תיאור מוצר | דגם |
|---|---|
| תקרות גבוהות 8" | RCDH200 |
| תקרות גבוהות 10" | RCDH250 |
| תקרות גבוהות 12" | RCDH300 |
| תקרות גבוהות 14" | RCDH350 |
| תקרות גבוהות 16" | RCDH400 |



## ג'ט טבעות- SJ

| תיאור מוצר | דגם |
|---|---|
| ג'ט טבעות 8" | SJ200 |
| ג'ט טבעות 10" | SJ250 |
| ג'ט טבעות 12" | SJ300 |
| ג'ט טבעות 14" | SJ350 |
| ג'ט טבעות 16" | SJ400 |
| ג'ט טבעות 20" | SJ450 |

## תקרות גבוהות מתכוונן

| תיאור מוצר | דגם |
|---|---|
| תקרתי עגול מתכוונן 12" | VSR-12 |
| תקרתי עגול מתכוונן 16" | VSR-16 |
| תקרתי עגול מתכוונן 20" | VSR-20 |

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## ממ"ד

| תיאור מוצר |
|---|
| ממ"ד 8" |
| ממ"ד 8" מסגרת 350*350 |
| ממ"ד 8" מסגרת 380*380 |

## יניקה / אספקה

| תיאור מוצר | דגם |
|---|---|
| אספקה / יניקה 4" | SV/EV-100 |
| אספקה / יניקה 5" | SV/EV-125 |
| אספקה / יניקה 6" | SV/EV-150 |
| אספקה / יניקה 8" | SV/EV-200 |

## תקרתי 4 כיוונים

| תיאור מוצר |
|---|
| תקרתי 6" |
| תקרתי 9" |
| תקרתי 12" |
| תקרתי 15" |
| תקרתי 18" |
| תקרתי 21" |

תקרתי 4 כיוונים

תריס ממ"ד

## מחליפי אריח

| תיאור מוצר |
|---|
| מחליף אריח 6" |
| מחליף אריח 9" |
| מחליף אריח 12" |
| מחליף אריח 15" |
| מחליף אריח 18" |

תריס ממ"ד מסגרת

תריס אספקה/יניקה

## לב אספקה

| תיאור מוצר |
|---|
| תקרתי + לב אספקה 9" |
| תקרתי + לב אספקה 12" |
| תקרתי + לב אספקה 15" |
| תקרתי + לב אספקה 18" |
| מחליף אריח + לב אספקה 9" |
| מחליף אריח + לב אספקה 12" |
| מחליף אריח + לב אספקה 15" |

לב אספקה

## מחליפי אריח

| תיאור מוצר | מינימום לחישוב |
|---|---|
| תקרתי | 0.085 מ"ר |
| תקרתי לב אספקה | 0.085 מ"ר |

מחליף אריח


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# לוח מערבולת

| דגם | מינימום לחישוב |
|---|---|
| SWCI-24 | יחידה |
| SWCI-28 | יחידה |
| SWCI-36 | יחידה |
| SWCI-48 | יחידה |
| SWCI-48/2 | יחידה |

לוח מערבולת

סלוט צר 2 חריצים

דקור

דקור פרימיום

דקור פרימיום שחור לבן

שחר 1 חריץ

ג'ט

תריס נגד גשם

תריס אספקה עגול

תריס יניקה עגול

תריס אספקה
שדרות שחורות

תריס אוויר חוזר
נפתח על ציר

תקרתי 4 כיוונים



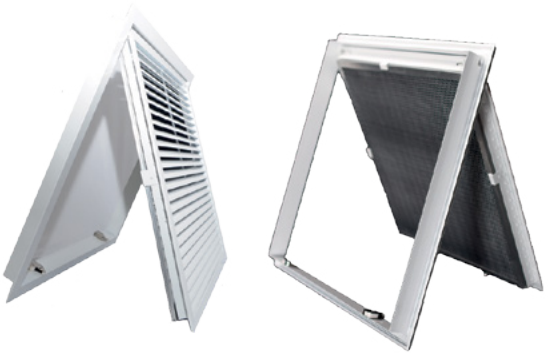
החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

## אוויר חוזר

| תיאור מוצר | מינימום לחישוב |
|---|---|
| תריס אוויר חוזר | 0.1 מ"ר |
| אוויר חוזר + ווסת | 0.1 מ"ר |
| אוויר חוזר קבוע + פילטר | 0.1 מ"ר |

## אוויר חוזר נפתח על ציר

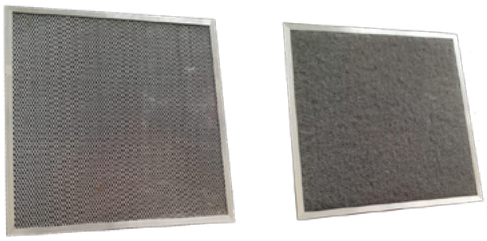
| תיאור מוצר | מינימום לחישוב |
|---|---|
| אוויר חוזר נפתח על ציר + פילטר | 0.2 מ"ר |
| אויר חוזר ליבה נשלפת ללא ציר עם פרפרים | 0.2 מ"ר |
| אוייר חוזר נ.ע.צ + פילטר 540 * 540 | יחידה |
| אוייר חוזר נ.ע.צ + פילטר 540 * 540 פינליין | יחידה |
| אויר חוזר פלסטיק עם פרפרים 540 * 540 | יחידה |
| אוייר חוזר נ.ע.צ + פילטר 545 * 545 - חוץ 600 | יחידה |
| אוייר חוזר נ.ע.צ + פילטר 550 * 550 - חוץ 605 | יחידה |
| אוייר חוזר נ.ע.צ + פילטר 590 * 590 | יחידה |
| אוייר חוזר נ.ע.צ + פילטר 595 * 595 | יחידה |
| אוייר חוזר 540 * 540 - חוץ 595 | יחידה |
| אוייר חוזר 540 פינליין | יחידה |
| אויר חוזר 545 * 545 - חוץ 600 | יחידה |
| אויר חוזר 595 * 595 | יחידה |
| אויר חוזר 140 * 140 | יחידה |
| אויר חוזר 190 * 190 | יחידה |



אוויר חוזר נפתח על ציר

## פילטר

| תיאור מוצר | מינימום לחישוב |
|---|---|
| פילטר חסין אש עם מסגרת | 0.2 מ"ר |



## אוויר חוזר נגד גשם

| תיאור מוצר | מינימום לחישוב |
|---|---|
| אויר חוזר נגד גשם להב רחב | 0.1 מ"ר |
| אויר חוזר נגד גשם להב צר | 0.1 מ"ר |
| מחיר לרשת | 0.25 מ"ר |
| אויר חוזר נגד גשם עגול 4" | יחידה |
| אויר חוזר נגד גשם עגול 6" | יחידה |
| אויר חוזר נגד גשם עגול 8" | יחידה |
| אויר חוזר נגד גשם עגול 10" | יחידה |
| אויר חוזר נגד גשם עגול 12" | יחידה |
| אויר חוזר נגד גשם עגול 14" | יחידה |
| אויר חוזר נגד גשם כיפתי 4" | יחידה |
| אויר חוזר נגד גשם כיפתי 6" | יחידה |
| אויר חוזר נגד גשם כיפתי 8" | יחידה |

תריס נגד גשם

תריס נגד גשם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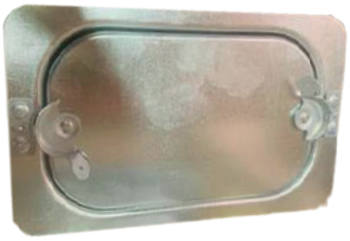
## דלת שירות - AD

| תיאור מוצר | מינימום לחישוב |
|---|---|
| דלת שירות | 0.2 מ"ר |
| דלת שירות ללא ציר | 0.1 מ"ר |
| דלת שירות 400*400 | יחידה |
| דלת שירות 500*500 | יחידה |
| דלת שירות 600*600 | יחידה |
| דלת שירות 540*540 | יחידה |

דלת שרות AD

## דלת שירות למדף אש - AC

| תיאור מוצר | מינימום לחישוב |
|---|---|
| דלת שירות מדף אש 295*205 | יחידה |
| דלת שירות מדף אש 295*430 | יחידה |



דף שרות למדף אש - AC

## דלת גבס - DG

| תיאור מוצר | מינימום לחישוב |
|---|---|
| דלת שרות גבס 400*400 | יחידה |
| דלת שרות גבס 500*500 | יחידה |
| דלת שרות גבס 600*600 | יחידה |

דלת גבס - DG

## ווסתים - OBD-VCD

| תיאור מוצר | מינימום לחישוב |
|---|---|
| ווסת כמות | 0.1 מ"ר |
| דמפר וויסות ידני גלגלי שיניים נסתרים | 0.25 מ"ר |
| דמפר וויסות גלגלי שניים נסתרים הכנה למנוע/מנוע נסתר | 0.25 מ"ר |
| מדף ויסות עגול 6" | יחידה |
| מדף ויסות עגול 8" | יחידה |
| מדף ויסות עגול 10" | יחידה |

ווסתים OBD-VCD

## אל חזור

| תיאור מוצר | מינימום לחישוב |
|---|---|
| תריס/דמפר אל חזור | 0.25 מ"ר |

*להב קצר בתוספת 30%

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RAM
Airflow

# פתרונות מיזוג אוויר מעוצבים

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